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Obstructions to the Hasse Principle for Curves

by

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Abstract

In this thesis, we examine the failure of the Hasse Principle for one-dimensional, complete, regular, geometrically integral schemes of finite type over a number field with genus 1. Via generalised reciprocity laws, one establishes an obstruction to the existence of global points on a scheme. We examine a promising method pioneered by Bright and Siksek that employs certain rational functions on the scheme and the law of Artin reciprocity to produce a Brauer-Manin obstruction. We derive an algorithm that allows for the computation of many examples in the case of 2-coverings. We also examine an example of a 3-covering in-depth, specifically the famous cubic curve of Selmer.

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1. INTRODUCTION

One of the fundamental motivations of arithmetic geometry is to find rational solutions to a system of equations with rational coefficients. For example, we might be interested in determining if a Diophantine equation, such as

$$x^2 + y^2 + 1 = 0, \tag{1.1}$$

is soluble in $\mathbb{Q}$, i.e. determine if there is some $(x, y) \in \mathbb{Q} \times \mathbb{Q}$ satisfying (1.1). However, one easily sees that (1.1) has no solutions in $\mathbb{Q}$ since it has no solutions in $\mathbb{R}$, which follows because squares of real numbers are always positive.

For many equations, it suffices to show that there are no solution in $\mathbb{Z}$. In this case, we can often prove the equations have no solutions modulo n for some $n \in \mathbb{Z}_{\geq 2}$. For example, consider the equation:

$$y^2 = -(x^2 - z^2)^2 + 24z^4. \tag{1.2}$$

First, we can multiply through by an appropriate integer, which means we can assume that x, y, z are integers and that there is no common factor dividing the numbers x, y, z . Then, one computes that it has no solutions modulo 9 except $(x, y, z) \equiv (0, 0, 0) \pmod{9}$, which means 9 divides each of x, y, z , contradicting the fact that x, y, z are coprime. We conclude there are no solutions to (1.2) in $\mathbb{Q}$.

More generally, given a finite algebraic field extension K of $\mathbb{Q}$, we search for solutions with values in K to equations with coefficients in K . If we suspect that there are no K -valued solutions, we may be able to prove this by showing that there are no local solutions. By which we mean: given a prime ideal $\mathfrak{p}$ in the ring of integers $\mathcal{O}_K$ of K , we can obtain a field K_v that is complete with respect to the valuation $v : K^* \rightarrow \mathbb{Z}$ that sends $\alpha \in K$ to the maximal integer n such that $\alpha \in \mathfrak{p}^n$ [Neu99, Section II.4 pp 123-124]. These fields are called p -adic fields, and there is a natural inclusion $K \subseteq K_v$.

Generalising the ideas outlined for equations (1.1) and (1.2), one may then be able to show that there are no K_v -valued solutions to the original system over K . Indeed, by multiplying the equations by an appropriate element of $\mathcal{O}_K$, this often reduces to checking if it admits a solution over a finite field via Hensel's lemma [Lan95, Proposition III.2.2 p 42], similar to equation (1.2). Furthermore, since the fields K are finite algebraic extensions of $\mathbb{Q}$, there are embeddings $K \hookrightarrow \mathbb{C}$, which determine absolute values on K that return one of the complete fields: $\mathbb{R}$ or $\mathbb{C}$ [Neu99, Theorem II.4.3 p 124]. Thus, we can also see how the proof of the lack of solubility to equation (1.1) also fits into this framework.

The embeddings $K \hookrightarrow \mathbb{C}$ and the valuations $v : K^* \rightarrow \mathbb{Z}$ form the places of a number field K [Neu99, Definition III.1.1 p 183]. If for every place, which we denote by v , there is a solution in K_v to the system but no global solution in K , we say that the system of equations violates the Hasse principle. On the positive side, there are classes of equations that satisfy the Hasse principle, i.e. it is sufficient to conclude a solution in K exists if we can find a solution for every place v . For example, the class of conic equations, equations where each term has degree 2 in three variables, is one of the simplest

examples known to satisfy the Hasse principle [Voi21, Theorem 14.6.9 p 233]. If we instead consider equations where each term has degree 3, so-called cubic curves, then the Hasse principle is violated for various examples, such as the Selmer curve [Sel51, p 205]:

$$3X^3 + 4Y^3 + 5Z^3 = 0.$$

This latter class of equations is of particular interest in the study of elliptic curves. To each elliptic curve E over K , we can associate a cubic curve with coefficients in K that has a solution in K [Sil09, Proposition III.3.1 p 59]. One also attaches a group law to E that adds and subtracts points in a particular way via a theorem of algebraic geometry [Sil09, Lemma III.3.3 and Proposition III.3.4 pp 61-62]. We are often interested in knowing the group structure of the K -valued points, and using certain cubic equations, we may be able to determine this structure. This comes down to checking if these cubic equations admit a point in K . If there is no point, this can often be decided by finding a single prime $\mathfrak{p}$ and showing that there are no solutions to the cubic equation to associated p -adic field K_v . However, as just discussed, this class is known to fail the Hasse principle, complicating this process. Therefore, it is of great interest to determine if a given equation that we suspect violates the Hasse principle actually does so.

In this thesis, we will prove and utilise a method developed by Bright and Siksek in [BS08] that is capable of proving that certain curves violate the Hasse principle. We explain the contents of the original paper in a more accessible way, elaborating on many ideas that were only briefly mentioned or alluded to. The core idea of the method is to employ a generalisation of quadratic reciprocity to show that there are no points on the curve. The method looks for rational functions that have a certain Galois extension L/K associated to them via class field theory. To explain this somewhat informally, we prove that there are group morphisms that take the points in the p -adic field K_v and map them to elements in the Galois group $G(L/K)$. There is an additional map that sends any point with values in K to $1 \in G(L/K)$ that factors through all these local maps. This creates room to show there are no global points using the interactions of the images of the local points in $G(L/K)$ via a generalised form of reciprocity.

The method is considerably useful for finding a proof that there is a violation. To this end, we develop algorithms that are capable of producing such a proof for certain classes of curves. In the original paper, it is shown that the function $f = \frac{x-\alpha}{x-\beta}$, with α, β rational numbers such that $\mathbb{Q}(\sqrt{f(\alpha)}) = \mathbb{Q}(\sqrt{f(\beta)})$, is always capable of producing a field extension $L \neq K$. Thus, we wrote an algorithm that can check if a 2-covering of an elliptic curve produces an obstruction via the functions f . We tested this method on 116 curves, using about 2300 functions in total.

To go one step further, we also apply the method to a cubic curve that is a 3-covering of an elliptic curve. We still use a function of the form $f = \frac{x-\alpha}{x-\beta}$, but the method is complicated in two ways. The first being that the ground field is no longer $\mathbb{Q}$ but $K = \mathbb{Q}(\rho)$, with ρ a primitive third root of unity. This field has ring of integers $\mathcal{O}_K = \mathbb{Z}[\rho]$. This ends up not being a major issue since $\mathbb{Z}[\rho]$ is also a unique factorisation domain, like $\mathbb{Z}$. The second complicating factor is that we now wish to compute $E(K_v)/3E(K_v)$. This is a more intricate process than the case of $n = 2$, but the cubic curve we examine admits a 3-isogeny, which will simplify the process. Thus, we set to work on implementing these computations in Magma, and we managed to produce a new Brauer-Manin obstruction.

We now outline the content of the chapters to come. In the first two sections of Chapter 2, we begin with a review of the concepts of completions and field extensions of p -adic fields. The theory is often similar, but it is not exactly the same as in the case of number fields; however, we are often awarded simpler and nicer results. This is followed by a section on infinite Galois theory; in particular we

need to assign these groups a topology to recover the fundamental theorem. Then Section 2.4 covers the basics of cohomology, with particular focus on Galois cohomology and Tate cohomology. This is followed by a review of scheme theory concepts in Section 2.5. Of particular importance, we define what a variety will be in this thesis. We then round off the preliminaries with Section 2.6, which is a brief review of elliptic curves. We will recall the Tate-Shafarevich group as well, which is of the utmost importance in the study of the failure of the Hasse principle.

In Chapter 3, we generalise the notion of quadratic reciprocity. We discuss the generalised law of reciprocity and local class field theory in Section 3.1. This is followed by a brief introduction into the theory of adèles and idèles in Section 3.2. We follow this with a key result: Artin reciprocity and its consequences in Section 3.3. We close this chapter with Section 3.5, which briefly touches on the Brauer-group of a field and the local-global principle arising in this case.

In Chapter 4, we discuss p-adic schemes and how they connect to varieties defined over number fields. We follow this with Section 4.2, which concerns n -coverings of elliptic curves, and their importance in the theory of descent. Then we discuss the Brauer group of a scheme and its connection to the function field $K(X)$ associated to X in Section 4.3. We touch on Lind’s counterexample and a method from Siksek that produces obstructions using quadratic reciprocity. This serves as the natural transition into the method of Bright and Siksek. In Section 4.4, We state the main theorem, and we discuss the technicalities of a moving lemma that will be essential for ensuring that the method they devised is actually useful for computations.

In Chapter 5, we devise all of the algorithms and computations to utilise the method of Bright and Siksek. We compute many examples for the case of 2-coverings, and we cover 2 examples in depth in Section 5.1. The remaining computations are found in the Github repository of [Roz]. In Section 5.2, we then cover an in-depth example for the case of a 3-covering in the form of Selmer’s curve: we discover a new obstruction of the Brauer-Manin type. Originally studied by Selmer in [Sel51], where he proved it had no global solutions via an argument from algebraic number theory. We find a rational function that will produce an obstruction of the Brauer-Manin type, which serves an elegant proof for the lack of global solubility.

In the appendices, we provide a brief overview of schemes and their connection to varieties, specifically via an equivalence of categories and the usage of the functor of points that recovers the geometry of varieties. The end of Appendix A covers the basics of models of curves over valued fields. In Appendix B, we build on the ideas of models, and talk about fibred surfaces. This is followed by a brief theorem on Néron models of elliptic curves and useful theorem for computing the cardinality of the Mordell-Weil group $E(K)/nE(K)$ for K a local field.

2. PRELIMINARIES

There are two important and deeply intertwined notions to be discussed in this thesis: Artin reciprocity and the Hasse principle. As mentioned in the introduction, we are interested in p -adic fields since we want to use them to study rational points to a system of equations. Beyond their usage in the Hasse principle, they will also play a key role in class field theory, and in particular, Artin reciprocity. This latter result also requires knowledge of Galois theory, especially infinite Galois groups and their topology.

Another important area of study to both class field theory and varieties is cohomology. It is used to determine rational points on elliptic curves, surjectivity of norm maps between certain modules, and determine certain algebras over a field up to a specific equivalence, among many things. Thus it is hard to understate its importance.

The modern number theorist should also have a solid understanding of scheme theory. That being said, knowledge of classical variety theory is sufficient to comprehend most of the material in this thesis. It is important to have some exposure to the theory of elliptic curves, but we will review their cohomology and other related concepts, such as Tate-Shafarevich groups.

Therefore, in the background, we will cover the basics of local field theory: p -adic completions and their extensions via [Neu99] and [Lan95]. In the sequel, we delve into profinite groups and cohomology using [Neu99], [Har17] and [NSW13]. Then this will be followed up by a discussion of scheme theory and its relation to classical variety theory using ideas and results from [LE02], [Mum04], [Vak25] and [Har77]. In the final section, we review the Tate-Shafarevich group of an elliptic curves from [Sil09].

2.1 Completions

In this section, we briefly examine p -adic completions and their relations to algebraic number fields, following [Neu99, Section II.4] and [Lan95, Chapter II]. The first step to discussing the Hasse principle is to determine what it means to have a local solution on variety over a number field K . To this end, we discuss completions with respect to absolute values. We have the following definition.

Definition. Given a field F , an **absolute value** is a map $|(-)| : F \rightarrow \mathbb{R}_{\geq 0}$ satisfying the following properties:

1. For all $x \in F$, $|x| = 0$ if and only if $x = 0$.
2. For all $x, y \in F$, $|xy| = |x||y|$.
3. For all $x, y \in F$, $|x + y| \leq |x| + |y|$.

If an absolute value satisfies $|a + b| \leq \max(|a|, |b|)$, it is said to be **non-Archimedean** and is called **Archimedean** otherwise.

Note that absolute values are also referred to as **valuations** as in [Neu99]. A valuation that is not associated to a discrete valuation is called an **Archimedean valuation** and is called **non-Archimedean valuation** otherwise.

Absolute values are also norm maps, so we view F as a metric space, which returns the usual metric topology given by open balls $B_\alpha(r) := \{x \in F \mid |x - \alpha| < r\}$. As a metric space, we know that F is also Hausdorff [Mun00, p. 129], i.e. distinct points have open neighbourhoods that have empty intersection. Moreover, we have that F , together with its metric, forms a topological ring: multiplication $\cdot : L \times L \rightarrow L$ and addition $+$: $L \times L \rightarrow L$ are continuous maps.

Let K be a number field. We recall the usual absolute value $|\cdot| : K \rightarrow \mathbb{R}_{\geq 0}$, which can either be restricted from $\mathbb{R}$ or $\mathbb{C}$ depending on whether $K/\mathbb{Q}$ is a real or complex extension. This valuation is Archimedean since, for example, we have that $|1 + 1| = |2| = 2 > \max(|1|, |1|)$. On the other hand, consider the ring of integers $\mathcal{O}_K \subseteq K$, the unique maximal order of K . Let $\mathfrak{p} \in \text{Spec } \mathcal{O}_K$ with $\mathfrak{p}|p$ for some rational prime p . Since $\mathcal{O}_K$ is a Dedekind domain, we have that $\mathcal{O}_{K,\mathfrak{p}}$ is a discrete valuation ring of K so that $\mathfrak{p}\mathcal{O}_{K,\mathfrak{p}} = \pi\mathcal{O}_{K,\mathfrak{p}}$ for some element $\pi \in \mathcal{O}_{K,\mathfrak{p}}$. Then we have that for each $\alpha \in K$, there exists a unique integer n such that $\alpha = u\pi^n$ in $\mathcal{O}_{K,\mathfrak{p}}$. This defines a valuation $v_{\mathfrak{p}} : K^* \rightarrow \mathbb{Z}$ in the usual sense as seen in [AM69, Chapter 5]. To this discrete valuation we associate an absolute value for a choice of real number $q \in \mathbb{R}$, and as in [Lan95, II.1 p 34], we select $q := p^{\frac{1}{v_{\mathfrak{p}}(p)}}$:

$$|\alpha|_{\mathfrak{p}} = q^{-v_{\mathfrak{p}}(\alpha)}. \quad (2.1)$$

Definition. - Let K be a number field. We define the **places of K** as

$$\mathfrak{M}(K) := \{\text{valuations on } K\} / \sim,$$

where we say that $|(-)|_1 \sim |(-)|_2$ if there exists $c \in \mathbb{R}_{>0}$ such that $|(-)|_1 \leq c|(-)|_2$.

Note that this coincides with the definition of equivalence given in [Neu99, Proposition II.3.3 p 117], which requires that $|\cdot|_1 = c|\cdot|_2$, by using that $\mathbb{R}_{\geq 0}$ has the Archimedean property from analysis. Using [Neu99, Proposition II.3.3 p 117], this is equivalent to the fact that the topologies defined from both absolute values are homeomorphic.

Then given a valuation on a number field K , we may restrict it to obtain a valuation on $\mathbb{Q}$. Thus, it turns out that all valuations on a number field are equivalent to either an absolute value defined on $\mathbb{R}$ or $\mathbb{C}$ or a p -adic absolute value [Neu99, Proposition II.3.7 p 119]. We may denote all valuations using v , where we will interchangeably use this notation to refer to both the $\mathfrak{p}$ -adic valuation or the associated absolute value $|\cdot|_{\mathfrak{p}}$ in the non-Archimedean case.

Now, we can complete K as a metric space with respect to a place v to obtain K_v . To do this, as in [Neu99, Section II.4 pp 123-124], let (α_i) be any sequence $\mathbb{N} \rightarrow K$ satisfying the Cauchy condition: for $\epsilon > 0$ there exists some $N \in \mathbb{N}$ such that for all $i, j \geq N$, $|\alpha_i - \alpha_j| < \epsilon$. We may form a ring R of these sequences with multiplication and addition defined componentwise. This ring has a maximal ideal $\mathfrak{m}$ containing all sequences that converge to zero. Then we have that $K_v \cong R/\mathfrak{m}$. We obtain a natural inclusion map $K \rightarrow K_v$ defined by $\alpha \mapsto (\alpha)$.

Definition. We call a field **local** if it is a field extension of a number field that is complete with respect to any valuation $v \in \mathfrak{M}(K)$.

We note that in [Neu99], a local field is actually one that is complete with respect to a discrete valuation. In other sources, such as [Voi21, Definition 12.2.15 p 181], a local field encompasses both Archimedean and non-Archimedean valued fields, which is the definition we opt to use.

If our valuation is Archimedean, it turns out that $K_v \cong \mathbb{R}$ or $\mathbb{C}$ depending on if v is equivalent to the valuation restricted from $\mathbb{R}$ and $\mathbb{C}$, respectively by [Neu99, Theorem II.4.3 p 124].

In the non-Archimedean case, we have a valuation v that can be extended to K_v and all such fields will be called **p -adic fields**. Let $\mathcal{O}_v := \{x \in K_v | v(x) \geq 0\}$ and $\mathfrak{p}_v := \{x \in K_v | v(x) \geq 1\}$. Then we have that $\mathcal{O}_v$ is the discrete valuation ring and unique maximal order of K_v , with maximal ideal $\mathfrak{p}_v$ [Lan95]. We give an explicit basis of K_v as follows. Since K_v is a metric space, it is also Hausdorff, so compact sets are open and closed. The prime ideal powers $\mathfrak{p}_v^r$ form a **fundamental system of neighbourhoods** of 0 since $B_0(p^{-r/v(p)}) = \{x \in K_v | |x|_{\mathfrak{p}_v} < p^{-r/v(p)}\} = \{x \in K_v | v(x) > r\} = \mathfrak{p}_v^{r+1}$. In particular, we have that the closed ball $\mathcal{O}_v$ is also an open subset of K_v , so that the sets $a + \mathcal{O}_v$ are open subsets of K_v , making it locally compact [Neu99, I.5.1 p 135].

2.2 Finite Extensions of p -Adic Fields

Suppose we take a p -adic field K and some irreducible polynomial $f \in K[x]$ and look at the algebraic extension $L = K[x]/(f(x))$. It turns out that L is also a complete field by [Neu99, Theorem II.4.8 p 131]. Let $\mathfrak{P}$ be the corresponding unique maximal ideal of the valuation ring of L , i.e. $\mathfrak{P}|\mathfrak{p}$. The absolute value $|(-)|_{\mathfrak{P}}$ from (2.1) extends $|(-)|_{\mathfrak{p}}$ and is the unique one up to equivalence by [Neu99, Theorem II.6.2 p 144].

Returning to the case that L/K is an extension of number fields given by an irreducible polynomial $f \in K[x]$, if $w : L^* \rightarrow \mathbb{Z}$ is a valuation extending $v : K^* \rightarrow \mathbb{Z}$, a natural question is when $L_w \cong K_v[x]/(f(x))$? It is possible that the minimal polynomial f of L/K will no longer be irreducible over K_v . From [Lan95], we admit the the following crucial isomorphism:

$$K[x]/(f(x)) \otimes_K K_v \cong \prod_{w|v} L_w =: L_v. \quad (2.2)$$

The left-hand side has $K[x]/(f(x)) \otimes_K K_v \cong K_v[x]/(f(x))$. As such we may determine this by purely considering the residue fields $K(\mathfrak{p})$ of the primes $\mathfrak{p} \in \text{Spec } \mathcal{O}_K$. From [Neukirch, Theorem II.6.2 p 144], we also know that $\mathcal{O}_w$ is the integral closure of $\mathcal{O}_v$ in L_w .

Definition. Let K be a number field and $v \in \mathfrak{M}(K)$ a non-Archimedean place. We call the place v **inert** if there is only one place $w|v$ such that $[L_w : K_v] = [L : K] = [\mathcal{O}_w/\mathfrak{P} : \mathcal{O}_v/\mathfrak{p}]$. If $w|v$ has $[L_w : K_v] \neq [\mathcal{O}_w/\mathfrak{P} : \mathcal{O}_v/\mathfrak{p}]$, then we say v **ramifies** in L . The place v is said to be **totally ramified** if $w|v$ has $[L_w : K_v] = [L : K]$, but there is an isomorphism of residue fields: $\mathcal{O}_w/\mathfrak{P} \cong \mathcal{O}_v/\mathfrak{p}$. Lastly, we call v **totally split** if the number of places $w|v$ is equal to $[L : K]$.

Now using Galois theory of number fields, we have that if $K/\mathbb{Q}$ is Galois, then

$$[K : \mathbb{Q}] = e_p f_p g_p,$$

where e_p is the ramification index of p in K , f_p is the residue class degree and g_p is the number of extensions, see [Ste08, equation 15-2 p 259].

We close this section off by introducing Hensel's lemma, but in its analytic incarnation. This theorem reduces the process of determining p -adic solubility of an integral polynomial $f \in \mathcal{O}_v[x]$ to checking if there exists an integer n such that $\bar{f} \in \mathcal{O}_v/\mathfrak{p}_v^n[x]$ has a solution.

Theorem 2.2.1. (Hensel's Lemma) Let K be a number field, $v \in \mathfrak{M}(K)$ and $f \in \mathcal{O}_v[x]$. Suppose there exists an $a_0 \in \mathcal{O}_v$ such that

$$v(f(a_0)) > 2v(f'(a_0)).$$

Then there exists an element $a \in \mathcal{O}_v$ such that $f(a) = 0$.

Proof. See [Lan95, Proposition III.2.2 p 42]. □

2.3 Infinite Galois Theory

Infinite Galois theory is fundamental in the study of the cohomology of fields and of curves. Therefore, we recall the basics from [Neu99, Sections IV.1-2 pp 261-274]. An infinite Galois extension L/K is defined as the splitting field of some set of polynomials in K , and an infinite Galois group $G(L/K)$ is the group of K -automorphisms of L . To this group, we assign it a canonical topology; it is important for generalising the results of Galois theory beyond finite extensions.

Definition. Let L/K be a Galois extension. There is a canonical topology on $G(L/K)$ called **the Krull topology** given as follows: the open subsets are defined to be the cosets $\sigma G(L/M)$ for M/K a finite subextension of L/K .

In fact, we may realise an (infinite) Galois group as a projective limit: $G(L/K) \cong \varprojlim_M G(M/K)$ for $M \subseteq L$ such that M/K is a finite extension [Neu99, IV.2 1 p 271]. Since the groups $G(M/K)$ are finite they are compact, and the projective limit of compact groups is also compact since products preserve compactness [Neu99, Proposition IV.2.3 p 266]. One also easily shows that $G(L/K)$ is Hausdorff as in [Neu99, Proposition IV.1.1 p 262]. Going forward, we will also often write G_K for the Galois group $G(K_s/K)$, where K_s denotes the separable closure of K . Recall, if K is perfect, then $K_s = \bar{K}$, the algebraic closure. For any finite extension $M \supseteq L \supseteq K$ of Galois groups we have

$$0 \longrightarrow G(M/L) \longrightarrow G(M/K) \longrightarrow G(L/K) \longrightarrow 0.$$

Then using the exactness of projective limits [Neu99, Proposition IV.2.7 p 269], we obtain

$$0 \longrightarrow G_L \longrightarrow G_K \longrightarrow G(L/K) \longrightarrow 0,$$

which shows that $G_K/G_L \cong G(L/K)$. Then the main theorem of Galois theory is modified to be the following bijection

$$\{M/K \text{ Galois subextensions of } L/K\} \longleftrightarrow \{\text{Closed subgroups of } G(L/K)\}$$

by [Neukirch, thm IV.1.2 p 263].

More generally we have the notion of a **profinite group**, which is a group that is Hausdorff and compact, with a fundamental system of neighbourhoods of 1 given by normal subgroups. This is characterised by the following proposition from [Neu99, Proposition IV.2.8 p 270]:

Proposition 2.3.1. A topological group G is profinite if and only

$$G \cong \varprojlim_N G/N$$

where N varies over all normal open subgroups of G .

2.4 Cohomology

In this section, we explore the basics of group cohomology, which will be important in the class field theory of Chapter 3 and much of the arithmetic geometry throughout the thesis. We will explain the connection between Galois cohomology and Tate cohomology, the latter which measures the extent to which norm maps between Galois modules fail to be surjective. We also obtain the

important notion of long exact sequences, which are typical of cohomology. We close the section with a discussion of cohomology of schemes. The literature we consulted in writing this section was [NSW13, Chapter I], [GS06] and [Har17].

Group Cohomology

Definition. Let G be a profinite group and M an abelian group such that there is a (left) action of G on M compatible with the group structure of M , in the sense that $(m_1 + m_2)^\sigma = m_1^\sigma + m_2^\sigma$ for $\sigma \in G$ and $m_i \in M$. We call M a **G -module**.

This yields a category of G -modules, where morphisms φ are group homomorphisms also compatible with the group action. It turns out that this category forms an abelian category, which means we admit all limits and colimits, there is a the snake lemma and the five lemma, etc. [Har17, Section 1.2 and Appendix A.3-A.4]. We take particular interest in the subgroup of M fixed by G , denoted M^G . We admit a functor $M \mapsto M^G$ between G -modules and abelian groups, which sends maps to their restriction $\varphi \mapsto \varphi|_{M^G}$. This functor is left-exact, meaning that given any exact sequence of G -modules:

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0,$$

we obtain a left-exact sequence in abelian groups:

$$0 \longrightarrow L^G \longrightarrow M^G \longrightarrow N^G.$$

by [NSW13, Section II.6 p 132].

Since $M \mapsto M^G$ is a left-exact functor and the category has enough injectives [Har17, 35.A (c) p 325], we are able to assign a right-derived functor $H^i(G, \) : G\text{mod} \rightarrow Ab$ that induces long exact sequences [Har17, Section 1.3 p 19]. In order to compute these cohomology groups, we now define the associated cochain complexes. Consider $C^n(G, M) := \{\varphi : G^n \rightarrow M\}$, which forms a group under addition; it has identity map $(g_0, \dots, g_{n-1}) \mapsto 0$. We define a map from [Har17, Theorem 1.27]:

$$d^n : C^{n-1}(G, M) \longrightarrow C^n(G, M)$$

$$d^n(\varphi)(g_1, \dots, g_n) = \varphi(g_2, \dots, g_n)^{g_1} + \sum_{i=1}^{n-1} (-1)^i \varphi(g_1, \dots, g_{i-1}, g_i g_{i+1}, \dots, g_n) + (-1)^n \varphi(g_1, \dots, g_{n-1})$$

Now each d^i is a group homomorphism. One may check we have the defining property of the cochain complex: $d^{n+1} \circ d^n = 0$. We therefore have a resolution which we denote $C^\bullet(G, M)$:

$$0 \longrightarrow C^0(G, M) \longrightarrow C^1(G, M) \longrightarrow C^2(G, M) \longrightarrow \dots$$

Therefore, we put

$$\hat{H}^i(G, M) := \frac{\ker(d^{i+1} : C^i(G, M) \rightarrow C^{i+1}(G, M))}{\text{im}(d^i : C^{i-1}(G, M) \rightarrow C^i(G, M))}$$

It turns out that the $\hat{H}^i(G, M)$ coincides with $H^i(G, M)$, the cohomology obtained from using the theory of right-derived functors by [Har17, Theorem 1.27 p 25]; going further we will denote $H^i(G, M) := \hat{H}^i(G, M)$. When $i = 0$, we can compute using the above definitions that $C^0 := \{\varphi : 1 \rightarrow M\} \cong M$ and $d^1(\varphi)(g) = m^g - m$. Hence, we have that

$$H^0(G, M) = \ker(d^1 : C^0(G, M) \longrightarrow C^1(G, M)) = M^G.$$

Likewise, we can show that

$$H^1(G, M) = Z^1(G, M)/B^1(G, M)$$

where

$$Z^1(G, M) := \{\xi : G \rightarrow M \mid \xi(\sigma\tau) = \xi(\sigma)^\tau + \xi(\tau)\}$$

is the **group of 1-cocycles** and

$$B^1 := \{\xi : G \rightarrow M \mid \xi = m^\sigma - m, \exists m \in M\}$$

is the group of **1-coboundaries**. As usual in the study of cohomology, given a short exact sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$, we obtain a long exact sequence (see [Har17] Theorem 1.17):

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(G, L) & \longrightarrow & H^0(G, M) & \longrightarrow & H^0(G, N) \\ & & & & & & \downarrow \delta \\ & & \hookrightarrow & H^1(G, L) & \longrightarrow & H^1(G, M) & \longrightarrow & H^1(G, N) \\ & & & & & & \downarrow \delta \\ & & \hookrightarrow & H^2(G, L) & \longrightarrow & H^2(G, M) & \longrightarrow & H^2(G, N) \longrightarrow \dots \end{array} \quad (2.3)$$

Take K to be a perfect field. Group cohomology is essential for studying rational points on a curve C/K , especially in the case that C is a group scheme. For example, there is a natural action of the Galois group G_K given on the group $E(\bar{K})$; for more details on this action we refer one to Appendix A. Given an isogeny $\varphi : E \rightarrow E'$ of degree n , there is the well-known Kummer sequence of elliptic curves:

$$0 \longrightarrow E(\bar{K})[\varphi] \longrightarrow E(\bar{K}) \xrightarrow{\varphi} E'(\bar{K}) \longrightarrow 0.$$

This induces an exact sequence by truncating the long exact sequence from (2.3):

$$0 \longrightarrow E(K)[\varphi] \longrightarrow E(K) \longrightarrow E'(K) \longrightarrow H^1(K, E[\varphi]) \longrightarrow H^1(K, E) \longrightarrow H^1(K, E'),$$

where we identify, for example, $H^0(K, E(\bar{K})) = E(K)$.

The following two definitions will be important for constructing counterexamples to the Hasse principle. We will see them primarily used in Chapter 4 to explain the so-called Brauer-Manin obstruction.

Definition. If G is profinite and $N \subseteq G$ is an open subgroup, for any G -module M , we have that M is naturally also an N module, and there is a canonical **restriction** map

$$\text{res} : H^i(G, M) \longrightarrow H^i(N, M).$$

If $N \subseteq G$ is normal, then there is an action of G/N on M^N and a canonical **inflation** map

$$\text{inf} : H^i(G/N, M^N) \longrightarrow H^i(G, M).$$

Definition. Given two G modules A, B for any $p, q \geq 0$ one obtains a formal map

$$H^p(G, A) \times H^q(G, B) \longrightarrow H^{p+q}(G, A \otimes_{\mathbb{Z}} B)$$

called a **cup-product pairing** obtained from the pairing

$$\begin{aligned} C^p(G, A) \times C^q(G, B) &\longrightarrow C^{p+q}(A \otimes_{\mathbb{Z}} B) \\ (a \cup b)(g_0, g_1, \dots, g_{p+q}) &= a(g_0, g_1, \dots, g_p) \otimes b(g_{p+1}, \dots, g_{p+q}) \end{aligned}$$

since $a \cup b$ indeed defines a $(p+q)$ -cocycle by [NSW13, Proposition 1.4.1 p 36].

We then have the following proposition which says that inflation maps and restriction maps are compatible with the cup product.

Proposition 2.4.1. Let $N \subset G$ be a subgroup of H . Let $a \in H^p(G, A)$ and $b \in H^q(G, B)$ then

$$\text{res}(a \cup b) = \text{res}(a) \cup \text{res}(b).$$

If N is normal, then for $a' \in H^p(G/N, A^N)$ and $b' \in H^q(G/N, B^N)$ we have

$$\text{inf}(a' \cup b') = \text{inf}(a') \cup \text{inf}(b').$$

Proof. This is clear from the definitions. □

Now we move onto a discussion of the basics of **Tate Cohomology**, for example as seen in [Neu99, Section IV.3]. One may also wish to consult [NSW13] for a more in depth explanation. Let L/K be a finite Galois extension and A_L a $G(L/K)$ -module and $A_K := H^0(G(L/K), A_L)$. Then there is a canonical norm map

$$\begin{aligned} N_{L/K} : A_L &\longrightarrow A_K \\ a &\longmapsto \prod_{\sigma \in G(L/K)} a^\sigma. \end{aligned}$$

Definition. With everything as defined above, we have the following groups:

$$H_T^i(G(L/K), A_L) := \begin{cases} A_K / N_{L/K} A_L & \text{if } i = 0 \\ H^n(G(L/K), A_L) & \text{if } i \geq 1 \end{cases},$$

so that the higher Tate cohomology groups agree with those from Galois cohomology. Further, using the notation from [Neu99], we define the following:

$$N_{L/K} A_L = \{a \in A_L \mid N_{L/K}(a) = 1\}$$

$$I_{G(L/K)} A_L := \langle a^{\sigma-1} \mid a \in N_{L/K} A_L, \sigma \in G(L/K) \rangle$$

and their quotient:

$$H_T^{-1}(G(L/K), A_L) := N_{L/K} A_L / I_{G(L/K)} A_L$$

This leads to the classic theorem of Hilbert:

Theorem 2.4.2. (Theorem 90) If L/K is a finite cyclic extension of fields, then

$$H_T^{-1}(G(L/K), L^*) = 1$$

Proof. See [Hil97, Section 54 Satz 90 p 272]. □

From [Neu99] we have the following proposition:

Proposition 2.4.3. If L/K is cyclic, then

$$H^1(G(L/K), A_L) \cong H_T^{-1}(G(L/K), A_L).$$

Proof. [Neu99, Proposition IV.3.7]. □

We also obtain the Class Field Axiom, which is important for generalised reciprocity in Chapter 3:

Definition. - Let K be a field and A a G_K -module. For any extension L/K , denote $A_L := A^{G_L}$. The **Class Field Axiom** is said to hold if for every finite extension L/K and every cyclic extension M/L , we have

$$\#H_T^i(G(M/L), A_M) = \begin{cases} [M : L], & \text{if } i = 0 \\ 1, & \text{if } i = -1 \end{cases}$$

Theorem 2.4.4. Let L/K an unramified extension of local fields, we have that

$$H_T^0(G(L/K), L^*) = 1,$$

and in particular,

$$N_{L/K} : \mathcal{O}_L^* \rightarrow \mathcal{O}_K^*$$

is surjective.

Proof. See [Neu99, Proposition IV.6.2 pp 299-300]. □

Using the theory of right-derived functors briefly mentioned above, we can determine more cohomologies relevant to algebraic geometry. The next simplest is the (Zariski) sheaf cohomology. One shows that the category $\text{Ab}(X)$ of sheaves of abelian groups has enough injectives [Har77, Corollary III.2.3 p 207] and that the global sections functor $X \mapsto \Gamma(X, \mathcal{F})$ is left-exact. Therefore, we obtain a right-derived functor $H^i(X, \mathcal{F})$ with similar cohomological properties as above. Regarding the computation of this cohomology, we have that for separated K -schemes X that are covered by affine opens $U = \text{Spec } A$ with A a finitely generated K -algebra, one can compute these groups using the Čech cohomology construction [Har77, Theorem III.4.5 p 222], which is similar to the case of Galois cohomology.

We also briefly mention étale cohomology, which is a type of union of Zariski cohomology and Galois cohomology. We replace the notion of Abelian sheaf on open subsets with that of an étale sheaf on étale morphisms, and the cohomology theory of right-derived functors still holds [Mil13, Section III.1 pp 82-94]. Then for any discrete G_K -module M there is a functor $G_K\text{-Mod} \rightarrow S((\text{Spec } K)_{\text{et}})$ by mapping $M \mapsto \mathcal{F}_M$, where $\mathcal{F}_M$ is an étale sheaf associated to M , and there is an isomorphism

$$H^i(K, M) \cong H_{\text{et}}^i(\text{Spec } K, \mathcal{F}_M)$$

for all $i \geq 0$ as in [CS21, Section 2.2.2 pp 52-53]. Though we emphasize that étale cohomology will play a small but important role in this thesis since it will relate to the failure of the Hasse principle via the so-called Brauer-Manin obstruction.

2.5 Scheme Theory

We now describe some of the scheme theory that will be necessary to know. If the reader is not familiar with schemes, but has studied varieties then there is an equivalence of categories that links certain schemes over algebraically closed fields to the usual K -varieties (see Appendix A for more details). In short, scheme theory subsumes variety theory, including the theory of elliptic curves, which we will also partially review in the next section.

There are many useful resources for learning about schemes; for example, the following are quite useful for seeing the details that link classical variety theory to scheme theory: [Vak25], [LE02] or [Mum04]. We also provide a highlight of the results that will be of use in this thesis in the Appendix A.

We begin by briefly reviewing fibre products of schemes. Let S be a scheme and X and Y schemes over S , i.e. there are scheme morphisms $X \rightarrow S$ and $Y \rightarrow S$. The category of schemes admits fibre products [Har77, Theorem II.3.3 p 87], which means there exists an S -scheme denoted $X \times_S Y$, which fits into the following cartesian diagram:

$$\begin{array}{ccc} X \times_S Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & S \end{array}$$

such that $X \times_S Y$ satisfies a universal property: every S -scheme T such that $T \rightarrow X$ and $T \rightarrow Y$ admits a unique map $T \rightarrow X \times_S Y$. In particular, we will be interested in the case that $S = \operatorname{Spec} K$ and $Y = \operatorname{Spec} L$ for K a perfect field and L/K a field extension.

We have the following definitions:

Definition. A scheme X is called **quasi-compact** if for every open covering, there exists a finite subcovering. We say a scheme morphism $f : X \rightarrow S$ is of **finite type** if for all affine opens $V \subseteq S$, $f^{-1}V$ is a quasi-compact topological space and every affine open $U \subseteq f^{-1}(V)$ implies $\mathcal{O}_X(U)$ is a finite type $\mathcal{O}_S(V)$ -algebra, i.e. there exists a variables $x_1, \dots, x_s$ such that $\mathcal{O}_S(V)[x_1, \dots, x_s] \rightarrow \mathcal{O}_X(U)$ is a surjective map. If we have the additional condition that there is a finitely generated ideal I such that $\mathcal{O}_S(V)[x_1, \dots, x_s]/I \cong \mathcal{O}_X(U)$, we say that $f : X \rightarrow S$ is of **finite presentation**.

We note that for noetherian rings R , the properties finite type and finite presentation are equivalent since ideals are finitely generated. If $X \rightarrow \operatorname{Spec} A$ with A a noetherian ring, it follows that finite type and finite presentation also coincide since localisation preserves the noetherian property [AM69, Proposition 7.3 p 80], the distinguished opens $\operatorname{Spec} A_f$ form a basis of the topology of $\operatorname{Spec} A$ and it suffices to check being of finite type and finite presentation on a single open cover [Mum04, Proposition II.3.1 p 122]. One also often says that a K -scheme X has property **P** if the morphism $X \rightarrow \operatorname{Spec} K$ has property **P**.

If our scheme is an affine variety, then being of finite presentation recovers the usual affine coordinate ring. One also recalls the well-known corollary of the (projective) Nullstellensatz from [Har77, Corollary I.1.4], which, in particular, establishes a bijective correspondence between irreducible (projective) affine varieties to (homogeneous) prime ideals in $k[x_1, \dots, x_n]$. With this in mind, we have the following definition:

Definition. A scheme X is called **reduced** if for all $x \in X$, $\mathcal{O}_{X,x}$ is a reduced ring, i.e. admits no nilpotents. A scheme X is also called **irreducible** if the underlying topological space is irreducible, which means that if $X = Z_1 \cup Z_2$ with Z_1, Z_2 closed subsets, then $X = Z_1$ or $X = Z_2$. A scheme is called **integral** if it is reduced and irreducible. A scheme X/K is **geometrically integral** if $\bar{X} = X \times_{\operatorname{Spec} K} \operatorname{Spec} \bar{K}$ is also an integral scheme,

With these definitions established, we can now define what we will consider a variety in this paper. For motivation as to why we will use the following definition, see Appendix A.

Definition. We call X/K a **variety** if $X \rightarrow \operatorname{Spec} K$ is geometrically integral and of finite type. Moreover, we call X/K a **curve** if for all $x \in X$, $\mathcal{O}_{X,x}$ is a ring of Krull dimension 1.

Note that from [Mum04, Corollary 1.7.2 pp 61-62], the requirement that local rings $\mathcal{O}_{X,x}$ have dimension 1 is equivalent to the condition that the maximum length of chains of closed irreducible subsets $Z \subsetneq X$ is at most 1 and attains this value at least once. Finally, we have the scheme notion of non-singular:

Definition. A point $x \in X$ is called **non-singular** or **regular** if the ring $\mathcal{O}_{X,x}$ is regular, i.e. the minimal number of generators of the maximal ideal $\mathfrak{m}_{X,x}$ is equal to the Krull dimension of $\mathcal{O}_{X,x}$ as in [AM69, Proposition 11.21 p 123]. The scheme X is called **non-singular** or **regular** if this holds for all $x \in X$.

There is also a notion of smoothness associated to scheme morphisms, which is in general not the same as a scheme being regular. The definition of smoothness can be found in Appendix A. Since we will be primarily interested in curves defined over perfect fields in this paper, which have irreducible components of dimension 1, then the classical notion of smoothness, that the Jacobian matrix has corank equal to 1, corresponds the notion of regularity or non-singular defined here, see [Vak25, definition 13.2.4 and Exercise 13.2.J p 363]. Therefore, going forward, we will sometimes interchangeably use these terms in the case that $X \rightarrow \operatorname{Spec} K$ with K perfect.

We also define a notion of completeness:

Definition. Let X/K be a K -variety. It is **complete** if for all K -varieties Y , $X \times_K Y \rightarrow Y$ is a closed map.

We have the following useful lemma that will help with some of the technical details later.

Lemma 2.5.1. If X/K is a complete curve, then X/K is projective.

Proof. One can prove this by following [LE02, Section 4.1 Exercise 1.16 p 125]. Note that Liu assumes the scheme X is normal, which means that for each $x \in X$, $\mathcal{O}_{X,x}$ is integrally closed. However, we assume that each ring $\mathcal{O}_{X,x}$ is a discrete valuation ring since the rings $\mathcal{O}_{X,x}$ are integral domains that are regular and of dimension 1, which implies they are integrally closed by [AM69, Proposition 9.2 p 95]. $\square$

We define an important class of varieties fundamental in algebraic geometry:

Definition. We call a variety X/K **projective** if there is a nonnegative integer n and a closed immersion $X \hookrightarrow \mathcal{P}^n$, See Appendix A.

The last thing to discuss is the genus of a curve. For a complete curve X/K , there is a canonical sheaf $\mathcal{K}_X$ (related to the canonical divisor from classical algebraic geometry) such that we admit Serre Duality [Vak25, Theorem 18.5.1 p 485]:

$$h^1(X, \mathcal{O}_X) = h^0(X, \mathcal{K}_X).$$

One may define **geometric genus** to be the quantity on the right-hand side and arithmetic genus to be that on the left. Hence, in light of this result, we simply say that such a curve X has genus 1 if the quantity $h^1(X, \mathcal{O}_X) = 1$.

Definition. Let X be an integral scheme. We define $\operatorname{Div}(X)$, the **group of divisors**, to be the formal group of the prime divisors of our scheme X , i.e. the free $\mathbb{Z}$ -module generated by closed subschemes $Z \subseteq X$ that have Krull dimension $\dim(\mathcal{O}_{X,z}) = 1$, where $z \in Z$ is the generic point

Since X is a curve, we have that all prime divisors correspond to the closed points $\mathcal{P} \in X$. To each closed point $\mathcal{P}$, we associate $\deg(\mathcal{P}) = [K(\mathcal{P}) : K]$. If $D = \sum_i a_i \mathcal{P}_i$, it follows that $\deg(D) = \sum_i a_i \deg(\mathcal{P})$ and we say a divisor is **rational** if each point in its support has $\deg(\mathcal{P}) = 1$.

Let X/K be a K -curve over a perfect field. From the functor of points description, we know that $X(K)$ is in bijection with the closed points admitting a morphism $K(\mathcal{P}) \rightarrow K$ that factors through X (see Appendix A). If $K = \bar{K}$ then we know that all closed points are given by such a map $K(\mathcal{P}) \rightarrow \bar{K}$. There is a natural action of G_K on $X(\bar{K})$ and, as a result, an induced action on $\text{Div}(\bar{X})$. We have the following map from [BS08, Lemma 4.3] linking the groups of divisors:

$$\begin{aligned} \text{Div}(X) &\longrightarrow \text{Div}(\bar{X}) \\ \mathcal{P} &\longmapsto \sum_{\sigma: K(\mathcal{P}) \rightarrow \bar{K}} \mathcal{P}_\sigma \end{aligned}$$

where each $\mathcal{P}_\sigma$ corresponds to a Galois conjugate of the roots of f .

Definition. We may define the subgroup of principal divisors $\text{Princ}(X)$ given by $\text{div } f = \sum_{\mathcal{P}} \text{ord}_{\mathcal{P}}(f) \mathcal{P}$ for $f \in K(X)^*$. Then we let $\text{Cl}(X) := \text{Div}(X)/\text{Princ}(X)$.

If X is a regular projective curve, then there is a group isomorphism $\text{Cl}(X) \cong \text{Pic}(X)$ sending $D \mapsto \mathcal{O}_X(D)$ by [Har77, Corollary II.6.16 p 145]. Therefore, going forward, we simply denote the class group as $\text{Pic}(X)$.

Proposition 2.5.2. Let K be a local field, then we have that there is a natural isomorphism $\text{Pic}(X) \cong H^0(K, \text{Pic}(\bar{X}))$ if there is a K -rational divisor of degree 1 on X . Furthermore, if we instead take K to be a number field, then the same holds if K is LSE.

Proof. See [Sik02, p 121] or [SP97, Section 3 pp 4-6]. □

2.6 Elliptic Curves

Elliptic curves are perhaps one of the simplest examples of the concept of abelian varieties, i.e. proper geometrically integral finite-type k -schemes that are also group schemes. We refer one to [Sil09] for much of the background theory of elliptic curves. In this section, we will primarily concern ourselves with homogeneous spaces and the Tate-Shafarevich group of elliptic curves.

The Selmer and Tate-Shafarevich Groups

Let E/K be an elliptic curve. That means there is a section $\mathcal{O} : \text{Spec } K \rightarrow E$ and $E \rightarrow \text{Spec } K$ is a regular complete curve of genus 1 such that the fibres $E_L = E \times_K L$ also have genus 1 for all extensions L/K . We recall that an elliptic curve over K is a group scheme, i.e. that for all k -schemes T , we have that $E(T)$ is a group [LE02, Section 7.4.4 pp 297-299]. Viewed as a variety, the group law is endowed via one of two equivalent ways: via the chord-tangent method or on $\text{Pic}^0(E)$ using the Riemann-Roch theorem, [Sil09, Proposition III.3.4]. We begin by defining a right action of our elliptic curve E/K on smooth curves C/K .

Definition. Let C/K be a smooth curve such that there exists a pairing

$$\mu : C \times E \longrightarrow C$$

satisfying the following criteria:

1. $\mu(p, \mathcal{O}) = p$

2. $\mu(\mu(p, P), Q) = \mu(p, P + Q)$ for all $p \in C(\bar{K})$ and $P, Q \in E(\bar{K})$
3. for all $p, q \in C(\bar{K})$ there is a unique $P \in E(\bar{K})$ satisfying $\mu(p, P) = q$

then we call C/K a **homogeneous space** for E/K .

As is standard, we replace μ with the notation $' + '$, and we define a map $' - ' : C \times C \rightarrow E$ by sending (p, q) to the unique point P from (3). The symbols are used since the usual properties of addition and subtraction indeed hold, see [Sil09, Lemma X.3.1 p 322].

We define an equivalence relation on the set of homogenous spaces $C \sim C'$ if there exists an isomorphism $\theta : C \rightarrow C'$ defined over K compatible with the action of E/K , i.e. $\theta(p + P) = \theta(p) + P$. In [Sil09, Proposition X.3.2 p 323], it is shown that the homogeneous spaces of an elliptic curve E are given by its twists, i.e. E'/K such that $E' \cong E$ over $\bar{K}$. Thus we can now define the Weil-Chatelet group:

Definition. We define

$$\text{WC}(E/K) = \{\text{Homogeneous spaces } C/K\} / \sim,$$

which is called the **Weil-Chatelet group** for E/K .

That is a group follows from the bijection

$$\begin{aligned} \text{WC}(E/K) &\xrightarrow{\sim} H^1(K, E) \\ [C/K] &\longmapsto [\theta : p_0^\sigma - p_0], \end{aligned} \tag{2.4}$$

where $p_0 \in C(\bar{K})$, by [Sil09, Theorem X.3.6. p 325]. Next, we discuss a fundamental result in the study of Jacobians $\text{Pic}^0(C)$ of genus 1 curves.

Theorem 2.6.1. For an elliptic curve E/K , we have a group isomorphism:

$$\begin{aligned} E(\bar{K}) &\xrightarrow{\sim} \text{Pic}^0(\bar{E}) \\ \mathcal{P} &\longmapsto [\mathcal{P} - \mathcal{O}]. \end{aligned}$$

For a homogeneous space C/K and a point $p_0 \in C(\bar{K})$, we also have:

$$\begin{aligned} \text{sum} : \text{Pic}^0(\bar{C}) &\xrightarrow{\sim} E(\bar{K}) \\ \left[\sum n_i q_i \right] &\longmapsto \sum [n_i] (q_i - p_0). \end{aligned}$$

Proof. See [Sil09, Proposition III.3.4 pp 61-63] for the first statement and [Sil09, Proposition X.3.8 pp 329-331] for the second. $\square$

This theorem is important in the computation of $\text{Pic}(C)$ since we will see later that $\text{Pic}(C) = \mathbb{Z} \oplus \text{Pic}^0(C)$. More generally, for a complete regular K -curve C , one assigns to it the Jacobian variety, and its computation can be useful in computing $C(K)$ as in [BS10].

We recall that in Section 2.4 that we had the following exact sequence:

$$0 \longrightarrow E(K)[\varphi] \longrightarrow E(K) \xrightarrow{\varphi} E'(K) \xrightarrow{\delta} H^1(K, E[\varphi]) \longrightarrow H^1(K, E) \xrightarrow{\varphi} H^1(K, E'),$$

where δ is the canonical connecting map. Then $\ker \delta = \text{im } \varphi$ implies $E'(K)/\varphi(E(K)) \rightarrow H^1(K, E[\varphi])$ is injective. We also see that the kernel of $\varphi : H^1(K, E) \rightarrow H^1(K, E')$ is the image of $H^1(K, E[\varphi])$ so that $H^1(K, E[\varphi]) \rightarrow H^1(K, E)[\varphi]$ is surjective. Hence we obtain

$$0 \longrightarrow E'(K)/\varphi(E(K)) \xrightarrow{\delta} H^1(K, E[\varphi]) \longrightarrow H^1(K, E)[\varphi] \longrightarrow 0.$$

We take a place v of K and replace K with K_v , obtaining a similar exact sequence:

$$0 \longrightarrow E'(K_v)/\varphi(E(K_v)) \xrightarrow{\delta} H^1(K_v, E[\varphi]) \xrightarrow{\varphi} H^1(K_v, E)[\varphi] \longrightarrow 0.$$

Then we obtain the following commutative diagram from [Sil09, X.4 p 332]:

$$\begin{array}{ccccccc} 0 & \longrightarrow & E'(K)/\varphi(E(K)) & \xrightarrow{\delta} & H^1(K, E[\varphi]) & \longrightarrow & H^1(K, E)[\varphi] \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \prod_{v \in \mathfrak{M}(K)} E'(K_v)/\varphi(E(K_v)) & \xrightarrow{\delta} & \prod_{v \in \mathfrak{M}(K)} H^1(K_v, E[\varphi]) & \longrightarrow & \prod_{v \in \mathfrak{M}(K)} H^1(K_v, E)[\varphi] \longrightarrow 0 \end{array}$$

The first down arrow is induced by the map sending a point $\mathcal{P} \in E'(K)$ to the unique point $\mathcal{P}_v \in E'(K_v)$ returned via the map of fields $K \rightarrow K_v$. For the second down arrow, it is given by restricting cocycles: $\xi \mapsto \xi|_{G_v}$ since $G_v \subseteq G_K$ is an open subgroup. Finally, the third down arrow is also induced by restriction.

The goal is to determine the image of $E'(K)/\varphi(E(K))$ in $H^1(K, E[\varphi])$ under δ . Using the first row of the diagram tells us this is the same as computing the kernel of $H^1(K, E[\varphi]) \rightarrow H^1(K, E)[\varphi]$. Next, using the isomorphism $H^1(K, E) \cong \text{WC}(E/K)$, this reduces to determining if certain homogeneous spaces are soluble. In line with intuition, the diagram tells us that this should be done locally. This prompts the following definition:

Definition. We define the φ -Selmer group of E/K as

$$S^{(\varphi)}(E/K) := \ker \left\{ H^1(K, E[\varphi]) \longrightarrow \prod_{v \in M_K} \text{WC}(E_v/K_v) \right\}$$

and the **Tate-Shafarevich group** of E/K as

$$\text{III}(E/K) := \ker \left\{ \text{WC}(E/K) \longrightarrow \prod_{v \in M_K} \text{WC}(E_v/K_v) \right\}.$$

From [Sil09, Theorem X.4.2 p 333], we know that the Selmer group is finite, and we obtain the following exact sequence:

$$0 \longrightarrow E'(K)/\varphi(E(K)) \xrightarrow{\delta} \text{Sel}^{(\varphi)}(E/K) \longrightarrow \text{III}(E/K)[\varphi] \longrightarrow 0.$$

And in particular if we take $\varphi = [m]$ and $E' = E$, we have that

$$0 \longrightarrow E(K)/mE(K) \xrightarrow{\delta} \text{Sel}^{(m)}(E/K) \longrightarrow \text{III}(E/K)[m] \longrightarrow 0.$$

This exact sequence plays a crucial role in determining the group $E(K)/mE(K)$: the elements of the Selmer group correspond to curves called m -coverings that are LSE, which we test for solubility over K . The curves with $X(K) \neq \emptyset$ return points on $E(K)$. The m -coverings that admit no K -rational points are the non-trivial elements of the Tate-Shafarevich group, i.e. Hasse principle violations. We discuss the notion of m -coverings in more detail in Section 4.2.

3. GENERALISED RECIPROCITY

In this chapter, we will cover generalised laws of reciprocity. The reader is likely familiar with the classical Gauss law of reciprocity and may also be familiar with the slightly more obscure cubic law of reciprocity. These are both encapsulated in the remarkably important result of class field theory called Artin reciprocity. To understand this latter type of reciprocity, it will be necessary to have some knowledge of adèles and idèles, which will be covered in Section 3.2. Then we show how Artin reciprocity can lead to cubic reciprocity. The results involving the cubic residue symbol will be illustrative of the theory and also useful in Chapter 5 involving 3-coverings of elliptic curves. We will also close the chapter with another generalisation of reciprocity related to Brauer groups.

3.1 Class Field Theory

For a full treatment of Class Field Theory, one may consult [Lang] or [Neukirch]. We summarise the main results here that will be useful. The fundamental result of class field theory, which is also often referred to as the **general reciprocity law** is given in the following theorem:

Theorem 3.1.1. (General Reciprocity Law) Let A be a G_K -module. For any Galois field extension L/K , there exists an isomorphism

$$r_{L/K} : G(L/K)^{ab} \longrightarrow A_K/N_{L/K}A_L,$$

where $G(L/K)^{ab}$ denotes the Galois group of the largest abelian subextension of L/K , if we have the following data:

1. There is a surjective continuous homomorphism $d_K : G_K \rightarrow \widehat{\mathbb{Z}}$, where $\widehat{\mathbb{Z}} := \varprojlim_n \mathbb{Z}/n\mathbb{Z}$ is the **Prüfer ring**.
2. There is a map $v : A_K \rightarrow \widehat{\mathbb{Z}}$, so that $Z := v(A_K)$ satisfies (i) $Z \supseteq \mathbb{Z}$, (ii) $Z/nZ \cong \mathbb{Z}/n\mathbb{Z}$ and (iii) $v(N_{L/K}A_L) = f_L Z$ where $f_L := [\widehat{\mathbb{Z}} : d_K(G_L)]$ (we call v a **Henselian valuation**).
3. The group A satisfies the Class Field Axiom as in Section 2.4.

The name of the map and its associated conditions in (2) become clear if we pick, for example, $A_K = K^*$ to be a p -adic field. The conditions above require that v is a discrete valuation in the usual sense. Then the quantity f_L also becomes the usual residue class degree $f_L = [\mathcal{O}_L/\mathfrak{P} : \mathcal{O}_K/\mathfrak{p}]$. One also usually assumes that the Galois group G_K given in this theory is formal, and not per se the usual one. However, we disregard this and assume that the usual properties of Galois extensions hold.

The proof of this result requires much background theory, which we will not indulge further than what was covered in the preliminaries Sections 2.3 and 2.4. In any case, one may find the proof of this result in [Neu99, Theorem 6.3 pp 300-302]. We also note that the notion of a Henselian valuation in general and the map $G_K \rightarrow \widehat{\mathbb{Z}}$ will only receive further mention in this thesis in relation to the above Theorem.

Next, we obtain the **norm residue map** $[\cdot, L/K] : A_K \rightarrow G(L/K)$ by taking the inverse of $r_{L/K}$ and lifting the map. As such $\ker([\cdot, L/K]) = N_{L/K}A_L$. With generalised reciprocity established, we assign A_K a topology that is compatible with the above isomorphism.

Definition. For every field K , we equip each G_K -module A_K with the **norm topology**, where there is a fundamental system of neighbourhoods of 1 given by $N_{L/K}A_L$ for L/K Galois extensions.

From [Neukirch, Proposition IV.6.6 p 304], we obtain the following proposition:

Proposition 3.1.2. The norm topology has the properties below:

1. The open subgroups of the norm topology correspond exactly to the closed subgroups of A_K of finite index.
2. If L/K is finite, then $N_{L/K} : A_L \rightarrow A_K$ is continuous.

Proof. See [Neukirch, prop IV.6.6 p 304]. □

With the topology on A_K established, we are now ready to state a fundamental result of class field theory, sometimes called the classification theorem.

Theorem 3.1.3. For A_K as in Theorem 3.1.1, we supply the following bijection:

$$\begin{aligned} \{\text{Finite abelian extensions } L/K\} &\longleftrightarrow \{\text{Normal subgroups of finite index in } A_K\} \\ L &\longmapsto N_{L/K}A_L \end{aligned}$$

such that

$$L_1 \subseteq L_2 \iff \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}$$

Proof. We follow the proof of [Neu99, Theorem IV.6.7 pp 304-305]. To show the map is well-defined, we need that $N_{L/K}A_L$ is normal and of finite index in A_K . The first fact is obvious, and the second follows from generalised reciprocity. The first two of the last three statements follow from transitivity of the norm and generalised reciprocity.

If $L \neq M$ are two distinct abelian extensions, but $N_L = N_M$ then $N_L \cap N_M = N_L = N_M$. This means that $N_{LM} = N_L \cap N_M = N_L$. Hence, $G(L/K) \cong G(ML/K)$ which is impossible since ML is a proper extension of L . On the other hand, if $\mathcal{N}$ is normal subgroup of finite index in A_K , then by definition of the norm topology, there exists a Galois extension L/K such that $N_{L/K}A_L \subseteq \mathcal{N}$. By selecting the minimal finite abelian extension M of L , then we have that $G(M/K) = G(L/K)^{ab}$. Hence, we can assume that L is abelian. Now $(\mathcal{N}, L/K) \subseteq G(L/K)$ is a subgroup so it corresponds to some subextension $G(L/L')$, and it satisfies $\mathcal{N} = \ker([\cdot, L'/K] : G(L/K) \rightarrow G(L'/K))$. This shows the map is surjective.

Finally, the last identity follows from the surjectivity of the map. □

We are interested in the cases that K is a local field or an idèle; the latter objects are discussed in Section 3.2. Returning to the case that $A_K = K_v^*$, we obtain the **local norm residue map**. For an unramified extension $w|v$, we had in theorem 2.4.4 that $H_T^0(G(L_w/K_v), \mathcal{O}_w^*) = 1$, so that all units $\alpha \in \mathcal{O}_v^*$ satisfy $[\alpha, L_w/K_v] = 1$. We let our valuation be $v : K^* \rightarrow \mathbb{Z}$ be the usual one, which is Henselian. There is also a surjective map $G(\bar{K}/K) \rightarrow \hat{\mathbb{Z}}$ arising from the canonical surjection $G(\bar{K}/K) \rightarrow G(\bar{K}/K)$ and the isomorphism $G(\bar{K}/K) \cong \hat{\mathbb{Z}}$, where $\bar{K}$ is the maximal unramified

extension of K [Neu99, Section V.1]. We admit from [Neu99, Theorem V.1.1 pp 317-318] the so-called **Local Class Field Axiom**: for L/K finite cyclic extension of local fields then

$$\#H_T^i(G(L/K), L^*) = \begin{cases} [L/K] & \text{if } i = 0 \\ 0 & \text{if } i = -1. \end{cases}$$

Thus, we have the following the classification theorem of local class field theory:

Theorem 3.1.4. There is a 1-1 correspondence

$$\left\{ \begin{array}{c} \text{Finite abelian extensions} \\ \text{of the local field } K \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Open subgroups } \mathcal{N} \\ \text{of finite index in } K^* \end{array} \right\}$$

$$L \longmapsto N_{L/K} L^*$$

such that

$$L_1 \subseteq L_2 \iff \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}$$

Proof. This follows from Theorem 3.1.3. □

3.2 Adèles and Idèles

We seek a method for tracking all the local field information of a given number field K simultaneously. To this end, we consider the ring of adèles A_K . The unit group of this ring I_K , called the idle group, and its quotient I_K/K^* are essential in the study of rational points of curves.

Definition. We define the following **restricted product**:

$$A_K = \prod_{v \in \mathfrak{M}(K)} 'K_v := \{(a_v) | v(a_v) \geq 0 \text{ for all but finitely many } v\} \subseteq \prod_{v \in \mathfrak{M}(K)} K_v.$$

Using component-wise multiplication and addition, we have that A_K is a ring and is called the **ring of adèles**, with elements (a_v) referred to as **adèles**.

Formally, this ring is an example of an **inductive limit**: by considering finite subsets of non-Archimedean places $S \subseteq \mathfrak{M}(K)$ and taking products

$$A_{K,S} := \prod_{v|\infty} K_v \times \prod_{v \in S} K_v \times \prod_{v \notin S} \mathcal{O}_{K_v},$$

an inclusion $S \subseteq S'$ induces an embedding $f_{S,S'} : A_{K,S} \hookrightarrow A_{K,S'}$, making this an **inductive (direct) system**. From [Neu99, definition VI.2.2], two elements $x \in A_{K,S}$ and $y \in A_{K,S'}$ have $[x] = [y]$ if there exists a subset $S'' \supseteq S, S'$ such that $f_{S,S''}(x) = f_{S',S''}(y)$. We claim that $A_K = \varinjlim A_{K,S}$. Indeed from the definition of A_K it is clear that it fits into the following diagram:

$$\begin{array}{ccc} A_{K,S} & \hookrightarrow & A_{K,S'} \\ & \searrow & \swarrow \\ & A_K & \end{array}$$

and it follows that there exists a unique map $\varinjlim A_{K,S} \rightarrow A_K$ by the universal property such that the canonical maps $\varphi_{S,K} : A_{K,S} \rightarrow \varinjlim A_{K,S}$ factor through A_K . Since for any subset S , $A_{K,S} \rightarrow A_K$

sends $x \mapsto x$, we thus have that $\varinjlim A_{K,S} \rightarrow A_K$ is defined by $[x] \mapsto x$ by picking some $A_{K,S}$ containing x . On the other hand, we have a natural map $A_K \rightarrow \varinjlim A_{K,S}$ by sending $x \mapsto [x]$ where $[x]$ is an equivalence class corresponding to an element of A_{K,S_x} such that S_x includes all places where $x \notin \mathcal{O}_v$. That these maps are inverses is clear.

A fundamental system of neighbourhoods for 0 in A_K is given by the sets $A_{K,S}$. We also obtain a diagonal embedding $K \hookrightarrow A_K$ with $a \mapsto (a)$ via the map $K \hookrightarrow K_v$.

Definition. We define the following restricted product of groups:

$$I_K := \prod_{v \in \mathfrak{M}(K)} {}'K_v^*,$$

called the **idèle group of K** .

As before, I_K is the inductive limit of the sets $I_{K,S} = \prod_{v \text{ arch.}} K_v^* \times \prod_{v \in S} K_v^* \times \prod_{v \notin S} \mathcal{O}_v^*$ as in [Voi21, 27.4.8 p 452], so the $I_{K,S}$ form a fundamental system of neighbourhoods of 1, [Lan95, VII Section 1 p 138]. I_K is also locally compact by [Bra48, Proposition I.2.1 p 5] since each $I_{K,S}$ is the product of the compact $\mathcal{O}_v^*$ for all but finitely many places. Note that the same holds for A_K by the same proposition.

From the embedding of K in A_K , we obtain the embedding $K^* \hookrightarrow I_K$ and we call the image of elements of K^* **principal idèles**. In [Neu99, Proposition VI.1.5 p 361], it is shown that $K^* \subseteq I_K$ is a discrete closed subgroup. Therefore, this leads to an important definition:

Definition. As K^* is discrete, it is a closed subgroup of I_K . Then we define the **idèle class group** as

$$C_K := I_K / K^*.$$

Since I_K is locally compact and Hausdorff and K^* is a discrete closed subgroup, then C_K is also locally compact and Hausdorff from [Neu99, Proposition VI.1.5]. For each inclusion of number fields $K \hookrightarrow L$, we obtain maps $K_v \hookrightarrow L_w$ where $w|v$. As such we have a natural map:

$$\begin{aligned} I_K &\longrightarrow I_L \\ (a_v) &\longmapsto (a'_w), \end{aligned}$$

where a'_w is the image of a_v in L_w . This means that we view $I_K \subseteq I_L$. We also obtain a norm map $\text{Norm}_{L/K} : I_L \rightarrow I_K$ induced from the norm maps $\text{Norm}_{w/v} : L_w \rightarrow K_v$. In addition to the usual norm properties, it also satisfies the following property for each component v :

$$\text{Norm}_{L/K}(\alpha)_v = \prod_{w|v} \text{Norm}_{w/v}(a_w).$$

Since this norm map sends principal idèles to principal idèles, we also obtain an induced norm map on idèle class groups $N_{L/K} : C_L \rightarrow C_K$.

Before continuing, we briefly comment on the topology of C_K . It has the natural topology induced via the quotient from I_K as above. As we will see, it will also have the norm topology as in Section 3.1. It turns out that the closed subgroups of finite index in the natural topology are precisely the open subgroups of finite index in the norm topology by [Neu99, Theorem VI.6.1. pp 395-396].

Definition. For every place $\mathfrak{p} \in \mathfrak{M}(K)$ let $U_{\mathfrak{p}}^{(0)} := \mathcal{O}_{v_{\mathfrak{p}}}^*$. For $n_{\mathfrak{p}} > 0$, we put

$$U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} := \begin{cases} 1 + \mathfrak{p}^{n_{\mathfrak{p}}}, & \text{if } \mathfrak{p} \nmid \infty \\ \mathbb{R}_{>0} \subseteq K_{\mathfrak{p}}^*, & \text{if } \mathfrak{p} \text{ is real} \\ \mathbb{C}^* = K_{\mathfrak{p}}^*, & \text{if } \mathfrak{p} \text{ is complex.} \end{cases}$$

For $\alpha_{\mathfrak{p}} \in K_{\mathfrak{p}}^*$ (including at the infinite places) we write

$$\alpha_{\mathfrak{p}} \equiv 1 \pmod{\mathfrak{p}^{n_{\mathfrak{p}}}} \iff \alpha_{\mathfrak{p}} \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}.$$

The unit groups of $U_v^{(n)} = 1 + \mathfrak{p}_v^n$ of a local field have a corresponding notion in adèle rings: the **replete ideals** of A_K are given by formal ideals $\mathfrak{m} \subseteq \mathcal{O}_K$:

$$\mathfrak{m} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{n_{\mathfrak{p}}} \times \prod_{\mathfrak{p} \mid \infty} e^{n_{\mathfrak{p}}}$$

where $n_{\mathfrak{p}} \in \mathbb{Z}$ for $\mathfrak{p} \nmid \infty$, $n_{\mathfrak{p}} \in \mathbb{R}_+^*$ for $\mathfrak{p} \mid \infty$ and e is Euler's number.

We define an important subgroup for determining the closed subgroups of our idèle class group:

Definition. We call the subgroup of C_K

$$C_K^{\mathfrak{m}} := I_K^{\mathfrak{m}} K^* / K^*$$

a **congruence subgroup** where

$$I_K^{\mathfrak{m}} := \prod_{\mathfrak{p}} U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$$

and $\mathfrak{m} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{n_{\mathfrak{p}}} \times \prod_{\mathfrak{p} \mid \infty} e^{n_{\mathfrak{p}}}$.

We have from [Neu99, Proposition VI.1.8 p 364], which says that open subgroups are those that contain one of these congruence subgroups.

Proposition 3.2.1. The closed subgroups of finite index of C_K are precisely the subgroups that contain a congruence subgroup $C_K^{\mathfrak{m}}$.

Proof. See [Neu99, Proposition VI.1.8 p 364]. □

For a subgroup $N_{L/K} C_L \subseteq C_K$, we let $(N_{L/K} C_L)^n := \{a^n \mid a \in N_{L/K} C_L\}$. We put this proposition to immediate use to prove the following proposition stated without proof in [BS08, Section 1]:

Proposition 3.2.2. Let X be a complete, regular curve over a field K . Let $f \in K(X)$. Then we have

$$\prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(C_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}}(f)} \subseteq C_K$$

is an open subgroup (in the norm topology) of C_K if and only if there are no non-trivial common factors of all the $\text{ord}_{\mathcal{P}}(f)$.

Proof. We denote $H := \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(C_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}}(f)}$. First, suppose that there is no common divisor of the vanishing orders of f . We need to find a Galois extension L/K such that $N_{L/K} C_L \subseteq H$. Using the above proposition we can alternatively show that there exists $C_K^{\mathfrak{m}} \subseteq H$ for some replete ideal $\mathfrak{m} = \prod_{v \nmid \infty} \mathfrak{p}_v^{n_v} \times \prod_{v \mid \infty} e^{n_v}$.

The subgroups $N_{K(\mathcal{P})/K}C_{K(\mathcal{P})}$ are open because there is always a finite Galois extension $L \supseteq K(\mathcal{P})$ by [DF04, Corollary 14.23 p 594], which implies $N_{L/K}C_L \subseteq N_{K(\mathcal{P})/K}C_{K(\mathcal{P})}$ by the tower properties of norms. In fact, since the support of f is finite, there exists a finite Galois extension $L \supseteq K(\mathcal{P})$ for all $\mathcal{P} \in \text{Supp}(f)$. Thus, we know there must be some group $C_K^{\mathfrak{m}} \subseteq N_{K(\mathcal{P})/K}C_{K(\mathcal{P})}$ for each $\mathcal{P}$.

Now we have that $I_K^{\mathfrak{m}} = \prod U_v^{(m_v)}$ for values $m_v \in \mathbb{Z}_{\geq 0}$. Since there is no common factor of the $\text{ord}_{\mathcal{P}}f$, there is a linear combination $\sum_{\mathcal{P}} a_{\mathcal{P}} \cdot \text{ord}_{\mathcal{P}}f = 1$. Since any $x \in K_v^*$ (resp. $x \in \mathcal{O}_v^*$) clearly has $x^{a_{\mathcal{P}} \cdot \text{ord}_{\mathcal{P}}f} \in K_v^{*\text{ord}_{\mathcal{P}}f}$ (resp. $x^{a_{\mathcal{P}} \cdot \text{ord}_{\mathcal{P}}f} \in \mathcal{O}_v^{*\text{ord}_{\mathcal{P}}f}$), we get that $\prod_{\mathcal{P}} K_v^{*\text{ord}_{\mathcal{P}}f} = K_v^*$ (resp. $\prod_{\mathcal{P}} \mathcal{O}_v^{*\text{ord}_{\mathcal{P}}f} = \mathcal{O}_v^*$). For a non-Archimedean place with $m_v > 0$ and $l \in \mathbb{Z}_{\geq 0}$, we may pick a large enough value n_v such that $(U_v^{(n_v)})^l = U_v^{(n_v + v(l))}$. Indeed, as in the proof of [Lan95, Proposition II.3.6], pick n_v satisfying $2n_v \geq n_v + v(l) + 1$ and let $x \in \mathcal{O}_v^*$, then

$$\begin{aligned} (1 + x\pi^{n_v})^l &\equiv 1 + lx\pi^{n_v} \pmod{\mathfrak{p}^{n_v + v(l) + 1}} \\ &\equiv 1 + ux\pi^{n_v + v(l)} \pmod{\mathfrak{p}^{n_v + v(l) + 1}}, \end{aligned}$$

where we write $l = u\pi^{v(l)}$. We know that $\mathfrak{p}^{n_v + v(l) + 1} \subseteq \mathfrak{p}^{n_v + v(l)}$ which implies the desired conclusion. We can also see that $U_v^{(m)} \cdot U_v^{(n)} = U_v^{(\min(m, n))}$. Hence for the non-Archimedean primes, we have for any finite product in i and integers l_i the following:

$$\prod_i \left(\prod_{v|\mathfrak{m}} U_v^{(n_v)} \right)^{l_i} = \prod_{v|\mathfrak{m}} \left(\prod_i (U_v^{(n_v)})^{l_i} \right) = \prod_{v|\mathfrak{m}} U_v^{(\min_i(n_v + v(l_i)))}.$$

Thus, for $v|\mathfrak{m}$, we choose n_v to be the smallest number satisfying $n_v \geq m_v$ such that $2n_v \geq n_v + v(\text{ord}_{\mathcal{P}}f) + 1$ for each $\mathcal{P}$, and we let $n_v = 0$ for $v \nmid \mathfrak{m}$. We write $\mathfrak{n} := \prod_{v \in \mathfrak{M}(K)} \mathfrak{p}^{n_v}$ and obtain via the facts just discussed that

$$\begin{aligned} \prod_{\mathcal{P} \in X^c} (I_K^{\mathfrak{m}})^{\text{ord}_{\mathcal{P}}f} &= \prod_{\mathcal{P} \in X^c} \left(\prod_{v \in \mathfrak{M}(K)} U_v^{(m_v)} \right)^{\text{ord}_{\mathcal{P}}f} \\ &= \prod_{v|\mathfrak{m}} \left(\prod_{\mathcal{P} \in X^c} (U_v^{(m_v)})^{\text{ord}_{\mathcal{P}}f} \right) \times \prod_{v|\infty} U_v^{(m_v)} \times \prod_{v \nmid \mathfrak{m}} U_v^{(0)} \\ &\supseteq \prod_{v|\mathfrak{m}} U_v^{(n_v)} \times \prod_{v|\infty} U_v^{(n_v)} \times \prod_{v \nmid \mathfrak{m}} U_v^{(0)} = I_K^{\mathfrak{n}}. \end{aligned}$$

Therefore, we conclude that $C_K^{\mathfrak{n}} \subseteq H$, which makes H open.

On the other hand, suppose that H is open, which is true if and only if H is closed of finite index in C_K by [Neu99, Theorem VI.6.1 p 395]. If there exists an integer l that divides $\text{ord}_{\mathcal{P}}(f)$ for all $\mathcal{P}$, then $H \subseteq C_K^l := \{\alpha^l | \alpha \in C_K\}$. This means that $|C_K : H| > |C_K : C_K^l| > \infty$. To see this, consider the following argument: first, $C_K^l = I_K^l K^*/K^*$, and we clearly have

$$0 \longrightarrow I_{K,S}^l \longrightarrow I_{K,S} \longrightarrow I_{K,S}/I_{K,S}^l \longrightarrow 0.$$

Use the exactness of the projective limit [Neu99, Proposition IV.2.7 p 269] to see that

$$0 \longrightarrow \varprojlim I_{K,S}^l \longrightarrow I_K \longrightarrow \varprojlim I_{K,S}/I_{K,S}^l \longrightarrow 0,$$

with $I_K^l \cong \varprojlim I_{K,S}^l$ (which is true by adapting the argument that showed adèles have $A_K \cong \varprojlim_S A_{K,S}$) so that $\varprojlim I_{K,S}/I_{K,S}^l \cong I_K/I_K^l$. The index $|I_{K,S} : I_{K,S}^l|$ is infinite since $I_{K,S}/I_{K,S}^l =$

$\prod_{v \in S} K_v^*/K_v^{*l} \times \prod_{v \notin S} \mathcal{O}_v^*/\mathcal{O}_v^{*l}$ is an infinite group. Therefore, we have that I_K/I_K^l is infinite since $I_{K,S}/I_{K,S}^l$ is an open subgroup of I_K/I_K^l (which is true since it is the inductive limit), which implies that C_K/C_K^l is also infinite. This contradicts the fact that H must have finite index in C_K . $\square$

This proposition will be of fundamental importance in the coming chapters; it will be useful for showing a curve has no rational points via theorem 4.4.1.

3.3 Artin Reciprocity

With the fundamentals of idèle class groups covered, we are now ready to explore the class field theory associated to them. We have that for Galois extensions L/K with group $G := \text{Gal}(L/K)$, $\sigma \in G$ induces a natural action on I_L such that $I_L^G = I_K$ by [Neu99, Proposition VI.2.1 p 369]. Moreover, C_L is a G -module and satisfies the similar property that $C_L^G = C_K$. Importantly, we admit the **global class field axiom**: for a cyclic extension L/K ,

$$\#H_T^i(G(L/K), C_L) = \begin{cases} [L : K] & \text{for } i = 0 \\ 1 & \text{for } i = -1. \end{cases}$$

For the remaining pieces of generalised reciprocity we need maps $d : G_{\mathbb{Q}} \rightarrow \hat{\mathbb{Z}}$ and maps $v : C_{\mathbb{Q}} \rightarrow \hat{\mathbb{Z}}$ such that d is continuous and surjective and v is Henselian. For the details regarding the existence of these maps, we recommend that the reader consults [Neu99, Section VI.5]. Now we define maps which turn out to give the norm residue symbol for idèle class groups. For finite extensions L/K , we have:

$$\begin{aligned} (\cdot, L/K) : I_K &\longrightarrow G(L/K) \\ (a, L/K) &:= \prod_v [\alpha_v, L_v/K_v], \end{aligned}$$

where the maps on the right-hand side are the norm residue maps from local class field theory. Note that these maps are functorial, i.e. $I_{K'} \rightarrow I_K$ commutes in a square with $G(L'/K') \rightarrow G(L/K)$ [Neu99, Proposition VI.5.2 p 387]. Then for infinite extensions L/K , we define the map:

$$\begin{aligned} (\cdot, L/K) : I_K &\longrightarrow G(L/K) \\ (\cdot, L/K)|_{L'} &= [\cdot, L'/K], \end{aligned}$$

where L' ranges over all finite subextensions L' of K in L . If L_1 and L_2 are two extensions such that neither is contained in the other, then there exists a finite subextension L_3 of L such that $L_1, L_2 \subseteq L_3$. Then, there are canonical restriction maps $G(L_3/K) \rightarrow G(L_i/K)$ for $i = 1, 2$ which will send $(\alpha, L_3/K) \mapsto (\alpha, L_i/K)$. Since $G(L/K) = \varprojlim_j G(L_j/K)$ with L_i/K finite subextension, it follows that $(\alpha, L_2/K) = (\alpha, L_3/K) = (\alpha, L_1/K)$ in $G(L/K)$.

Now we admit the general reciprocity law for idèles class groups, which is called the **Artin Reciprocity Law**:

Theorem 3.3.1. For every Galois extension L/K of finite algebraic number fields, we have an isomorphism

$$r_{L/K} : G(L/K)^{ab} \longrightarrow C_K/N_{L/K}C_L,$$

and the inverse map yields a surjective homomorphism $[\cdot, L/K] : C_K \rightarrow G(L/K)^{ab}$, which turns out to be the global norm residue map from above by Corollary 3.3.3 below.

Proposition 3.3.2. We admit the following commutative square:

$$\begin{array}{ccc} K_v^* & \xrightarrow{[\cdot, L_w/K_v]} & G(L_w/K_v) \\ \downarrow & \searrow \theta_v & \downarrow \\ C_K & \xrightarrow{[\cdot, L/K]} & G(L/K) \end{array}$$

Proof. See [Neu99, Proposition VI.5.6 pp 391-393]. The short version is that one proves it in the case that L is unramified or equal to $K(\sqrt{-1})$. Then the argument reduces to these cases. $\square$

The map $K_v^*/N_{w/v}L_w \rightarrow G(L/K)$ induced by θ_v will play an important role in Theorem 4.4.1. We also have an important corollary, generalising the law of Hilbert reciprocity:

Corollary 3.3.3. For L/K an abelian extension, if $\alpha = (\alpha_v) \in I_K$, then

$$[\alpha, L/K] = \prod_v [\alpha_v, L_v/K_v],$$

and if $\alpha \in K^*$ is a principal idèle, then

$$\prod_v [\alpha_v, L_v/K_v] = 1. \quad (3.1)$$

Proof. Both equalities clearly follow from Proposition 3.3.2. $\square$

As in the case of local fields, we also have the following:

Theorem 3.3.4. There is 1-1 correspondence between finite abelian extensions L/K and the closed subgroups of finite index in C_K given by

$$L \mapsto N_{L/K}C_L$$

satisfying

$$L_1 \subseteq L_2 \iff \mathcal{N}_{L_1} \subseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}.$$

We call the extension L associated to a closed subgroup $\mathcal{N}$ the **class field** of $\mathcal{N}$.

Proof. In consideration of the general result Theorem 3.1.3 and the comments previously made on the topologies of C_K , the statement is proven. $\square$

3.4 Power Symbols and Reciprocity

First, we explain how we can use the Hilbert symbol to compute the Artin symbol in the case that $L_w \subseteq K_v(\sqrt[n]{K_v^*})$ is a degree n extension such that $\mu_n \subseteq K_v$. We have from local class field theory the following proposition:

Proposition 3.4.1. Let $a, b \in K_v^*$. Then for $L = K_v(\sqrt[n]{b})$ we have

$$[a, L/K_v] \sqrt[n]{b} = \left(\frac{a, b}{\mathfrak{p}_v} \right) \sqrt[n]{b},$$

where we recall that the left-hand side is the usual Galois action of $(a, L/K_v) \in \text{Gal}(L/K_v)$ on $\sqrt[n]{b}$ and the right-hand side is multiplication.

Proof. See [Neu99, Proposition V.3.1]. □

Theorem 3.4.2. (Hilbert reciprocity) Let K be a number field such that $\mu_n \subseteq K$. For $a, b \in K^*$, we have that

$$\prod_v \left(\frac{a, b}{\mathfrak{p}_v} \right) = 1$$

Proof. As in [Neu99, p. VI.8.1]. Using Proposition 3.4.1 and Corollary 3.3.3 of Artin reciprocity yields that

$$\prod_v \left(\frac{a, b}{\mathfrak{p}_v} \right) \sqrt[n]{b} = \prod_v [a, L/K_v] \sqrt[n]{b} = [a, L/K] \sqrt[n]{b} = 1(\sqrt[n]{b}) = \sqrt[n]{b}.$$

Therefore, we have that

$$\prod_v \left(\frac{a, b}{\mathfrak{p}_v} \right) = 1. \quad \square$$

We will now see the connection between Artin Reciprocity and the classical power symbols.

Definition. Let K be a number field, $a \in \mathcal{O}_K$ and $\mathfrak{p}$ a prime ideal with valuation v . If $\mathfrak{p} \nmid p$ and $(n, p) = 1$, we define the **n -th power residue symbol** for $a \in \mathcal{O}_v^*$ as

$$\left(\frac{a}{\mathfrak{p}} \right) := \left(\frac{\pi, a}{\mathfrak{p}_v} \right)$$

where $\pi \in \mathcal{O}_v$ is a uniformiser. This is independent of the choice π since the extension $K(\sqrt[n]{a})/K$ is unramified.

Note that this extends the classical legendre symbol, thanks to the following proposition:

Proposition 3.4.3. Let p be the characteristic of $\mathcal{O}_v/\mathfrak{p}_v$. For $\gcd(n, p) = 1$ and $a \in \mathcal{O}_v^*$, then

$$\left(\frac{a}{\mathfrak{p}} \right) = 1 \iff a \text{ is } n\text{-th power mod } \mathfrak{p}_v$$

Proof. See [Neu99, V.3.5 p 336]. □

We also recover, for example, cubic reciprocity as well. Take $K = \mathbb{Q}(\rho)$, then $\mathcal{O}_K = \mathbb{Z}[\rho]$. This is a unique factorisation domain since it is principal ideal domain with trivial class group, so each prime ideal $\mathfrak{p} = \pi\mathbb{Z}[\rho]$ for some $\pi \in \mathbb{Z}[\rho]$ and each element $x = \varepsilon \prod_i \pi_i^{e_i}$ with $\varepsilon \in \mathbb{Z}[\rho]^*$. We also have that the unit group $\mathbb{Z}[\rho]^* = \{\pm 1, \pm \rho, \pm \rho^2\}$.

The primes of $\mathbb{Z}[\rho]$ are, up to multiplication by a unit: $\lambda := (1 - \rho)|3$, rational primes $q \equiv 2 \pmod 3$ and conjugate factors $\pi_p \bar{\pi}_p = p$ of a rational prime $p \equiv 1 \pmod 3$ [IR90, Proposition 9.1.4 p 110]. Going forward, let $\pi \nmid 3$ be any prime. Via multiplication of an appropriate unit, we can also always uniquely put any $\pi = a + b\rho$ in a form with $a \equiv -1 \pmod 3$ and $b \equiv 0 \pmod 3$, the so-called **primary form** by [IR90, Proposition 9.3.5 p 113]. For any element $x \in \mathbb{Z}[\rho]$ with $v_\pi(x) = 1$, we also have that

$$x^{\frac{N(\pi)-1}{3}} \equiv 1, \rho, \text{ or } \rho^2 \pmod \pi$$

for $N : K \rightarrow \mathbb{Q}$ the usual norm map by [IR90, Proposition 9.3.1 p 112]. Thus we have the following proposition:

Proposition 3.4.4. For $x \in \mathbb{Z}[\rho]$, the cubic character modulo π satisfies

$$\left(\frac{x}{\pi}\right) \equiv x^{\frac{N(\pi)-1}{3}} \pmod{\pi}$$

when $\gcd(\pi, \lambda) = 1$ and $\gcd(x, \pi) = 1$.

Proof. The extension $K_\pi(\sqrt[3]{x})/K_\pi$ is unramified since the polynomial $T^3 - x$ splits in some extension of $\mathcal{O}_\pi/\pi$, which implies the solutions Hensel lift to an unramified extension of $K(\sqrt[3]{x})$ by [Neu99, Lemma V.3.3 p 335], and it is cyclic. We then use that unramified extensions satisfy the following congruence:

$$\left(\frac{x}{\pi}\right) = \left(\frac{\pi, x}{\pi\mathcal{O}_\pi}\right) = \frac{(\pi, K_\pi(\sqrt[3]{x})/K) \sqrt[3]{x}}{\sqrt[3]{x}} \equiv x^{\frac{N(\pi)-1}{3}} \pmod{\pi\mathcal{O}_\pi},$$

by Proposition 3.4.1 and [Neu99, Proposition IV.5.7 p 295] for the congruence. $\square$

Theorem 3.4.5. The cubic character modulo π for any primary prime satisfies the following properties:

1. We admit the **cubic law of reciprocity** for two distinct primary primes π, π'

$$\left(\frac{\pi}{\pi'}\right) = \left(\frac{\pi'}{\pi}\right).$$

Furthermore, if π and $\bar{\pi}$ are conjugates then $(\frac{\pi}{\bar{\pi}}) = (\frac{\bar{\pi}}{\pi}) = 1$.

2. If $\alpha \equiv \beta \pmod{\pi}$, then $(\frac{\alpha}{\pi}) = (\frac{\beta}{\pi})$, in particular the period of this symbol is π .
3. $(\frac{3}{\pi}) = \rho^{2b/3}$, where $\pi = a + b\rho$.
4. $(\frac{\lambda}{\pi}) = (\rho^{m+n})$ for $a = 3m - 1$ and $b = 3n$.

Proof. We sketch the proof of (1) by using Artin reciprocity. For the other properties one can consults: [IR90, Proposition 9.3.3 p 112] for (2) and [IR90, Exercises 9.24-9.26 p 136] for (4). Then (4) implies (3).

Let, τ, π be two primary primes of K . Now we have that $K_v(\sqrt[3]{\pi})$ is unramified for any $v \neq \pi, \lambda$ using the Kummer-Dedekind theorem. Hence, for a non-Archimedean prime $v \neq \tau, \pi$, we have by Theorem 4.4.3 the Hilbert symbol is trivial here. Hence, from Hilbert reciprocity we have

$$\left(\frac{\tau, \pi}{\lambda\mathcal{O}_K}\right) \left(\frac{\tau, \pi}{\pi\mathcal{O}_K}\right) \left(\frac{\tau, \pi}{\tau\mathcal{O}_K}\right) \left(\frac{\tau, \pi}{\infty}\right) = \left(\frac{\bar{\pi}}{\tau}\right) \left(\frac{\tau}{\pi}\right) \left(\frac{\tau, \pi}{\lambda\mathcal{O}_K}\right) = 1$$

where we use $(\frac{\tau, \pi}{\pi\mathcal{O}_K}) = (\frac{\pi, \tau}{\pi\mathcal{O}_K})^{-1}$ from [Neu99, Proposition V.3.2 p 334] and that ∞ corresponds to 2 conjugate complex valuations, so that the Hilbert symbol is also trivial there. Finally, the Hilbert symbol $(\frac{\tau, \pi}{\lambda\mathcal{O}_K}) = 1$, which can be seen by following the computations of [Cox14, Exercise 8.9 pp 176-177]. Thus we obtain cubic reciprocity. $\square$

3.5 Brauer Groups

We finally discuss one more manifestation of reciprocity arising from the so-called Brauer group of a field. Let A be a K -algebra, i.e. a (not necessarily commutative) ring such that there is a morphism $K \rightarrow A$, with $K \subseteq Z(A)$ in the centre. We recall that A is called **simple** if the only two-sided ideals are the trivial ones, 0 and A . We also refer to A as **central** if the centre of A , i.e. the elements x, y such that $xy = yx$, is K . Then we have the following important definition:

Definition. Let K be a field. The **Brauer group** $\text{Br}(K)$ is the group formed by the central simple K -algebras with finite dimension as K -vector spaces under tensor products up to the following equivalence: $A \sim A'$ if there exist n, n' such that $M_n(A) \cong M_{n'}(A')$, where $M_n(A)$ denotes the $n \times n$ matrix group with coefficients in A .

From [Voi21, Corollary 3.5.8 p 42], the only $\mathbb{R}$ -algebras up to isomorphism are $\mathbb{R}, \mathbb{C}$ and the quaternion algebra $(-1, -1)$. We have that $\mathbb{C}$ is clearly not central, so that $\text{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$. Then $\text{Br}(\mathbb{C}) = 0$ since any $\mathbb{C}$ -algebra is also an $\mathbb{R}$ -algebra, but $(-1, -1)$ is split so that $M_2(\mathbb{C}) \cong (-1, -1)$. Now let K be a number field and v any place; we also have from [CS21, Definition 12.1.7] the map

$$\text{inv}_v : \text{Br}(K_v) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

that is an isomorphism for any non-Archimedean place by [Har17, Theorem 8.9. pp 125-126] or [CS21, Theorem 12.1.8 p 243].

Then we have that there is a natural map $\text{Br}(K) \rightarrow \text{Br}(K_v)$ given by sending $A \mapsto A \otimes_K K_v$. Then we have the promised generalisation of reciprocity from [NSW13, Theorem 8.1.17 p 437]:

Theorem 3.5.1. (Local-Global Principle) There is an exact sequence

$$0 \longrightarrow \text{Br}(K) \longrightarrow \bigoplus_{v \in \mathfrak{M}(K)} \text{Br}(K_v) \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0,$$

where $(Q_v)_v \mapsto \sum_v \text{inv}_v(Q_v)$.

Proof. This follows from the exact sequence: if L/K a cyclic extension then

$$0 \longrightarrow H^2(G(L/K), L^*) \longrightarrow H^2(G(L/K), I_L) \longrightarrow \frac{1}{[L : K]} \mathbb{Z}/\mathbb{Z} \longrightarrow 0$$

from [NSW13, Proposition 8.1.15 pp 435-437]. □

We now give a little bit of motivation for why this is referred to as a generalisation of reciprocity. If we for example take $K = \mathbb{Q}$ and a quaternion algebra (a, b) over $\mathbb{Q}$, then we map it to the algebra $(a, b)_v \cong (a, b) \otimes_{\mathbb{Q}} \mathbb{Q}_v$ and we obtain that

$$\sum_v \text{inv}_v(a, b)_v = 0.$$

By [Voi21, p. 8.2.2], we have that any quaternion algebra over a field F satisfies $(a, b) \otimes_F (a, b) \cong M_4(F)$, so that $(a, b) \in \text{Br}(F)[2]$. In particular, we have $\text{inv}_v(a, b)_v \in \{0, 1/2\} \subseteq \mathbb{Q}/\mathbb{Z}$. Then by [Voi21, Theorem 5.4.4], $(a, b)_v \cong M_2(\mathbb{Q}_v)$ if and only if $a \in N_{\mathbb{Q}_v(\sqrt{a})}(\mathbb{Q}(\sqrt{b})^*)$. Therefore, we have recovered Hilbert reciprocity:

$$\prod_v \left(\frac{a, b}{\mathfrak{p}_v} \right) = 1.$$

We also note that the Local-Global Principle of Theorem 3.5.1 implies that conics that are LSE satisfy the Hasse principle since every conic can be diagonalised and assigned a quaternion algebra over K by [CS21, Remark 1.1.11 p 16]. Therefore, we see that the Hasse principle for conics and the law of Hilbert reciprocity are inextricably linked via this single result.

We have the following important characterisation of Brauer groups:

Theorem 3.5.2. There is an isomorphism $H^2(K, K_s) \cong \text{Br}(K_s)$, where K_s is the separable closure of K . In particular, for any field K with characteristic 0, we have that $H^2(K, \bar{K}) \cong \text{Br}(K)$.

Proof. See [CS21, Theorem 1.3.5. p 29]. □

This isomorphism serves as a bridge between the classical notion of the Brauer group and its generalisation via étale cohomology since one may identify $H^2(K, K_s) \cong H_{\text{ét}}^2(\text{Spec } K, \mathbb{G}_m)$, where $\mathbb{G}_m$ represents the functor that sends a scheme $X \mapsto \mathcal{O}_X(X)^*$.

4. OBSTRUCTIONS VIA RATIONAL FUNCTIONS

In this chapter, we prove the method from [BS08] that is useful for showing certain curves violate the Hasse principle. We begin by connecting scheme theory over number fields to that over p -adic fields. In particular, we determine an explicit map between the groups of formal divisors between a K -curve and its fibred K_v -curve, using some results originally found in [BS08]. We then look at n -coverings of elliptic curves in depth, following the notes of [Sto10]. We move on to a brief description of the Brauer-Manin obstruction, following the text [CS21]. Then we state the main theorem of [BS08]. This will require some other background theory to prove, which in particular, requires theorems from [Sik02] and also some technical lemmas found in the original paper [BS08]. Finally, we close the chapter by showing that the obstruction obtained from Theorem 4.4.1 is of the Brauer-Manin type.

4.1 The Hasse Principle and Curves over p -Adic Fields

In this thesis, we take much interest in the p -adic points of a scheme, i.e. $X(K_v)$ for some valuation $v \in \mathfrak{M}(K)$ since the Hasse principle states that $X(K) \neq \emptyset$ when $X(K_v) \neq \emptyset$ for all $v \in \mathfrak{M}(K)$ (including the Archimedean primes).

Definition. We say a K -variety X over a number field is **locally soluble everywhere**, abbreviated LSE, if for all $v \in \mathfrak{M}(K)$, $X(K_v) \neq \emptyset$. We say a K -variety X violates the **Hasse principle** if X is LSE but $X(K) = \emptyset$.

We explain the connection between curves X defined over a number field K and its fibre product X_v defined over a local field K_v . Let X be a K -scheme, with K a number field. We have a natural inclusion $K \rightarrow K_v$ which gives a map between $\mathrm{Spec} K_v \rightarrow \mathrm{Spec} K$ via the equivalence of categories between rings and affine schemes (see Appendix A). We recall from Section 2.5, the fibre product $X_v := X \times_{\mathrm{Spec} K} \mathrm{Spec} K_v$ is equipped with a canonical projection map $\pi : X_v \rightarrow X$ fitting into the following cartesian diagram:

$$\begin{array}{ccc} X_v & \xrightarrow{\pi} & X \\ \downarrow & & \downarrow \\ \mathrm{Spec} K_v & \longrightarrow & \mathrm{Spec} K \end{array}$$

Given an invertible sheaf $\mathcal{L}$ over X , we obtain another invertible sheaf $\pi^*\mathcal{L}$ of X_v via the pullback. In particular, there is a well-defined map $\pi^* : \mathrm{Pic}(X) \rightarrow \mathrm{Pic}(X_v)$ by [Har77, Exercise II.6.8 p 148], which means that there is a map $\pi^* : \mathrm{Div} X \rightarrow \mathrm{Div} X_v$.

We now assume that X is a K -variety. The scheme $X_v \rightarrow \mathrm{Spec} K$ is not a morphism of finite type, so X_v is not a variety over K . To see this, we have that if A is a finite type K_v -algebra, then $K_v[x_0, \dots, x_s] \rightarrow A$ is surjective for some $s \in \mathbb{N}$. Hence, it is sufficient to show that K_v is not a finitely generated K -algebra, but this follows from [AM69, Proposition 7.9 p 82], which says that

finitely generated K -algebras that are fields are finite extensions of K . However, we know from Cantor's diagonal argument [Can91] applied to the p -adic's that K_v is transcendental over K .

For any morphism $\text{Spec } K_v \rightarrow X_v$, i.e. a point $\mathcal{P} \in X_v(K_v)$, there is a map $\text{Spec } K_v \rightarrow X$ via composition with $X_v \rightarrow X$. On the other hand, any morphism $\text{Spec } K_v \rightarrow X$ satisfies the universal property of the fibre product, so there is a unique map $\text{Spec } K_v \rightarrow X_v$. Putting these together, we have that $X(K_v) \cong X_v(K_v)$ as sets. Therefore, the study of K_v points on X is naturally done on X_v , where X_v is indeed a K_v -variety. Thus, we may really think of the points of our scheme X_v in the usual sense: as being closed points with p -adic coordinates.

Let $\mathcal{P} \in X/K$ and let $w \in \mathfrak{M}(K(\mathcal{P}))$ such that $w|v$. From Section 2.2, there is an isomorphism

$$K_v \otimes_K K(\mathcal{P}) \cong \prod_{w|v} K(\mathcal{P})_w \quad (4.1)$$

as a direct product of K_v -algebras. It follows that

$$\text{Spec}(K_v \otimes_K K(\mathcal{P})) = \coprod_{w|v} \text{Spec}(K(\mathcal{P})_w),$$

such that each point $\mathcal{Q}_w \in \text{Spec}(K(\mathcal{P})_w)$. We obtain the following result from [BS08, Lemma 4.3], originally proven for a more general setting, but we opt to prove it explicitly in the p -adic case.

Proposition 4.1.1. The projection map $\pi : X_v \rightarrow X$ is unramified (maps uniformisers to uniformisers) and for every closed point $\mathcal{P} \in X$, each point $\mathcal{Q} \in \pi^{-1}(\mathcal{P})$ is also a closed point of X_v .

Proof. The only prime ideals $\mathfrak{p}$ of our algebra $\prod_{w|v} K(\mathcal{P})_w$ are $(0) \times \prod_{j \neq i} K(\mathcal{P})_{w_j}$ where at the i -th place we have the zero ideal, which implies that for any such $\mathfrak{p}$, $\dim(\mathfrak{p}/\mathfrak{p}^2) = 0$. Therefore, each of $\mathcal{O}_{K_v \otimes_K K(\mathcal{P}), \mathfrak{p}}$ is regular ring, which makes the fibre $\text{Spec}(K_v \otimes_K K(\mathcal{P}))$ a regular scheme.

Let $\mathcal{Q}_w$ be one of the points of $\pi^{-1}\{\mathcal{P}\} \cong \text{Spec}(K_v \otimes_K K(\mathcal{P}))$, using the functor of points view (see Appendix A), this means that there is a canonical morphism $K_v \otimes_K K(\mathcal{P}) \rightarrow K_v(\mathcal{Q}_w)$, which, after enumerating the $w|v$ as w_j , gives a surjective map using (4.1):

$$\prod_j K(\mathcal{P})_{w_j} \longrightarrow K_v(\mathcal{Q}_{w_i}).$$

Let $\mathfrak{p}_0$ denote the kernel of this map. Then since limits commute, we have

$$\mathfrak{p}_0 = \prod_j \ker \left(K(\mathcal{P})_{w_j} \longrightarrow K_v(\mathcal{Q}_{w_i}) \right) = (0) \times \prod_{j \neq i} K(\mathcal{P})_{w_j}.$$

Therefore, we have $K_v(\mathcal{Q}_{w_i}) \cong (K_v \otimes_K K(\mathcal{P}))_{\mathfrak{p}_0} / \mathfrak{p}_0 = K(\mathcal{P})_{w_i}$. We claim that $\mathcal{O}_{X_v, \mathcal{Q}_w} = (\mathcal{O}_{X, \mathcal{P}} \otimes_K K_v)_{\mathfrak{p}}$, where $\mathfrak{p}$ is the inverse image of $\mathfrak{p}_0$ under $\mathcal{O}_{X, \mathcal{P}} \otimes_K K_v \rightarrow K(\mathcal{P}) \otimes_K K_v$. Since taking tensors is right-exact, there is a surjection $\mathcal{O}_{X, \mathcal{P}} \otimes_K K_v \rightarrow K_v(\mathcal{Q}_w)$, and so $\mathfrak{p}$ is a maximal ideal. Thus $(\mathcal{O}_{X, \mathcal{P}} \otimes_K K_v)_{\mathfrak{p}}$ is local ring with residue field $K_v(\mathcal{Q}_w)$. On the other hand, from the construction of fibre products, we have that for any $U \subseteq X$ open that $\pi^{-1}U = U \times_K K_v$ [Har77, Theorem II.3.3 pp 87-88]. Let $\mathcal{P} \in \text{Spec } A \subseteq X$, then $\mathcal{Q}_w \in \text{Spec}(A \otimes_K K_v)$. Let $A_{\mathcal{P}}$ denote the localisation of A at the maximal ideal corresponding to $\mathcal{P}$. This implies that $(\mathcal{O}_{X, \mathcal{P}} \otimes_K K_v)_{\mathfrak{p}} = (A_{\mathcal{P}} \otimes_K K_v)_{\mathfrak{p}} \cong \mathcal{O}_{X_v, \mathcal{Q}_w}$.

Therefore, using the following exact sequence:

$$0 \longrightarrow \mathfrak{m}_{X, \mathcal{P}} \longrightarrow \mathcal{O}_{X, \mathcal{P}} \longrightarrow K(\mathcal{P}) \longrightarrow 0,$$

together with the exactness of both $(-) \otimes_K K_v$ – which is exact by [AM69, Exercise 2.27] and using that ideals of K are (0) and itself – and localisation by our prime ideal $\mathfrak{p}_0$, in combination with the identities $(\mathcal{O}_{X,\mathcal{P}} \otimes_K K_v)_{\mathfrak{p}} = \mathcal{O}_{X_v,\mathcal{Q}_w}$ and $(\mathfrak{m}_{X,\mathcal{P}} \otimes_K K_v)_{\mathfrak{p}} = \mathfrak{m}_{X,\mathcal{P}} \mathcal{O}_{X_v,\mathcal{Q}_w}$ yield a short exact sequence:

$$0 \longrightarrow \mathfrak{m}_{X,\mathcal{P}} \mathcal{O}_{X_v,\mathcal{Q}_w} \longrightarrow \mathcal{O}_{X_v,\mathcal{Q}_w} \longrightarrow (K(\mathcal{P}) \otimes_K K_v)_{\mathfrak{p}_0} \longrightarrow 0.$$

Thus, $\mathcal{O}_{X_v,\mathcal{Q}_w} / \mathfrak{m}_{X,\mathcal{P}} \mathcal{O}_{X_v,\mathcal{Q}_w} \cong (K(\mathcal{P}) \otimes_K K_v)_{\mathfrak{p}_0} = K(\mathcal{P})_w$. Now the right-hand side of this equation is a field. Therefore, we conclude that $\mathfrak{m}_{X,\mathcal{P}} \mathcal{O}_{X_v,\mathcal{Q}_w} = \mathfrak{m}_{X_v,\mathcal{Q}_w}$. In particular, we obtain that any uniformiser of $\mathcal{P} \in X$ remains a uniformiser in X_v .

Since X_v is also a curve over K_v , we have that X_v decomposes as a topological space into $X_v^c \cup \eta$, with η the generic point. Thus, since the points $\mathcal{Q}_w$ have a residue field $K_v(\mathcal{Q}_w)$ that is a finite extension of K_v it follows that $\mathcal{Q}_w$ is closed. $\square$

We may realise the map $\pi^* : \text{Pic}(X) \rightarrow \text{Pic}(X_v)$ in terms of the class groups of X and X_v using the previous proposition.

Corollary 4.1.2. The induced pullback map on divisors is given as

$$\pi^* : \text{Pic}(X) \longrightarrow \text{Pic}(X_v) \tag{4.2}$$

$$[\mathcal{P}] \longmapsto \sum_{w|v} [\mathcal{Q}_w] \tag{4.3}$$

where $w \in \mathfrak{M}(K(\mathcal{P}))$ such that $w|v$.

Proof. For $f \in K(X)$, if $\text{ord}_{\mathcal{P}}(f) = n$, then $\text{ord}_{\mathcal{Q}_w}(\pi^* f) = n$. In particular, $\pi^*(\text{div } f) = \text{div } \pi^* f$ is a principal divisor. Therefore, the map is well-defined. $\square$

4.2 n -Coverings of Elliptic Curves

We now come to an important definition in the context of the theory of descent. We take the following definition from [Sto10, p 9].

Definition. An n -**covering** of an elliptic curve E/K is a pair (X, φ) with X/K a regular curve admitting a $\bar{K}$ -isomorphism $X \dashrightarrow E$ and $\varphi : X \rightarrow E$ a K -rational morphism making the following diagram commute:

$$\begin{array}{ccc} X & \dashrightarrow & E \\ & \searrow \varphi & \downarrow [n] \\ & & E \end{array}$$

Then (X, φ) is called a **twist** of the map $[n] : E \rightarrow E$ and, we note that φ is a map of degree n^2 , i.e. $[K(X) : \varphi^* K(E)] = n^2$.

Given an n -covering (X, φ) , let f denote the $\bar{K}$ -isomorphism from the above diagram. The curves X are homogeneous spaces of E since we may define a pairing

$$\begin{aligned} + : X \times E &\longrightarrow X \\ (p, \mathcal{P}) &\longmapsto f^{-1}(f(p) + \mathcal{P}). \end{aligned}$$

Indeed it is K -rational: we have for all $\mathcal{Q} \in E$ that

$$(\tau_{\mathcal{Q}} \circ f)^{-1}(\tau_{\mathcal{Q}} \circ f(p) + \mathcal{P}) = f^{-1}(f(p) + \mathcal{Q} + \mathcal{P} - \mathcal{Q}) = f^{-1}(f(p) + \mathcal{P}),$$

which implies $+ : E \times C \rightarrow C$ is K -rational since for all $\sigma \in G_K$, f^σ is given by a translation of f .

For $p \in X$ it satisfies

$$p + \mathcal{O} = f^{-1}(f(p) + \mathcal{O}) = f^{-1}f(p) = p.$$

For $P, Q \in E$, we also have

$$(p + P) + Q = f^{-1}(f(f^{-1}(f(p) + P)) + Q) = f^{-1}(f(p) + P + Q) = p + (P + Q).$$

For the third property, for $p, q \in X$, since $f : X \rightarrow E$ is an isomorphism, there exists a unique $Q \in E$ such that $f^{-1}Q = q$. Since p is fixed, and $P \mapsto P$ is clearly an automorphism on E , we have that $P \mapsto f(p) + P$ defines a scheme isomorphism $E \rightarrow E$, which implies that there exists a unique P such that $f(p) + P = Q$.

Thus as a homogeneous space, we inherit a relation on n -coverings as well: $(X_1, \varphi_1) \sim (X_2, \varphi_2)$ if we have the following commutative diagram of scheme morphisms

$$\begin{array}{ccc} X_1 & \xrightarrow{\varphi_1} & E \\ \downarrow \wr & & \downarrow \wr \\ X_2 & \xrightarrow{\varphi_2} & E \end{array}$$

where the first down arrow is K -rational. We will now show that the soluble n -coverings X are in bijection with the points of $E(K)/nE(K)$. In particular, as discussed in [Sto10, pp 9-10], we can recover the connecting homomorphism:

$$\delta : E(K)/nE(K) \longrightarrow H^1(K, E[n])$$

from the notion of n -coverings. Indeed, this is done by first taking any point $\mathcal{P} \in E(K)$ and considering the n -covering (E, φ) with $\varphi(\mathcal{Q}) := n\mathcal{Q} + \mathcal{P}$. That this is an n -covering follows since we know there exists an $\mathcal{R}$ such that $n\mathcal{R} = \mathcal{P}$. This implies that $\varphi(\mathcal{Q}) = n(\mathcal{Q} + \mathcal{R})$. Then we have that $E \rightarrow E, \mathcal{Q} \mapsto \mathcal{Q} + \mathcal{R}$ is a scheme isomorphism. Also mentioned in [Sto10], the n -covering is the same for any choice of $\mathcal{P} \bmod nE(K)$: if $\mathcal{P} = \mathcal{Q} + n\mathcal{R}$ for $\mathcal{P}, \mathcal{Q}, \mathcal{R} \in E(K)$, we define the n -covering $\varphi'(\mathcal{Q}) = n\mathcal{Q} + \mathcal{P} + n\mathcal{R}$ for $\mathcal{P}, \mathcal{R} \in E(K)$, then $\varphi = \tau_{\mathcal{R}} \circ \varphi'$, so that $[(E, \varphi)] = [(E, \varphi')]$.

Conversely, we let $\varphi : X \rightarrow E$ be an n -covering such that $X(K) \neq \emptyset$. As a homogeneous space admitting a rational section, there is an K -isomorphism $E \xrightarrow{\sim} X$ which sends $\mathcal{O} \mapsto \mathcal{Q}$ for some $\mathcal{Q} \in X(K)$. Then we have the composition $E \xrightarrow{\sim} X \xrightarrow{\varphi} E$ which will map $\mathcal{O} \mapsto \mathcal{P}$ for some $\mathcal{P} \in E(K)$. In [Sil09, Proposition X.5.1], it is proven that

$$E \times \text{Aut}(E) \longrightarrow \text{Isom}(E), \quad (P, \alpha) \longmapsto \tau_P \circ \alpha$$

is a bijection. Hence, we have that the above composition is $\mathcal{Q} \mapsto n\mathcal{Q} + \mathcal{P}$. Therefore, we furnish the discussed bijection with the map from above:

$$\begin{aligned} E(K)/nE(K) &\longleftrightarrow \{\text{Soluble } n\text{-coverings of } E/K\} / \sim \\ \mathcal{P} &\longmapsto (E, \mathcal{Q} \mapsto n\mathcal{Q} + \mathcal{P}). \end{aligned}$$

Then we define the following injective map:

$$\begin{aligned} E(K)/nE(K) &\longrightarrow H^1(K, E[n]) \\ \mathcal{P} &\longmapsto [\xi : \sigma \mapsto \mathcal{Q}^\sigma - \mathcal{Q}], \end{aligned}$$

such that $\varphi(\mathcal{Q} - \mathcal{R}) = \mathcal{P}$, with $\mathcal{R}$ as above, i.e. $n\mathcal{Q} = \mathcal{P}$. We check this is well-defined. Since $n(\mathcal{Q}^\sigma - \mathcal{Q}) = \mathcal{P}^\sigma - \mathcal{P} = \mathcal{O}$, it follows that $\mathcal{Q}^\sigma - \mathcal{Q} \in E[n]$. If we define the n -covering $\varphi'(\mathcal{Q}) := n\mathcal{Q} + \mathcal{P} + n\mathcal{R}$ for $\mathcal{P}, \mathcal{R} \in E(K)$, then $\varphi = \tau_{\mathcal{R}} \circ \varphi'$, so that $[(E, \varphi)] = [(E, \varphi')]$.

Finally, we show that the insoluble n -coverings are elements of order dividing n in $\text{WC}(E/K)$. In particular, the LSE n -coverings that satisfy $X(K) = \emptyset$ are elements of order dividing n in the Tate-Shafarevich group.

Proposition 4.2.1. There is a surjective group morphism:

$$\{n\text{-coverings of } E/K\} / \sim \longrightarrow \text{WC}(E/K)[n].$$

Proof. First, we have that φ is K -rational, so that $\varphi(p^\sigma) = \varphi(p)^\sigma$, which implies that $\varphi(p^\sigma) = \mathcal{O}$ for p with $\varphi(p) = \mathcal{O}$. Now, using that $p^\sigma - p = f(p^\sigma) - f(p)$, we have

$$n(p^\sigma - p) = n(f(p^\sigma) - f(p)) = \varphi(p^\sigma) - \varphi(p) = \mathcal{O}.$$

From the map from (2.4), we have that the $X \mapsto [\xi : \sigma \mapsto (p^\sigma - p)]$, so that $n\xi(\sigma) = \mathcal{O}$, which implies that $X \in \text{WC}(E/K)[n]$. Thus, we obtain a well-defined map.

Let $X \in \text{WC}(E/K)[n]$. Then, by (2.4), for any $p \in X$, there exists $\xi \in H^1(K, E)$ such that $\xi : \sigma \mapsto p^\sigma - p$ with $n\xi(\sigma) = \mathcal{O}$ for all $\sigma \in G(\bar{K}/K)$. As a homogeneous space, by taking the inverse in [Sil09, Proposition X.3.2 p 323], there exists an isomorphism $g : X \rightarrow E$ such that $g(q) = q - p$ for all $q \in X$. Hence, we have that

$$([n] \circ g(q))^\sigma = n(g(q)^\sigma - g(p)^\sigma) = n((q^\sigma - p) - (p^\sigma - p)) = n(q^\sigma - p) = [n] \circ g(q^\sigma),$$

which implies that the map is K -rational and must be an n -covering. $\square$

Hence, we have the short exact sequence:

$$0 \longrightarrow E(K)/nE(K) \longrightarrow H^1(K, E[n]) \longrightarrow \{n\text{-coverings of } E/K\} / \sim \longrightarrow 0$$

2-Coverings

For the case $n = 2$, we wish to compute

$$E(K)/2E(K) \rightarrow H^1(K, E[2]),$$

which may classically be done using descent by 2-isogeny in the event that $\#E(K)[2] \neq 1$. We note that in the computation section, we devise an algorithm for computing this over p -adic fields. In any case, the homogeneous spaces that arise from these computations are hyperelliptic curves:

$$C : y^2 = x^4 + ax^2 + b.$$

by [Sil09, Proposition X.4.9]. If $X(K)[2] = \emptyset$, 2-descent is still possible (see [Sim02] and Section 5.1 below), and it turns out that in this case that the two-coverings that arise are also determined by hyperelliptic curves. In general, we have from [Sto10, p. 12] that the 2-coverings always correspond to hyperelliptic curves.

3-Coverings

In [Sto10], it is also shown that the curves arising from 3-descent can always be embedded in $\mathbb{P}^2$ as a plane cubic. For example the famous Selmer curve

$$C : 3X^3 + 4Y^3 + 5Z^3 = 0$$

is a 3-covering of

$$E : y^2 = x^3 + 24300$$

and turns out to be a nontrivial element of the Tate-Shafarevich group; see Chapter 5 below for more details on this example.

4.3 The Brauer-Manin Obstruction

Now we associate a Brauer group $\text{Br}(X)$ to a scheme X . Mimicking the core idea of the Brauer group of field, one may define $\text{Br}_{Az}(X)$ to be the $\mathcal{O}_X$ -algebras A such that for all points $x \in X$, the stalk A_x is an **Azumaya algebra** over $\mathcal{O}_{X,x}$ up to a certain equivalence [CS21, Section 3.1 p 76]. This is the definition used by Grothendieck in [Gro66a, p 200]. However, the definition used here is derived from étale cohomology, and there is an injection $\text{Br}_{Az}(X) \rightarrow \text{Br}(X)$ from [CS21, Theorem 3.3.1 p 79]. Moreover, our assumption that X is a regular, complete curve over K will actually imply that $\text{Br}_{Az}(X) \cong \text{Br}(X)$ by [CS21, Lemma 3.5.2 and Theorem 3.3.2]. Therefore, the results we list below also apply to the Azumaya-Brauer group.

For any $A \in \text{Br}(X)$, there is a commutative diagram:

$$\begin{array}{ccccc} X(K) & \longrightarrow & X(A_K) & & \\ \text{ev}_A \downarrow & & \downarrow & \searrow \text{ev}_A & \\ \text{Br}(K) & \longrightarrow & \bigoplus_{v \in \mathfrak{M}(K)} \text{Br}(K_v) & \xrightarrow{\text{inv}_v} & \mathbb{Q}/\mathbb{Z} \end{array}$$

where the bottom line is the exact sequence of Theorem 3.5.1. Following the left side and bottom of the diagram, we have that the rational points $\mathcal{P} \in X(K)$ satisfy:

$$\sum_v \text{inv}_v(\text{ev}_A(\mathcal{P})) = 0 \in \mathbb{Q}/\mathbb{Z}$$

for all $A \in \text{Br}(X)$. For proper curves, we have the isomorphism $X(A_K) \cong \prod_{v \in \mathfrak{M}(K)} X(K_v)$ from [CS21, Section 12.1 p 246]. If X is LSE, then there is the following necessary criterion for the existence of rational points: if $X(K) \neq \emptyset$, then there exists $\mathcal{P}_v \in X(K_v)$ such that for all $A \in \text{Br}(X)$ we have

$$\sum_{v \in \mathfrak{M}(K)} \text{ev}_A(\mathcal{P}_v) = 0,$$

where $\text{ev}_A : X(K_v) \rightarrow \text{Br}(K_v)$ is the induced map from the section $\text{Spec } K_v \rightarrow X$ from [CS21, (3.1) p 77]. Now we define the **Brauer-Manin** set:

$$X(A_K)^{\text{Br}} := X(A_K)^{\text{Br}(X)} := \left\{ (\mathcal{P}_v) \in X(A_K) \left| \sum_{v \in \mathfrak{M}(K)} \text{inv}_v(\text{ev}_A(\mathcal{P}_v)) = 0 \text{ for all } A \in \text{Br}(X) \right. \right\}$$

This brings us to the idea behind **the Brauer-Manin obstruction**: find a finite subset $B \subseteq \text{Br}(X)$ such that if

$$X(A_K)^B := \left\{ (\mathcal{P}_v) \in X(A_K) \left| \sum_{v \in \mathfrak{M}(K)} \text{inv}_v(\text{ev}_A(\mathcal{P}_v)) = 0 \text{ for all } A \in B \right. \right\}$$

then $X(A_K)^B = \emptyset$, which would then imply that $X(K) = \emptyset$. To this end, one desires to construct elements of $\text{Br}(X)$ such that this phenomenon occurs. One natural way to do this is to use the Brauer group of the function field $K(X)$. In particular, from [CS21, Theorem 3.5.4 pp 88-89], if X is a geometrically integral regular K -scheme (which is enough for our purposes), then there is a natural injection

$$\text{Br}(X) \longrightarrow \text{Br}(K(X))$$

induced from étale cohomology. The image of $\text{Br}(X)$ in $\text{Br}(K(X))$ is characterised by the following exact sequence from [CS21, Proposition 3.6.1 pp 89-90]:

$$0 \longrightarrow \text{Br}(X) \longrightarrow \text{Br}(K(X)) \longrightarrow \bigoplus_{\mathcal{P} \in X^c} H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z}) \longrightarrow 0. \quad (4.4)$$

We exhibit how these definitions are used via the classical example from Lind and Reichardt that

$$U : 2y^2 = x^4 - 17$$

is a hyperelliptic curve that violates the Hasse principle [LPS25]. Lind's original proof that $U(\mathbb{Q}) = \emptyset$ is elementary and utilises quadratic reciprocity [Cas66]. For completeness, we reproduce the argument here. We write this in its weighted homogeneous form $2Y^2 = X^4 - 17Z^4$ and may assume that $\gcd(X, Z) = 1$. For any prime $p|Y$, we have that $(\frac{17}{p}) = 1$. Then $17 \equiv 1 \pmod{4}$ so that $(\frac{17}{p}) = (\frac{p}{17}) = 1$, which in turn implies that $(\frac{Y}{17}) = 1$. This would imply that $2 \in \mathbb{F}_{17}^{*4}$, which is false.

In [CS21, p. 5.2.11], it is shown that for the smooth projectivisation X of Lind's affine curve $2y^2 = x^4 - 17$ that the quaternion algebra $A := (y, 17) \in \text{Br}(K(X))$ corresponds to an element of $\text{Br}(X)$. Then in [CS21, section 12.5 p 260], it is demonstrated that for any $(\mathcal{P}_v) \in X(A_{\mathbb{Q}})$

$$\sum_{v \in \mathfrak{M}(\mathbb{Q})} \text{inv}_v(\text{ev}_A(\mathcal{P}_v)) = \frac{1}{2} \in \mathbb{Q}/\mathbb{Z},$$

which therefore implies a Brauer-Manin obstruction for the choice $B = \{A\} \subseteq \text{Br}(X)$. One shows that for any place $v \neq 17$, $\text{inv}_v(\text{ev}_A(\mathcal{P}_v)) = 0$, and one shows that $\text{inv}_{17}(\text{ev}_A(\mathcal{P}_{17})) = 1/2$ [CS21]. In [Sti11, 5.2.2 p 1383], this is shown to relate to Lind's original proof involving quadratic reciprocity; if one assumed that $\text{inv}_{17}(\text{ev}_A(\mathcal{P}_{17})) = 0$, this would imply that $(\frac{y}{17}) = 1$, which in turn implies $2 \in \mathbb{F}_{17}^{*4}$.

Siksek's Method for Generating Congruences

The core idea of Lind's counterexample was to attach the congruence condition $(\frac{Y}{17}) = 1$ via quadratic reciprocity to the curve. Siksek set out to use the Gauss law of reciprocity to systematically produce similar types of congruence conditions on the points of a scheme. Consider a hyperelliptic curve

$$y^2 = f(x),$$

with $f(x) \in \mathbb{Z}[x]$. We may also view it as a weighted homogeneous curve $Y^2 = F(X, Z)$, where $F(X, Z)$ is homogeneous. In Appendix A, we show this is an affine open in its smooth projectivisation. Let I be an interval of $\mathbb{R}$ such that $f(x) \geq 0$ for all $x \in I$. We suppose that $X, Z \in \mathbb{Z}$ satisfy the following conditions:

$$\gcd(X, Z) = 1, \quad Z \geq 1 \quad \text{and} \quad \frac{X}{Z} \in I.$$

The idea that the Siksek came up with was to select pairs of integers (α, β) such that $F(\alpha, \beta) = \gamma\delta^2$ with γ square-free and then deduce congruence restrictions on X and Z using the linear term $\beta X - \alpha Z$. Indeed he derived the following theorem:

Theorem 4.3.1. Let $(\alpha, \beta, \delta, \gamma)$ be a tuple such that $\gcd(\alpha, \beta) = 1$ and $F(\alpha, \beta) = \gamma\delta^2$ with γ square-free and not equal to 0 or 1. Let

$$N = \begin{cases} |\gamma| & \text{if } \gamma \equiv 1 \pmod{4} \\ 4|\gamma| & \text{if } \gamma \equiv 2, 3 \pmod{4}. \end{cases}$$

Let I be an interval of $\mathbb{R}$ such that $f(x) \neq 0$ for all $x \in I$. If the tuple (X, Z, Y) corresponds to a point of $Y^2 = F(X, Z)$ and X and Z satisfy $\gcd(X, Z) = 1$, $Z \geq 1$ and $\frac{X}{Z} \in I$ then we have that there exist integers h, P such that

1. γ is a square modulo h .
2. For each $p|N$, $v_p(P) \leq v_p(\beta X - \alpha Z)$.
3. For at most one prime $q|2\delta$, $v_q(P) = 1$ and otherwise $v_q(P) = 0$.

Then we have

$$\beta X + \alpha Z \equiv \zeta Ph \pmod{PN},$$

where $\zeta = 1$ if there exists $w \in I$ such that $\beta w - \alpha \geq 0$ and $\zeta = -1$ otherwise.

Proof. See [Sik01] for the complete proof. We sketch the proof and briefly explain how this ties into the theory of quadratic reciprocity. The idea of the proof is to write

$$|\beta X - \alpha Z| = p_1^{r_1} \dots p_l^{r_l} M,$$

where we split the primes $p_i|N$ from the rest. Quadratic reciprocity implies the existence of a subgroup $H \subseteq (\mathbb{Z}/N\mathbb{Z})^*$ of index 2, where elements $a \equiv q \pmod{N} \in H$ (for some prime q by Dirichlet's theorem on progressions) if and only if $(\frac{N}{q}) = 1$. Using that this subgroup has index 2 and that $p^r|\beta X - \alpha Z$ implies $\gamma\delta^2$ is a square modulo p^r by [Sik01, Lemma 2.1], one shows that either $M \pmod{N} \in H$ or that $M \equiv hq \pmod{Nq}$ for some prime $q|2\delta$. This implies that

$$|\beta X - \alpha Z| \equiv Ph \pmod{PN}$$

where $P = q^j \prod_i p_i^{r_i}$ with $j = 0$ or 1 depending on M . Finally, one removes the absolute bars, and if one must multiply by -1 to make it nonnegative, this implies that $\gamma \geq 0$, which means that $-1 \pmod{N} \in H$ so that this permutes the elements of H . Then the result stands. $\square$

In [Sik01, Section 4], he showed the hyperelliptic curve

$$y^2 = -4x^4 + 4x^3 + 92x^2 - 104x - 727$$

violates the Hasse principle using the linear terms Z and $16X + 53Z$ and Theorem 4.3.1. We will revisit this example in Section 5.1 using Theorem 4.4.1 in the next section.

4.4 Artin Reciprocity and the Hasse Principle

Let X/K be a complete and regular curve. Let $f \in K(X)^*$ be a nonzero rational function. We recall that we showed in Proposition 3.2.2 that the subgroup

$$\prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(C_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}} f} \subseteq C_K$$

was open in the norm topology of C_K . Using the bijection of Theorem 3.3.4, there exists an abelian extension L/K such that

$$N_{L/K} C_L \cong \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(C_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}} f}.$$

Our main theorem in this thesis is the following from [BS08, Section 1]:

Theorem 4.4.1. (Bright & Siksek) Let X be a complete, regular curve over a number field K . Let $f \in K(X)^*$ non-constant such that integers $\text{ord}_{\mathcal{P}}(f)$ do not share a common factor as $\mathcal{P}$ runs through the support of f . Let L be the class field associated to f . The following hold:

1. For $v \in \mathfrak{M}(K)$, there exists a unique homomorphism

$$\varphi_v : \text{Pic}(X_v) \longrightarrow G(L/K)$$

such that for any closed point $\mathcal{Q}$ of X_v

$$\varphi_v([\mathcal{Q}]) = \theta_v(N_{K_v(\mathcal{Q})/K_v}(f(\mathcal{Q})))$$

where $\theta_v : K_v/N_{w/v}L_w \rightarrow G(L/K)$ is induced from the canonical map from Artin reciprocity.

2. There exists a finite set of places $B \subseteq \mathfrak{M}(K)$ such that for any $v \notin B$, we have $\varphi_v = 1$.
3. Under the diagonal map $\Delta : \text{Pic}(X) \rightarrow \prod_{v \in B} \text{Pic}(X_v)$, we have

$$\Delta(\text{Pic}(X)) \subseteq \ker \left(\prod_{v \in B} \text{Pic}(X_v) \xrightarrow{\prod \varphi_v} \text{Gal}(L/K) \right)$$

i.e. we have the following commutative diagram:

$$\begin{array}{ccc} & \prod_{v \in B} \text{Pic}(X_v) & \\ \nearrow & & \searrow \prod_v \varphi_v \\ \text{Pic}(X) & \xrightarrow{D \mapsto 1} & G(L/K) \end{array}$$

The elements of $\text{Pic}(X_v)/n\text{Pic}(X_v)$ have a well-defined degree modulo n . Let $[D_1]$ and $[D_2]$ be two divisors from $\text{Pic}(X)$ that map to the same class modulo $n\text{Pic}(X)$. Now this means that $[D_1] = [D_2] + n[D]$ for some divisor D . Since principal divisors over a projective scheme always have degree zero, we have that $\deg D_1 = \deg D_2 + n \deg D$, so their images have the same degree modulo n . Therefore, we can define the subset $(\prod_{v \in B} \text{Pic}(X_v)/n\text{Pic}(X_v))_r$ of **divisors of degree r modulo n** . With this in mind, we derive from the theorem the following useful corollary seen in [BS08, Corollary 1.1]:

Corollary 4.4.2. For n an integer, there is an induced diagram:

$$\begin{array}{ccc} & \prod_{v \in B} \text{Pic}(X_v)/n\text{Pic}(X_v) & \\ \nearrow & & \searrow \prod_v \varphi_v \\ \text{Pic}(X)/n\text{Pic}(X) & \xrightarrow{D \mapsto 1} & G(L/K)/G(L/K)^n \end{array}$$

that in particular has for any integer r :

$$\left(\prod_{v \in B} \text{Pic}(X_v)/n\text{Pic}(X_v) \right)_r \longrightarrow G(L/K)/G(L/K)^n \quad (4.5)$$

such that $(\text{Pic}(X)/n\text{Pic}(X))_r$ is in the kernel of (4.5). In particular, if this kernel is empty, then there are no divisors of degree of $r, r+n, r+2n, \dots$ and so on.

Proof. The map φ_v from Theorem 4.4.1 (a) is a group morphism, so we obtain a natural induced map:

$$\text{Pic } X_v/n\text{Pic } X_v \longrightarrow \text{Gal}(L/K)/\text{Gal}(L/K)^n$$

fitting into the following commutative diagram:

$$\begin{array}{ccc} \text{Pic}(X_v) & \longrightarrow & \text{Pic}(X_v)/n\text{Pic}(X_v) \\ \downarrow \varphi_v & & \downarrow \\ G(L/K) & \longrightarrow & G(L/K)/G(L/K)^n \end{array}$$

Therefore, from Theorem 4.4.1 (c), we have under the induced diagonal map $\Delta : \text{Pic}(X) \rightarrow \prod_{v \in B} \text{Pic}(X_v)/n\text{Pic}(X_v)$ that

$$\Delta(\text{Pic } X) \subseteq \ker \left(\prod_{v \in B} \text{Pic } X_v/n\text{Pic } X_v \longrightarrow \text{Gal}(L/K)/\text{Gal}(L/K)^n \right).$$

We saw above that degree modulo n is well-defined. It remains to prove that a divisor $D \in \text{Pic}(X)/n\text{Pic}(X)$ with degree r modulo n , has image in $\text{Pic}(X_v)/n\text{Pic}(X_v)$ of degree r modulo n . Then from Corollary 4.1.2, we have that $\mathcal{P} \mapsto \sum_{w|v} \mathcal{Q}_w$ such that $[K(\mathcal{P}) : K] = \sum_{w|v} [K_v(\mathcal{Q}_w) : K_v]$, which therefore implies that degree is preserved under the diagonal map. $\square$

We will need several essential ingredients to prove Theorem 4.4.1.

1. A technical lemma that ensures that $\text{Pic}(X_v)$ is generated by points $\mathcal{P}$ such that $f(\mathcal{P})$ is defined and $v(N_{K_v(\mathcal{P})/K_v} f(\mathcal{P})) = 0$.
2. A map $\text{Pic}(X_v) \rightarrow K_v/N_{w/v}L_w$ for $w \in \mathfrak{M}(L)$ with $w|v$.

The Moving Lemma

We begin with the first ingredient for the proof of Theorem 4.4.1, which we will refer to as the **moving lemma**. We emphasize that the following lemma plays mostly a technical role in this thesis, and it will not have any use beyond proving the main theorem.

There is a more typical notion of a moving lemma, which says that given a projective K -variety X for a field K and a function $f \in K(X)$, then any class of $\text{Pic}(X)$ is represented by a Weil divisor such

that it has no overlap with the support of f [LE02, Corollary 9.1.10 p p 379]. The moving lemma we prove in this subsection, originally proven in [BS08], is a stronger version of this. If we assume that our curve $X \rightarrow \operatorname{Spec} K$, with K a number field, has a model that satisfies a few good reduction criteria, we can find generators $\mathcal{P}$ of $\operatorname{Pic}(X_v)$ so that we essentially have $v(f(\mathcal{P})) = 0$.

To our curve X_v/K_v , we can associate a **projective model** $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_v$, see Appendix A for the details on its existence and possible constructions. This scheme is projective, so it admits a closed immersion $\mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_v}^n$. Informally, we think about our projective model as corresponding to a system of equations in integral coefficients. For example, given an elliptic curve E/K over a number field, there is the projective curve $\mathcal{X}/\mathcal{O}_K$ defined by an integral weierstrass equation $y^2 + a_1xy + a_3 = f(x)$ such that the discriminant $\Delta(\mathcal{X})$ is minimal with respect to all p -adic valuations of K [Sil09, VII.1 p 186]. Via the inclusions $\mathcal{O}_K \rightarrow \mathcal{O}_v$ and $K \rightarrow K_v$, the curves $\mathcal{X}_v := \mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_v$ are then projective models of $E_v = E \times_K K_v$.

We now discuss good reduction. Take K to be a local non-Archimedean field, R the unique discrete valuation ring and κ its residue field. Following Appendix A, a curve $X \rightarrow \operatorname{Spec} K$ has **good reduction** if there exists a projective model $\mathcal{X} \rightarrow \operatorname{Spec} R$ such that the special fibre $\mathcal{X}_\kappa := \mathcal{X} \times_{\operatorname{Spec} R} \operatorname{Spec} \kappa$ is regular, i.e. $\mathcal{X} \rightarrow \operatorname{Spec} R$ is smooth. From [Sta25, tag 0C2U], since our schemes X are regular projective K -curves, then there is always a regular model $\mathcal{X}$. However, a model $\mathcal{X}$ might be regular, but that does not imply it is smooth since the residue fields κ are not flat R -algebras. For elliptic curves and abelian varieties more generally, there is a flat scheme $\mathcal{N}$ called a Néron model that is smooth, but this scheme is not always projective.

Since $\mathcal{X}$ is proper, to each closed point $\mathcal{P} \in X \subseteq \mathcal{X}$, we associate a unique closed point $\tilde{\mathcal{P}} \in \mathcal{X}$ such that $\operatorname{Spec} K(\mathcal{P}) \rightarrow X$ has $*$ $\mapsto \mathcal{P}$ if and only if $\operatorname{Spec} \mathcal{O}_{K(\mathcal{P})} \rightarrow \mathcal{X}$ with $(0) \mapsto \mathcal{P}$ and $\mathfrak{P} \mapsto \tilde{\mathcal{P}}$ by Theorem A.5.3 from Appendix A. Going forward, we denote $\mathcal{O}_{\mathcal{P}} := \mathcal{O}_{\mathcal{X}, \mathcal{P}} \cong \mathcal{O}_{X, \mathcal{P}}$ and $\mathcal{O}_{\tilde{\mathcal{P}}} := \mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}$.

We have from the results in Appendix B, that $\dim \mathcal{X} = 2$, which makes it a so-called **fibred surface**. As noted in Appendix A, it also has the same function field as X . From Theorem B.0.1, we have that $\mathcal{X}_\kappa$ has dimension 1, and if it is geometrically irreducible, then it is a projective curve. Let Y_i denote the irreducible components of the special fibre, then these are prime divisors of $\mathcal{X}$. This means that there are valuations $\operatorname{ord}_{Y_i} : K(X)^* \rightarrow \mathbb{Z}$ such that their associated valuation rings $\mathcal{O}_{Y_i}$ dominate $\mathcal{O}_v$. From Theorem B.0.1, these valuations ord_{Y_i} are the only valuations that extend v arising from the prime divisors of $\mathcal{X}$. If the special fibre is irreducible, there is a single valuation that must be the unique extended valuation of $v : K(X)^* \rightarrow \mathbb{Z}$.

Different from the usual notion of good reduction, we will assume that the projective model $\mathcal{X}$ and $f \in K(X)$ satisfy the following good reduction criteria:

1. For each of the irreducible components Y_i of the special fibre, $\operatorname{ord}_{Y_i}(f) = 0$. If $\mathcal{X}_\kappa$ is irreducible, then this means that $v(f) = 0$, where we write $v : K(X)^* \rightarrow \mathbb{Z}$ to mean the extended valuation.
2. For each closed point $\tilde{\mathcal{P}} \in \mathcal{X}$ such that either $f \notin \mathcal{O}_{\tilde{\mathcal{P}}}$ or $f \in \mathfrak{m}_{\tilde{\mathcal{P}}}$, $\mathcal{X}$ is smooth at $\tilde{\mathcal{P}}$, i.e. $\mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}$ is regular.

We note that by [LE02, Theorem 4.3.36 p 142] that assumption (2) implies $\mathcal{X}$ is regular at $\tilde{\mathcal{P}}$. We now have the following important result, the so-called moving lemma, originally proven in [BS08, lemma 3.1].

Theorem 4.4.3. (Moving Lemma) Let X be a regular, complete curve over a non-Archimedean local field K , $f \in K(X)$ and $\mathcal{X}$ a projective model satisfying criteria (1) and (2). Then $\operatorname{Pic}(X)$ is

generated by classes of points $[\mathcal{P}]$ such that f is regular and nonzero at $\tilde{\mathcal{P}}$.

The moving lemma essentially tells us that in the situation that K is a number field and X a curve over K , then for almost all $v \in \mathfrak{M}(K)$, the generators of $\text{Pic}(X_v)$ will be points $\mathcal{P}$ such that $f(\mathcal{P}) \neq 0$ and such that $v(N_{K_v(\mathcal{P})/K_v}(f(\mathcal{P}))) = 0$. Before we prove this, we state a useful lemma from commutative algebra:

Lemma 4.4.4. (Prime Avoidance Lemma) Let A be a ring. Given ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_s$ of A , all of which, except at most 2, are prime, and another ideal J such that $J \not\subseteq \mathfrak{p}_i$ for each i , then there exists $x \in J$ such that $x \notin \mathfrak{p}_i$.

Proof. See [Sta25, tag 00DS] (see [Dav67], also seen in Exercise 16.8 in Matsumura) $\square$

Proof of Theorem 4.4.3. The following proof is based on that from the original paper [BS08]. First, we claim that there are a finite number of points $\tilde{\mathcal{P}}_1, \dots, \tilde{\mathcal{P}}_s \in \mathcal{X}_\kappa \subseteq \mathcal{X}$ such that f has nonzero vanishing behaviour. There exists an open affine $U \subseteq \mathcal{X}$ such that $f \in \Gamma(U, \mathcal{O}_\mathcal{X})$. Then $\text{ord}_{Y_i}(f) > 0$ if and only if $U \cap Y_i \subseteq V(f) \subseteq U$. Hence (1) implies that $Y_i \cap U$ and $V(f)$ only intersect in finitely many primes since $Y_i \cap U$ has dimension 1. Thus the claim follows. Let $\mathcal{P} \in X$ be a point such that its extended point $\tilde{\mathcal{P}}$. It follows from the pigeonhole principle that $\tilde{\mathcal{P}}$ is one of the $\tilde{\mathcal{P}}_i$, and we may take without loss of generality $\tilde{\mathcal{P}} = \tilde{\mathcal{P}}_1$.

Now the goal is to find a function $h \in K_v(\mathcal{X})$ such that

- $\text{ord}_Y(h) = 1$, where $Y = \overline{\{\mathcal{P}\}}$.
- $\text{ord}_D(h) = 0$ for any prime divisor $D \neq Y \in \text{Div } \mathcal{X}$ that contains $\tilde{\mathcal{P}}$.
- $\text{ord}_{\tilde{\mathcal{P}}_i}(h) = 0$ for $i \geq 2$, where $\text{ord}_{\tilde{\mathcal{P}}_i} : \text{Frac}(\mathcal{O}_{\mathcal{X}_\kappa, \tilde{\mathcal{P}}_i})^* \rightarrow \mathbb{Z}$ is a well-defined valuation since we assume that points $\tilde{\mathcal{P}}_i$ are smooth.

We will see that these criterion imply that $[\mathcal{P}] = [\mathcal{P} - \text{div } h]$ indeed satisfies the theorem.

We will reduce to the case that $\mathcal{X}$ is affine. The argument we employ is based on that of [Gro61, Corollary 4.5.4 p 85]. In Appendix A, we define invertible sheaves $\mathcal{O}(n)$ over $\mathbb{P}_{\mathcal{O}_v}^n$ that are generated globally by the homogeneous elements of $\mathcal{O}_v[x_0, \dots, x_n]$ of degree m . Then since $i : \mathcal{X} \rightarrow \mathbb{P}_{\mathcal{O}_v}^n$ is a closed immersion, there is a canonical isomorphism $\mathcal{X} \cong \text{Proj}(\bigoplus_{n \geq 0} \Gamma(\mathcal{X}, \mathcal{O}_\mathcal{X}(n)))$ [Vak25, Exercise 15.6.G]. Thus, each $\mathcal{P}_i$ corresponds to a graded homogeneous maximal ideal $\mathfrak{p}_i$ of S . let $Z := \{\mathcal{P}_1, \dots, \mathcal{P}_n\}$ be the corresponding closed subset of $\mathcal{X}$ and let $U \supseteq Z$ be an open subset $U \neq \mathcal{X}$. Then the complement of U is closed and corresponds to a homogeneous graded ideal I such that $V(I) \cap Z = \emptyset$. Hence, we have that $I \not\subseteq \mathfrak{p}_i$. Therefore, using the prime avoidance lemma, there exists a homogeneous element $h \in I$ such that $h \notin \mathfrak{p}_i$. Hence, we conclude that $Z \subseteq \mathcal{X}_h := \{x \in \mathcal{X} | h_x \notin \mathfrak{m}_{\mathcal{X}, x}\}$. That $\mathcal{X}_h$ is affine follows from the fact that $\mathcal{X}_h \cong \text{Spec}(S_h)_0$ by [Vak25, Exercise 4.5.1].

Let $\mathfrak{m}_i \in \text{Spec } A$ be the maximal ideals such that $A_{\mathfrak{m}_i} \cong \mathcal{O}_{\tilde{\mathcal{P}}_i}$ and $\mathcal{I}$ the prime ideal with $A_{\mathcal{I}} \cong \mathcal{O}_{\mathcal{P}}$, where we also have that $\mathcal{I} \subseteq \mathfrak{m}_1$. We must have that $\mathcal{I} \not\subseteq \mathfrak{m}_i$ with $i \neq 1$ since the closed point $\tilde{\mathcal{P}}$ associated to $\mathcal{P}$ is unique by Theorem A.5.3.

We recall that since $\mathcal{X}$ is proper, then $\mathcal{P} \in X$ such that $\text{Spec } L \rightarrow X$ with $* \mapsto \mathcal{P}$ has $\mathcal{P} \mapsto \tilde{\mathcal{P}}$ such that $\tilde{\mathcal{P}} \in \mathcal{X}$ with $\text{Spec } \mathcal{O}_L \rightarrow \mathcal{X}$ with $\mathfrak{P}_L \mapsto \tilde{\mathcal{P}}$ and $(0) \mapsto \mathcal{P}$. Therefore, we have the following

diagram:

$$\begin{array}{ccccc} \mathrm{Spec} L & \longrightarrow & \mathrm{Spec} \mathcal{O}_{\mathcal{P}} & \xrightarrow{\sim} & \mathrm{Spec} A_{\mathcal{I}} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec} \mathcal{O}_L & \longrightarrow & \mathrm{Spec} \mathcal{O}_{\tilde{\mathcal{P}}} & \xrightarrow{\sim} & \mathrm{Spec} A_{\mathfrak{m}_1} \end{array}$$

where the bottom left arrow canonically maps $(0) \mapsto \ker(\mathcal{O}_{\tilde{\mathcal{P}}} \rightarrow \mathcal{O}_L)$. The top left horizontal map sends $(0) \mapsto \mathfrak{m}_{\mathcal{P}}$. Via the universal property of localisation, the rightmost down arrow sends $\mathcal{I}A_{\mathcal{I}} \mapsto \mathcal{I}A_{\mathfrak{m}_1}$. Therefore, we conclude that $\mathcal{I}A_{\mathfrak{m}_1} = \ker(\mathcal{O}_{\tilde{\mathcal{P}}} \rightarrow \mathcal{O}_L)$, and it follows that $A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1} \cong \mathcal{O}_L$, which is a DVR and, in particular, a regular ring.

We now show that $\mathcal{I} \not\subseteq \mathfrak{m}_1^2$. Suppose on the contrary that $\mathcal{I} \subseteq \mathfrak{m}_1^2$. Let $\mathfrak{n}_1 := \mathfrak{m}_1 A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1}$ be the maximal ideal of $A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1}$. Then

$$\mathfrak{n}_1/\mathfrak{n}_1^2 = (\mathfrak{m}_1 A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1})/(\mathfrak{m}_1^2 A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1}) \cong \mathfrak{m}_1 A_{\mathfrak{m}_1}/\mathfrak{m}_1^2 A_{\mathfrak{m}_1}.$$

Hence, $\dim_{\kappa_L}(\mathfrak{n}_1/\mathfrak{n}_1^2) = \dim_{\kappa_L}(\mathfrak{m}_1 A_{\mathfrak{m}_1}/\mathfrak{m}_1^2 A_{\mathfrak{m}_1})$ as κ_L -vectors spaces. Since $A_{\mathfrak{m}_1} \cong \mathcal{O}_{\tilde{\mathcal{P}}}$ is regular, then

$$\dim_{\kappa_L}(\mathfrak{m}_1 A_{\mathfrak{m}_1}/\mathfrak{m}_1^2 A_{\mathfrak{m}_1}) = \dim(A_{\mathfrak{m}_1}),$$

where we are equating the vector space and Krull dimensions by [AM69, Theorem 11.22 p 123]. However, we have $\mathcal{I}A_{\mathfrak{m}_1} \subsetneq \mathfrak{m}_1 A_{\mathfrak{m}_1}$ is a prime ideal of $A_{\mathfrak{m}_1}$, which is a local ring, so that

$$\dim(A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1}) < \dim(A_{\mathfrak{m}_1}).$$

However, we must have that $\dim_{\kappa_L}(\mathfrak{n}_1/\mathfrak{n}_1^2) = \dim(A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1})$ since $A_{\mathfrak{m}_1}/\mathcal{I}A_{\mathfrak{m}_1} \cong \mathcal{O}_L$ is a DVR, so in particular regular. Therefore, we obtain a contradiction.

Using another application of the prime avoidance lemma with $\mathfrak{p}_i = \mathfrak{m}_i$ for $i \neq 1$ and $\mathfrak{p}_1 = \mathfrak{m}_1^2$, we obtain an $h \in \mathcal{I}$ with $h \notin \mathfrak{m}_i$ for $i \neq 1$ and $h \notin \mathfrak{m}_1^2$. Then we have that $\mathrm{ord}_{\tilde{\mathcal{P}}_i}(h) = 0$ at each of these points. Finally, the fact that $h \notin \mathfrak{m}_1^2$ implies that $h \notin \mathfrak{m}_1^2 A_{\mathfrak{m}_1}$ so that $\mathrm{ord}_{\mathcal{P}}(h) = 1$. Note that if $Y = \overline{\{\mathcal{P}\}}$, then we have that $\mathrm{ord}_Y = \mathrm{ord}_{\mathcal{P}}$, hence $\mathrm{ord}_Y(h) = 1$ as well.

Let D be any prime divisor in $\mathrm{Div}(\mathcal{X})$ with $\tilde{\mathcal{P}} \in D$ such that it is not equal to $\overline{\{\mathcal{P}\}}$. We claim that $\mathrm{ord}_D(h) = 0$. As a closed subscheme of X , there is a closed immersion $i : D \rightarrow \mathrm{Spec} A$ and a sheaf of ideals $\mathcal{I}_D$ such that

$$0 \longrightarrow \mathcal{I}_D \longrightarrow \mathcal{O}_{\mathrm{Spec} A} \longrightarrow i_* \mathcal{O}_D \longrightarrow 0.$$

Then we have that $\mathcal{I}_{\tilde{\mathcal{P}}}$ and $\mathcal{I}_{D, \tilde{\mathcal{P}}}$ are both proper ideals of $\mathcal{O}_{\tilde{\mathcal{P}}}$. Since $\mathcal{O}_{\tilde{\mathcal{P}}}$ is regular, it is a unique factorisation domain. If $h \in \mathcal{I}_{\tilde{\mathcal{P}}} = (t)$ and $h \in \mathcal{I}_{D, \tilde{\mathcal{P}}} = (t')$ for some $t, t' \in \mathcal{O}_{\tilde{\mathcal{P}}}$, then $h = tt'u \in \mathfrak{m}_1^2 A_{\mathfrak{m}_1}$ for some $u \in \mathcal{O}_{\tilde{\mathcal{P}}}$, which is impossible. Hence, $\mathrm{ord}_D(h) = 0$. This condition now implies that h will not produce any new vanishing behaviour at $\tilde{\mathcal{P}}_1 \in \mathcal{X}_{\kappa}$.

Finally, we have that $[\overline{\{\mathcal{P}\}}] = [\overline{\{\mathcal{P}\}} - \mathrm{div} h]$ in $\mathrm{Pic}(\mathcal{X})$. Then since $X \rightarrow \mathcal{X}$ is an open immersion, we have that $\mathrm{Pic}(\mathcal{X}) \rightarrow \mathrm{Pic}(X)$ will send $[\overline{\{\mathcal{P}\}}] \mapsto [\mathcal{P}]$, which means that $[\mathcal{P} - \mathrm{div} h]$ will indeed satisfy the above criteria. $\square$

Picard Descent

To prove our main theorem, we first define a subgroup of K^* related to our function f as in [Sik02]:

$$G_f(K) := \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(K(\mathcal{P})^*)^{\mathrm{ord}_{\mathcal{P}}(f)}.$$

This is a well-defined product since we have that the support of any rational function on a projective variety is finite [Har77, Lemma II.6.1 p 131], so that for almost all $\mathcal{P}$, $\text{ord}_{\mathcal{P}}(f) = 0$. We note for the choice $K = K_v$ a local field that

$$G_f(K_v) = \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(K_v(\mathcal{P})^*)^{\text{ord}_{\mathcal{P}} f} = \prod_{\mathcal{P} \in X^c} \prod_{w|v} N_{w/v}(K(\mathcal{P})_w^*)^{\text{ord}_{\mathcal{P}} f},$$

which follows from the identity from $K_v \otimes_K K(\mathcal{P}) = \prod_{w|v} K(\mathcal{Q}_w)$ and the fact that for any field F and A an F -algebra with $A = \prod_i A_i$ with A_i finite F -algebras implies that

$$N_{A/F} A = \prod_i N_{A_i/F} A_i \quad (4.6)$$

by [CF67, Corollary in II.8 on p 55]. We also define the K_v -subgroup for a closed point $\mathcal{P} \in X$:

$$G_{\mathcal{P}}(K) := N_{K_v \otimes_K K(\mathcal{P})/K_v}(K_v \otimes_K K(\mathcal{P}))^* = \prod_{w|v} N_{K_v(\mathcal{Q}_w)/K_v}(K_v(\mathcal{Q}_w)^*)$$

so that

$$G_f(K_v) = \prod_{\mathcal{P} \in X} G_{\mathcal{P}}(K_v)^{\text{ord}_{\mathcal{P}} f}. \quad (4.7)$$

Then we have the following from [Sik02, Theorem 1]:

Theorem 4.4.5. (Siksek) Let X be a smooth curve over a perfect field K and $f \in K(X)$ a non-constant function. Then there is a unique homomorphism:

$$\varphi_f : \text{Pic}(X) \longrightarrow K^*/G_f(K)$$

such that if $\mathcal{D} := \sum_{i \in I} n_i \mathcal{P}_i \in \text{Div } X$ has that $\text{Supp } \mathcal{D}$ is disjoint from the zeros and poles of f then

$$\varphi_f([\mathcal{D}]) = \prod_{i \in I} N_{K(\mathcal{P}_i)/K}(f(\mathcal{P}_i))^{n_i}.$$

To prove this, we recall **Weil's reciprocity law** (for example in [Sil09, Exercise 2.11 p 39]), which says that for a smooth, complete curve X over a field K that functions $f, g \in K(X)$ with disjoint support satisfy

$$f(\text{div } g) = g(\text{div } f),$$

where $f(\text{div } g) := \prod_{\mathcal{P} \in \text{Supp } g} N_{K(\mathcal{P})/K}(f(\mathcal{P}))^{\text{ord}_{\mathcal{P}} g}$.

Proof. Based on the proof of [Sik02]. Let S be a finite subset of $\text{Div}(X)$. We denote $\text{Cl}_S := \text{Div}_S X / \text{Princ}_S X$ to be the class of divisors that have disjoint support from S . Now we will take S to be the support of f . Suppose that $[\mathcal{D}_1] = [\mathcal{D}_2]$ in $\text{Cl}_S X$, i.e. $\mathcal{D}_1 = \mathcal{D}_2 + \text{div } g$ for some $g \in K(X)$. If $\mathcal{D}_1$ and $\mathcal{D}_2$ have disjoint support from f , then so too must g . Thus we have that

$$\begin{aligned} \varphi_f(\mathcal{D}_1) &= \varphi_f(\mathcal{D}_2 + \text{div } g) \\ &= \varphi_f(\mathcal{D}_2) \varphi_f(\text{div } g) && \text{(def. of } \varphi_f \text{ and prop. of exponents)} \\ &= \varphi_f(\mathcal{D}_2) f(\text{div } g) \\ &= \varphi_f(\mathcal{D}_2) g(\text{div } f) && \text{(Weil reciprocity)} \end{aligned}$$

We have that $g(\text{div } f) \in G_f(K)$; therefore, we conclude that $\varphi_f([\mathcal{D}_1]) = \varphi_f([\mathcal{D}_2])$, so that the map is well-defined. Lastly, we have an isomorphism $\text{Cl}_S X \cong \text{Cl } X$; this follows from the moving lemma, which says we can always find a representative of our divisor class that is not a zero or pole of a given function f . $\square$

Let K now be a number field and let L be the class field belonging to f . We have the following result from [BS08, Lemma 4.7]:

Lemma 4.4.6. - For $v \in \mathfrak{M}(K)$, let $w \in \mathfrak{M}(L)$ be a place lying over v . Then $G_f(K_v) \subseteq \text{Norm}_{w/v}(L_w^*)$

Proof. We follow the original proof from [BS08]. We have by definition of L that

$$N_{L/K}(C_L) = \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(C_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}} f}.$$

We may pull this expression back from the class group $C_K := I_K/K^*$ to I_K to obtain

$$K^* N_{L/K}(I_L) = K^* \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(I_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}} f}.$$

Using [Neu99, Proposition VI.2.2], we have that for $\alpha = (\alpha_w) \in I_{K(\mathcal{P})}$ that

$$(N_{K(\mathcal{P})/K}(\alpha))_v = \prod_{w|v} N_{K(\mathcal{P})_w/K_v}(\alpha_w). \quad (4.8)$$

Therefore, we have that

$$\begin{aligned} K_v^* \cap N_{K(\mathcal{P})/K}(I_{K(\mathcal{P})}) &= \prod_{w|v} N_{K(\mathcal{P})_w/K_v}(K(\mathcal{P})_w^*) \\ &= G_{\mathcal{P}}(K_v) \end{aligned}$$

For F, G, H abelian groups contained in some larger ambient group, $G \cap FH \supseteq G \cap H$ as $FH \supseteq H$. It follows that

$$K_v^* \cap (K^* N_{L/K}(I_L)) = K_v^* \cap K^* \prod_{\mathcal{P} \in X^c} N_{K(\mathcal{P})/K}(I_{K(\mathcal{P})})^{\text{ord}_{\mathcal{P}} f} \supseteq \prod_{\mathcal{P} \in X^c} G_{\mathcal{P}}(K_v)^{\text{ord}_{\mathcal{P}} f} = G_f(K_v),$$

where the last equality is (4.7). Finally, we have that

$$N_{L/K} C_L \cap K_v^* = N_{w/v} L_w^*$$

from [Neu99, Corollary VI.5.8 p 394], which follows from Artin Reciprocity; this implies that $N_{w/v}(L_w^*) = K_v^* \cap (K^* N_{L/K}(I_L))$ and therefore the claim follows. $\square$

Now we are ready to prove the main theorem, which is done originally in [BS08, Section 5].

Proof of Theorem 4.4.1. (1) We have that X_v is a scheme over K_v . By theorem 4.4.5 we have a homomorphism

$$\varphi_f : \text{Pic } X_v \longrightarrow K_v^*/G_f(K_v).$$

Using lemma 4.4.6, we obtain an induced map

$$\varphi_f : \text{Pic } X_v \longrightarrow K_v^*/N_{w/v} L_w,$$

which by abuse of notation we also denote φ_f . Then we recall that we had the map $\theta_v : K_v \rightarrow G(L/K)$ from Proposition 3.3.2. We obtain an induced map on the quotient by the kernel $N_{w/v} L_w$: $\hat{\theta}_v : K_v/N_{w/v} L_w \rightarrow G(L/K)$ and we compose it with φ_f to obtain

$$\varphi_v := \hat{\theta}_v \circ \varphi_f : \text{Pic } X_v \longrightarrow G(L/K).$$

By the definition of the map φ_f , we see that for any closed point $\mathcal{Q} \in X_v$

$$\varphi_v([\mathcal{Q}]) = \hat{\theta}_v(N_{K_v(\mathcal{Q})/K_v}(f(\mathcal{Q}))),$$

as desired.

(2) To construct our set B , we need to gather the primes v such that $\varphi_v = \theta \circ \varphi_f \neq 1$. Our choice for B will be noncanonical, as such, in practice, we will often derive other criterion for what the set B is. Since we will be evaluating the Artin symbol θ_v , we need to consider the primes that are not well-behaved or where $f(\mathcal{P}) \neq 0$ vanishes in $\mathcal{O}_v/\mathfrak{p}_v$. Specifically, we will consider:

1. The Archimedean places.
2. The places v that ramify in L/K .
3. The places v such that $\mathcal{X}_v$ has bad reduction.
4. The places v such that $\text{ord}_Y(f) \neq 0$ for at least one component Y of the special fibre of $\mathcal{X}_\kappa \subseteq \mathcal{X}_v$.

First, the set B is finite. We know that there are only finitely primes satisfying (1) and (2) from algebraic number theory. For (3), a projective curve over a number field has finitely many places of bad reduction by [LE02, Proposition 10.1.21 p 462]. Finally for (4), the irreducible components Y correspond to prime divisors of $\mathcal{X}$ and we know from [Har77, Lemma II.6.1 p 131] that f has finite support in $\text{Div } \mathcal{X}$.

The failure of assumption (3) means that $\mathcal{X}_v$ has good reduction, so it satisfies criterion (2) of the moving lemma (Theorem 4.4.3). Then assumption (4) means that $\mathcal{X}$ satisfies criterion (1) of the moving lemma. Thus, we know that $\text{Pic}(X_v)$ is generated by points $\mathcal{P}$ such that $f \in \mathcal{O}_{\tilde{\mathcal{P}}}$ and $f(\tilde{\mathcal{P}}) \neq 0$ at them. Hence, we have that $N_{\kappa(\tilde{\mathcal{P}})/\kappa}(f(\tilde{\mathcal{P}})) \in \kappa^*$. From Proposition A.5.4, we then have that $f(\tilde{\mathcal{P}}) = 0$ if and only if $v_{\mathcal{P}}(f(\mathcal{P})) > 0$, where $v_{\mathcal{P}} : K_v(\mathcal{P}) \rightarrow \mathbb{Z}$ is the usual valuation on $K_v(\mathcal{P})$. It follows that $f(\mathcal{P}) \in \mathcal{O}_{K(\mathcal{P})}^*$, hence $N_{K_v(\mathcal{P})/K_v}f(\mathcal{P}) \in \mathcal{O}_v^*$. Finally, assumption (1) implies v is unramified, and so $N_{w/v} : \mathcal{O}_w^* \rightarrow \mathcal{O}_v^*$ is surjective by Theorem 2.4.4. Therefore, the local reciprocity law implies that $(N_{K_v(\mathcal{P})/K_v}f(\mathcal{P}), L_w/K_v) = 1$.

(3) We now prove that $\text{Pic}(X) \subseteq \ker(\prod_{v \in \mathfrak{M}(K)} \varphi_v)$. From (b), we know that $v \notin B$ implies that $\varphi_v = 1$. Let $D \in \text{Pic}(X)$ be a divisor, then since each φ_v is a homomorphism, it is only necessary to assume that $D = \mathcal{P} \in X^c$ is a closed point, and these satisfy $\mathcal{P}^\sigma = \mathcal{P}$ for all $\sigma \in G(\bar{K}/K)$ (but note that this does not imply $K(\mathcal{P}) = K$). Moreover, using the moving lemma (Theorem 4.4.3), we can assume that $\mathcal{P} \notin \text{Supp}(f)$. Since our function f has coordinates in K , then we have that σ acts as

$$(f(\mathcal{P}))^\sigma = f^\sigma(\mathcal{P}^\sigma) = f(\mathcal{P}),$$

so that $f(\mathcal{P}) \in K$. For each v , $P \mapsto \sum_{w|v} Q_w$ and the diagonal map is

$$\mathcal{P} \mapsto \left(\sum_{w|v} Q_w \right)_v.$$

Thus we have that

$$\prod_{v \in \mathfrak{M}(K)} \varphi_v \left(\left[\sum_{w|v} Q_w \right] \right) = \prod_{v \in \mathfrak{M}(K)} \prod_{w|v} \theta_v(N_{w/v}(f(Q_w))) = \prod_{v \in \mathfrak{M}(K)} \theta_v \left(\prod_{w|v} N_w(f(Q_w)) \right) = \prod_{v \in \mathfrak{M}(K)} \theta_v(f(\mathcal{P})) = 1$$

where we use that $\prod_{w|v} N_{w/v}(f(Q_w)) = f(\mathcal{P})$ from (4.8) and that $f(\mathcal{P}) \in K^*$ is a principal idèle making Artin reciprocity go through. $\square$

The Brauer-Manin obstruction

The following results come from [BS08, Sections 6-7], which we summarise here. We would also like to thank Rob de Jeu for explaining some of the details of the following theorem.

Theorem 4.4.7. (Bright & Siksek) The obstruction arising from Theorem 4.4.1 is of the Brauer-Manin type. In particular, let $f \in K(X)$ and L/K be the associated field extension. For any character $\chi : G_K \rightarrow \mathbb{Q}/\mathbb{Z}$ arising from a character $\tilde{\chi} : G(L/K) \rightarrow \mathbb{Q}/\mathbb{Z}$ via the inflation map of Section 2.4, we have that $\mathcal{A}_\chi := f \cup \delta\chi$ will lift to an element of $\text{Br}(X)$ satisfying

$$\prod_{v \in \mathfrak{M}(K)} \theta_v(f(\mathcal{P}_v)) = 1 \quad \text{if and only if} \quad \sum_v \text{inv}_v(\mathcal{A}_\chi(\mathcal{P}_v)) = 0 \quad (4.9)$$

for any adelic point $(\mathcal{P}_v) \in X(A_K)$

Proof. We provide a proof sketch. Let $F = K(X)$ be the function field of X and $G_F = G(\bar{F}/F)$ the usual absolute Galois group. There is a cup-product pairing as in Section 2.4:

$$\begin{aligned} H^0(F, \bar{F}^*) \times H^2(F, \mathbb{Z}) &\longrightarrow H^2(F, \bar{F}^*) \\ (f, \xi) &\longmapsto f \cup \xi, \end{aligned} \quad (4.10)$$

which we will show gives rise to an element of $\text{Br}(X)$ via the following exact sequence mentioned in Section 4.3:

$$0 \longrightarrow \text{Br}(X) \longrightarrow \text{Br}(F) \longrightarrow \bigoplus_{\mathcal{P} \in X^c} H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z}) \longrightarrow 0. \quad (4.11)$$

We first note that $\bar{K}F = \bar{K}(\bar{X})$ since $\bar{X} = X \times_{\text{Spec } K} \text{Spec } \bar{K}$. Now $G_{\bar{K}F}$ is a normal subgroup of G_F since $\bar{K}F$ is the splitting field of $K[T] \subseteq F[T]$. Indeed, clearly $K[T]$ still splits in $\bar{K}F$. On the other hand, if $g \in F[T] \setminus K[T]$ is irreducible, it remains irreducible in $\bar{K}F$ since $\bar{K}$ and F are linearly independent as $\bar{K} \cap F = K$. This last statement is true since X is geometrically integral. Next, via the action of $G_F/G_{\bar{K}F} = G(\bar{K}F/F)$ on $\bar{K}F/F$, we have $G(\bar{K}F/F)$ restricts to an action on $\bar{K} \subseteq \bar{K}F$. We claim this restricted action of $G(\bar{K}F/F)$ on $\bar{K}/K$ returns the usual action of G_K on $\bar{K}/K$.

We will prove the claim if we can show that $G(\bar{K}F/F) \cong G(\bar{K}/K)$ as profinite groups. We define the restriction map $r : G(\bar{K}F/F) \rightarrow G(\bar{K}/K)$, $\sigma \mapsto \sigma|_{\bar{K}}$ and its inverse $e : G(\bar{K}/K) \rightarrow G(\bar{K}F/F)$ by $\sigma \mapsto \tilde{\sigma}$ where $\tilde{\sigma}|_{\bar{K}} = \sigma$ and $\tilde{\sigma}|_F = \text{id}_F$, which agree on the overlap $\bar{K} \cap F = K$. We claim both maps are continuous. For the first map, let M/K be a finite Galois extension; we have $r^{-1}G(\bar{K}/M) = G(\bar{K}F/MF)$. For the second map, if $M \subseteq \bar{K}F$ normal then $M \cap \bar{K}$ is also normal and has $e^{-1}G(\bar{K}F/M) = G(\bar{K}/M \cap \bar{K})$. These facts follow from the linear independence of $\bar{K}$ and F .

Now we also have the another cup-product pairing:

$$H^0(G(\bar{K}F/F), (\bar{K}F)^*) \times H^2(G(\bar{K}F/F), \mathbb{Z}) \xrightarrow{\cup} H^2(G(\bar{K}F/F), (\bar{K}F)^*).$$

We will select a character $\chi : G(\bar{K}F/F) \rightarrow \mathbb{Q}/\mathbb{Z}$ that comes from a character $\tilde{\chi} : G(L/K) \rightarrow \mathbb{Q}/\mathbb{Z}$ and use the connecting homomorphism $\delta : H^1(G(\bar{K}F/F), \mathbb{Q}/\mathbb{Z}) \rightarrow H^2(G(\bar{K}F/F), \mathbb{Z})$ to obtain an element of $H^2(G(\bar{K}F/F), \mathbb{Z})$. To do this, we use the inflation map $\text{inf} : H^1(G(L/K), \mathbb{Q}/\mathbb{Z}) \rightarrow H^1(K, \mathbb{Q}/\mathbb{Z})$, which is compatible with cup-products by Proposition 2.4.1. Then, in light of the isomorphism $G(\bar{K}F/F) \cong G_K$, we obtain a character $\chi \in H^1(G(\bar{K}F/F), \mathbb{Q}/\mathbb{Z})$.

We raise f and $\delta\chi$ to elements of $H^2(F, \bar{K}F)$ and $H^2(F, \mathbb{Z})$, respectively via inflation maps. Finally, viewing f as an element of $\bar{F}$, we can then take our cup-product pairing to be the one from (4.10). We note that via [GS06, Proposition 4.7.3 pp 110-111] that $f \cup \delta\chi \mapsto (\frac{f\chi}{\bar{F}}) \in \text{Br}(F)$ is a cyclic algebra. As a brief aside, we will see in Chapter 5 that this fact proves that the method of [BS08] indeed generalises the original proof by Lind in Section 4.3.

Now our goal is to show that the morphism $\text{Br}(F) \rightarrow \bigoplus_{\mathcal{P}} H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z})$ of (4.11) sends $f \cup \delta\chi \mapsto 0$. In [BS08, Lemma 6.2 p 12], they proved that there is an exact sequence

$$0 \longrightarrow H^1(K, M) \longrightarrow H^2(K, \mathbb{Z}) \xrightarrow{(f)} H^2(K, D),$$

for some G_K -modules M and D , where the latter group satisfies

$$H^2(K, D) = \bigoplus_{\mathcal{P} \in \text{Supp}(f)} H^2(K, \text{Ind}_{K(\mathcal{P})/K} \mathbb{Z}).$$

Then from [NSW13, Proposition 1.6.4 p 62], we have that

$$H^2(K, \text{Ind}_{K(\mathcal{P})/K} \mathbb{Z}) = H^2(K(\mathcal{P}), \mathbb{Z}),$$

which implies that

$$H^2(K, D) = \bigoplus_{\mathcal{P} \in \text{supp}(f)} H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z}) \subseteq \bigoplus_{\mathcal{P} \in X} H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z}).$$

The first equality follows from the isomorphism $\delta : H^1(K(\mathcal{P}), \mathbb{Q}/\mathbb{Z}) \rightarrow H^2(K(\mathcal{P}), \mathbb{Z})$ coming from the exact sequence of $G_{K(\mathcal{P})}$ -modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0,$$

where $H^1(K(\mathcal{P}), \mathbb{Z}) = 0$ by [CS21, Proposition 3.5.3 p 88]. In order to show that $f \cup \delta\chi \in \text{Br}(X)$, it is sufficient to show that $f \cup \delta\chi = 0$ in $H^2(K, D)$. By [BS08], since χ is arising from a character $\tilde{\chi} : G(L/K) \rightarrow \mathbb{Q}/\mathbb{Z}$, it must be that this indeed holds.

Finally, suppose that there is a point $\mathcal{P} \in X(K)$ and let $(\mathcal{P}_v)_v$ be its adelic image in $X(A_K)$. Using [NSW13, Proposition 8.1.11], we have that

$$\chi(\theta_v(f(\mathcal{P}_v))) = \text{inv}(\mathcal{A}_\chi(\mathcal{P}_v)).$$

This implies (4.9). □

5. COMPUTATIONS AND RESULTS

In this chapter, we discuss the computations we implemented in Magma, which can be found in the GitHub repository of [Roz]. We then discuss algorithms to compute $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$ and some of the details of the corresponding Magma code, which is found in [Roz]. We then illustrate how the complete computations work for two hyperelliptic curves. The code we have written can compute many examples as in [Roz]. Finally, we close off with a procedure to compute $E(K_v)/3E(K_v)$ for $K = \mathbb{Q}(\rho)$ with ρ a primitive third root of unity, and we produce a new Brauer-Manin obstruction using 4.4.1.

5.1 2-Coverings of Elliptic Curves

In this section, we describe the various algorithms we implemented for computing the obstruction of Theorem 4.4.1. We wrote Magma code that allows one to compute $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$ for any elliptic curve $E/\mathbb{Q}$ (or $E/\mathbb{Q}_p$ as well). This algorithm works regardless of whether or not $E(\mathbb{Q}_2)[2] \neq \emptyset$. We also wrote a large search algorithm that produces all functions $f = \frac{x+\alpha}{x+\beta}$ up to given bounds. Finally, given a two-covering C of an elliptic curve E , we have produced an algorithm for computing the isomorphism $C_p \rightarrow E_p$. As such, our program simply takes as input a hyperelliptic curve and some precision parameters and outputs a small table of values corresponding to the Hilbert symbols evaluated at the primes finitely many primes $p \in B$ of Theorem 4.4.1.

Computer Specifications

We begin this section by noting the computer specifications used for all computations to come. Everything was accomplished on a personal laptop with a 6 core AMD processor and 8 GB of memory. Thus it is likely that with superior computing technology, one can accelerate some of the slower programs we have written.

5.1.1 Harmful Functions

To put the obstruction of the Theorem 4.4.1 to use, we must find rational functions so that the associated field extension L/K is not trivial. Mimicking the language of [BS08], we call such functions **harmful**.

As mentioned in Section 2.6, the 2-coverings of an elliptic curve are defined by hyperelliptic curves. Let X be the usual normalisation of the hyperelliptic curve,

$$y^2 = h(x) := a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0.$$

We wish to find $\alpha, \beta \in \mathbb{Q}$ such that $h(\alpha), h(\beta) \notin \mathbb{Q}^{*2}$ and $\mathbb{Q}(\sqrt{h(\alpha)}) = \mathbb{Q}(\sqrt{h(\beta)})$. Then as in [BS08], we will choose the rational function

$$f = \frac{x - \alpha}{x - \beta}$$

to find an abelian extension L/K that is anti-Hasse. First we compute that the divisor of f is given as

$$\operatorname{div}(f) = \mathcal{P}_\alpha - \mathcal{P}_\beta,$$

where $\mathcal{P}_\alpha = (\alpha, \sqrt{h(\alpha)}) + (\alpha, -\sqrt{h(\alpha)}) \in \operatorname{Div}(\bar{X})$ or equivalently $\mathcal{P}_\alpha = (\alpha, y^2 - h(\alpha)) \in \operatorname{Div}(X)$. We obtain an identical result for $\mathcal{P}_\beta$. We write let D be the discriminant of the field $\mathbb{Q}(\sqrt{\alpha}) = \mathbb{Q}(\sqrt{\beta})$. Then we have that $\gcd(\operatorname{ord}_{\mathcal{P}_\alpha} f, \operatorname{ord}_{\mathcal{P}_\beta} f) = 1$, so that C_L is an open subgroup, and it satisfies

$$N_{L/K} C_L = N_\alpha(C_{\mathbb{Q}(\mathcal{P}_\alpha)}) N_\beta(C_{\mathbb{Q}(\mathcal{P}_\beta)})^{-1} = N(C_{\mathbb{Q}(\sqrt{D})}) N(C_{\mathbb{Q}(\sqrt{D})}) = N(C_{\mathbb{Q}(\sqrt{D})})$$

where we use the equivalence $\mathcal{N}_{L_1} \mathcal{N}_{L_2} = \mathcal{N}_{L_1 \cap L_2}$ from Theorem 3.3.4. Thus we have that $L = \sqrt{D}$ is the field associated to f .

From the main theorem, we had that there was a set $B \subseteq \mathfrak{M}(K)$ such that

$$\prod_{v \in B} \theta_v(f(\mathcal{P})) = 1$$

In the proof of the main theorem, we made a choice of primes that would be included in the set B . We now make a different selection for B ; we select the following places of $\mathbb{Q}$:

1. the primes that ramify in L (and 2 if it is not split)
2. the primes in the denominators of α and β (that are not split)
3. primes dividing the numerator or the denominator of $h(\alpha), h(\beta)$ (that are not split)
4. ∞ only if L is not complex

We exclude split primes since in that case $L_w \cong K_v$ for all $w|v$, so that the norm map is an isomorphism (surjective) and $(a, L_w/K_v) = 1$ for all $a \in K_v$ by Artin reciprocity.

Proposition 5.1.1. Let $\mathcal{P} \in \operatorname{Pic}(X_v)/2\operatorname{Pic}(X_v)$. Then for all $v \notin B$, $\varphi_v(f(\mathcal{P})) = 1$ for φ_v as defined in Theorem 4.4.1.

Proof. From our assumptions, we find that v is non-Archimedean and is coming from a prime p that is odd and inert in $L = \mathbb{Q}(\sqrt{D})$. Now $D \notin \mathbb{Q}_p^{*2}$, which means that $(\frac{D}{p}) = -1$. Moreover, since p is unramified, $N_{L/\mathbb{Q}}(\mathcal{O}_L^*) = \mathbb{Z}_p^*$ by Theorem 2.4.4. Which implies that if we can show that $f(\mathcal{P}) \in \mathbb{Z}_p^*$, then $\theta_v(f(\mathcal{P})) = 1$ by Artin reciprocity. Thus, it remains to show that $v_p(f(\mathcal{P})) = 0$. Now we have that rearranging our function to isolate x gives

$$x = \frac{\beta f - \alpha}{(f - 1)}$$

We substitute this into the hyperelliptic curve equation and simplify:

$$y^2(f-1)^4 = a_4(\beta f - \alpha)^4 + a_3(\beta f - \alpha)^3(f-1) + a_2(\beta f - \alpha)^2(f-1)^2 + a_1(\beta f - \alpha)(f-1)^3 + a_0(f-1)^4$$

If $v(f) > 0$, then the right-hand side is integral since the denominators of α and β are coprime to p , which makes the left-hand side and, in particular, y also integral. Therefore, we reduce modulo p and obtain

$$y^2 \equiv a_4\alpha^4 + a_3\alpha^3 + a_2\alpha^2 + a_1\alpha + a_0 \pmod{p}$$

The right-hand side is $h(\alpha)$, so that we have that $(\frac{h(\alpha)}{p}) = 1$ since $\gcd(h(\alpha), p) = 1$, which contradicts the assumption that p is inert. On the other hand, if $v(f) < 0$, then we multiply through by f^{-4} and obtain

$$y^2(1 - f^{-1})^4 = a_4(\beta - \alpha f^{-1})^4 + a_3(\beta - \alpha f^{-1})^3(1 - f^{-1}) + a_2(\beta - \alpha f^{-1})^2(1 - f^{-1})^2 + a_1(\beta - \alpha f^{-1})(1 - f^{-1})^3 + a_0(1 - f^{-1})^4.$$

We likewise obtain that

$$y^2 \equiv a_4\beta^4 + a_3\beta^3 + a_2\beta^2 + a_1\beta + a_0 \pmod{p},$$

from which we have that $(\frac{h(\beta)}{p}) = 1$, again a contradiction. $\square$

Computing $\text{Pic}(X_v)/2\text{Pic}(X_v)$

Now the first step is to devise a method to compute $\text{Pic}(X_v)/n\text{Pic}(X_v)$ for any $v \in \mathfrak{M}(K)$. Let X be an n -covering an elliptic curve E . Then we have the following proposition:

Proposition 5.1.2. Let K be a number field and $v \in \mathfrak{M}(K)$. The Picard group of an n -covering X/K of an elliptic curve E/K satisfies

$$\text{Pic}(X_v)/n\text{Pic}(X_v) \cong \mathbb{Z}/n\mathbb{Z} \oplus E(K_v)/nE(K_v).$$

Proof. Since X_v is a homogeneous space of E_v , we have the canonical isomorphism $E_v(\bar{K}_v) \cong \text{Pic}^0(\bar{X}_v)$ from Theorem 2.6.1. Therefore it follows that $\text{Pic}(\bar{X}_v) \cong \mathbb{Z} \oplus E(\bar{K}_v)$. Since we assume that X_v admits a rational solution $\mathcal{P}$, which means $X_v \cong E_v$ over K_v , we have that $\text{Pic}(\bar{X}_v) \cong \text{Pic}(\bar{X})^{G(\bar{K}_v/K_v)}$ by Proposition 2.5.2, and so $\text{Pic}(X_v) \cong \mathbb{Z} \oplus E_v(K_v)$. And the statement follows. $\square$

We also have the following useful result from [Sch96]:

Proposition 5.1.3. (Shaefer) For p prime and $v|p$, let $r := \text{ord}_p(n)$, we have that

$$\#E(K_v)/nE(K_v) = p^{r[K_v:\mathbb{Q}_p]} \cdot \#E(K_v)[n].$$

Proof. The proof requires knowledge of Néron models of elliptic curves. See Appendix B for some of the details or [Sch96] for the complete proof. $\square$

5.1.2 2-Descent

Computing $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$ via 2-isogeny

First, we assume that we admit a 2-torsion point $P \in E(\mathbb{Q}_p)$ so that we can assume that $E : y^2 = x(x^2 + ax + b)$. Then we obtain a 2-isogeny:

$$\varphi = \left(\frac{y^2}{x^2}, \frac{y(b - x^2)}{x^2} \right)$$

to the elliptic curve $E' : v^2 = u(u^2 + a'u + b)$, with $a' = -2a$ and $b' = a^2 - 4b$. We have the following short exact sequence following from the usual Kummer sequence:

$$0 \longrightarrow E'(\mathbb{Q}_p)/\varphi'(E(\mathbb{Q}_p)) \longrightarrow H^1(\mathbb{Q}_p, E[\varphi]),$$

where the latter map sends $Q' = \varphi(Q)$ to $\sigma \mapsto Q^\sigma - Q$. Since we have that $E[\varphi] \cong \mu_2$ via $P \mapsto -1$ as G -modules, it follows that $H^1(\mathbb{Q}_p, E[\varphi]) \rightarrow H^1(\mathbb{Q}_p, \mu_2)$ is an isomorphism, and we have that $\sigma \mapsto Q^\sigma - Q$ either maps to $\sigma \mapsto -1$ or $\sigma \mapsto 1$ depending on whether Q is rational or not. Lastly, there is a canonical isomorphism identifying $H^1(\mathbb{Q}_2, \mu_2)$ with $\mathbb{Q}_2/\mathbb{Q}_2^*$. Putting these remarks together, we know the image of Q' in $\mathbb{Q}_2/\mathbb{Q}_2^*$ based on whether the action of σ fixes Q or not. Hence we determine algebraic extensions that x and y lie in. We assume that $Q' \neq \mathcal{O}, (0, 0)$, in particular $u \neq 0$. Using the definition of our isogeny map φ and the equation for E , we have that

$$u = \frac{x^2 + ax + b}{x}.$$

Thus,

$$x = \frac{u - a \pm \sqrt{u^2 - 2au + a^2 - 4b}}{2} = \frac{u - a \pm \sqrt{1/u} \cdot v}{2},$$

so that $x \in \mathbb{Q}_p(\sqrt{u})$. Then we have that $y^2 = ux^2$, which implies that $y \in \mathbb{Q}_p(\sqrt{u})$. Hence we map $Q' \mapsto u$. If $Q' = (0, 0)$, then the formulas tell us that $y = 0, x \neq 0$ since the preimage must be affine. We will thus need to solve $x^2 + ax + b = 0$, so that $x \in \mathbb{Q}(\sqrt{a^2 - 4b}) = \mathbb{Q}_p(\sqrt{b'})$, which means that $(0, 0) \mapsto b'$. Therefore, combining the above computations, we map

$$\begin{aligned} E'(\mathbb{Q}_p)/\varphi(E(\mathbb{Q}_p)) &\longrightarrow \mathbb{Q}_p^*/\mathbb{Q}_p^{*2} \\ (0, 0) &\longmapsto b' \\ (u, v) &\longmapsto u \end{aligned}$$

Thus using a point $(u, v) \neq (0, 0)$, this implies that $u = \delta z^2$ for some $z \in \mathbb{Q}_p^{*2}$. Hence

$$v^2 = \delta z^2(\delta^2 z^4 + a'\delta z^2 + b').$$

If we let $t = v/z$, then

$$t^2 = \delta^3 z^4 + a'\delta^2 z^2 + b'\delta, \tag{5.1}$$

so we are left to test the solubility of this hyperelliptic curve over $\mathbb{Q}_p$ for each $\delta \in \mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$. The procedure is repeated using the dual isogeny $\hat{\varphi}$:

$$\begin{aligned} E(\mathbb{Q}_p)/\hat{\varphi}(E'(\mathbb{Q}_p)) &\longrightarrow \mathbb{Q}_p^*/\mathbb{Q}_p^{*2} \\ (0, 0) &\longmapsto b \\ (x, y) &\longmapsto x \end{aligned}$$

We have the following exact sequence, which is useful for computing $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$:

$$0 \rightarrow E'(\mathbb{Q}_p)[\hat{\varphi}]/\varphi(E(\mathbb{Q}_p)[2]) \rightarrow E'(\mathbb{Q}_p)/\varphi(E(\mathbb{Q}_p)) \rightarrow E(\mathbb{Q}_p)/2E(\mathbb{Q}_p) \rightarrow E(\mathbb{Q}_p)/\hat{\varphi}(E'(\mathbb{Q}_p)) \rightarrow 0.$$

Using the the value of $\mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$ associated to each point, we can completely determine the group structure of $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$. We begin by considering the various possible cases. We have $b' \in \mathbb{Q}_p^{*2}$ if and only if there exists a $Q \in E(\mathbb{Q}_p)[2]$ with $Q \neq (0, 0)$. Hence $\#E(\mathbb{Q}_p) = 4$ if and only if $b' \in \mathbb{Q}_p^{*2}$. Then, from Proposition 5.1.3, we have that

$$\#E(\mathbb{Q}_p)/2E(\mathbb{Q}_p) = \begin{cases} 8 & p = 2, b' \in \mathbb{Q}_p^{*2} \\ 4 & p = 2, b' \notin \mathbb{Q}_p^{*2} \\ 4 & p \neq 2, b' \in \mathbb{Q}_p^{*2} \\ 2 & p \neq 2, b' \notin \mathbb{Q}_p^{*2}. \end{cases}$$

Let us begin by assuming we have run the above descent method for both isogenies and we return points $\mathcal{P}_1, \dots, \mathcal{P}_s \in E(\mathbb{Q}_p)$ and $\mathcal{Q}_1, \dots, \mathcal{Q}_t \in E'(\mathbb{Q}_p)$ with $s, t \leq 4$. We also test the existence of the points $P_i \in E(\mathbb{Q}_p)[2]$ and $Q_i \in E'(\mathbb{Q}_p)[2]$ by checking if the polynomials

$$x^2 + ax + b = 0$$

and

$$x^2 + a'x + b' = 0$$

admit rational roots over $\mathbb{Q}_p$. Let us denote d_i and e_i for the corresponding values in $\mathbb{Q}_p/\mathbb{Q}_p^{*2}$ to the P_i and Q_i , respectively. For a given $\mathcal{Q}_i$, if the image of $\mathcal{Q}_i$ is equal to one of the e_i modulo squares, then we know that this point is “unnecessary” since either (1) it will map to the class $[\mathcal{O}]$ in case $x(\mathcal{Q}_i) = b'$ or (2) $\mathcal{Q}_i \mapsto (0, 0)$ when $x(\mathcal{Q}_i) = e_i \neq b$ and we simply pick $(0, 0) \in E(\mathbb{Q}_p)$ to be the generator in this case. However, there is valuable information to be gleaned here: we can use this information to generate relations between these points. Indeed, if

$$x(\mathcal{Q}_i)x(\mathcal{Q}_j) = b' \in \mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$$

then $[\mathcal{Q}_i] + [\mathcal{Q}_j] = [(0, 0)] \in E'(\mathbb{Q}_p)/\varphi(E(\mathbb{Q}_p))$ via the homomorphism property, which implies that

$$[\hat{\varphi}(\mathcal{Q}_i)] = [\hat{\varphi}(\mathcal{Q}_j)] \in E(\mathbb{Q}_p)/2E(\mathbb{Q}_p).$$

If

$$x(\mathcal{Q}_i)x(\mathcal{Q}_j) = e_i \neq b \in \mathbb{Q}_p^*/\mathbb{Q}_p^{*2},$$

then $[\mathcal{Q}_i] + [\mathcal{Q}_j] = [\mathcal{Q}_i]$, so that

$$[\hat{\varphi}(\mathcal{Q}_i)] + [\hat{\varphi}(\mathcal{Q}_j)] = [(0, 0)] \in E(\mathbb{Q}_p)/2E(\mathbb{Q}_p).$$

Thus, we can discard the image of one of the generators $\mathcal{Q}_i$ and keep the other along with $(0, 0) \in E(\mathbb{Q}_p)$. We repeat this process until the relations are eliminated, i.e. none of the remaining points $\mathcal{Q}_i$ satisfy (1) $\mathcal{Q}_i = \mathcal{Q}_k$ and (2) $\mathcal{Q}_i + \mathcal{Q}_j = \mathcal{Q}_k$.

The next question we ask is if any of the 2-torsion points P_i are generators of $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$. If the corresponding value d_i is a generator of $E(\mathbb{Q}_p)/\hat{\varphi}(E'(\mathbb{Q}_p))$, then the answer is positive. As above, we perform a similar test to remove any relations: if $[\mathcal{P}_i] + [\mathcal{P}_j] = [\mathcal{P}_i]$ then we discard one of the $\mathcal{P}_i$. We note that it is unnecessary to test if $\varphi(\mathcal{Q}_i)$ satisfies relations since we know from the above exact sequence that these points are 0 in $E(\mathbb{Q}_p)/\hat{\varphi}(E'(\mathbb{Q}_p))$, and the basis of this group of course has linearly independent elements. Hence we may join the bases together, and we will have computed $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$.

We implemented the above algorithm in Magma, and for example, it computes a basis for the elliptic curve

$$E : y^2 = x^3 + 272x$$

over $\mathbb{Q}_2$ in 0.094 seconds.

Computing $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$ without 2-torsion

We follow the broad outline done in [SS97] and the expanded useful computations in [Sim02], which the latter author carried out for elliptic curves E/K with K a number field. Thus we claim that method generalises quite nicely to the case of p -adic numbers.

We begin by putting our Jacobian E in short Weierstrass form:

$$y^2 = x^3 + Ax + B.$$

Suppose that the polynomial $x^3 + Ax + B$ is irreducible over $\mathbb{Q}_p$ and let

$$K_p := \mathbb{Q}_p[x]/(x^3 + Ax + B) = \mathbb{Q}_p[\theta].$$

There is a well-defined map that is a homomorphism of groups:

$$\begin{aligned} E(\mathbb{Q}_p)/2E(\mathbb{Q}_p) &\longrightarrow K_p^*/K_p^{*2} \\ (a, b) &\longmapsto a - \theta \\ \mathcal{O} &\longmapsto 1 \end{aligned}$$

see [Cas91, Chapter 15 Lemma 1 pp 67-68] for the proof that this is a homomorphism. Crucially, one has that the group K_p^*/K_p^{*2} is finite: $K_p^* \cong \mathbb{Z} \times \mathcal{O}_p^*$ and for $u \in \mathcal{O}_p^*$ the polynomial $x^2 - u$ has a root if $|1 - u|$ sufficiently small by [Lan95, Proposition II.2.1 p 42].

Now K_p is a vector space of dimension 3 over $\mathbb{Q}_p$, and we can explicitly compute the norm map: for any $\alpha \in K_p$ we have $\xi \mapsto \alpha\xi$ is a linear map and has an associated matrix M_α . Then we have that

$$\begin{aligned} N &:= N_{K_p/\mathbb{Q}_p} : K_p^* \longrightarrow \mathbb{Q}_p^* \\ \alpha &\longmapsto \det M_\alpha. \end{aligned}$$

We show for any $a \in \mathbb{Q}_p$ that $N(a - \theta) = f(a)$. Then we will obtain an induced map $N : K_p^*/K_p^{*2} \rightarrow \mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$. For any $\alpha \in K_p$, we may write $\alpha = a + b\theta + c\theta^2$ where $a, b, c \in \mathbb{Q}_p$, and thus if we map the basis elements $1, \theta, \theta^2$, we obtain images:

$$\begin{aligned} \alpha &= a + b\theta + c\theta^2 \\ \alpha\theta &= -cB + (-cA + a)\theta + b\theta^2 \\ \alpha\theta^2 &= -bB + (-bA - cB)\theta + (a - cA)\theta^2, \end{aligned}$$

and choosing the vectors e_1, e_2, e_3 to represent $1, \theta, \theta^2$, respectively, yields the following matrix:

$$M_\alpha = \begin{bmatrix} a & -cB & -bB \\ b & a - cA & -bA - cB \\ c & b & a - cA \end{bmatrix}$$

And so

$$\det M_\alpha = a^3 - 2a^2cA + ac^2A^2 + ab^2A + 3abcB + c^3B^2 - b^3B - c^2bAB.$$

If $b = -1$ and $c = 0$, i.e. $\alpha = a - \theta$, then $\det M_\alpha = a^3 + aA + B = f(a)$. Let $x \in \mathbb{Q}_p$, if $f(x) \in \mathbb{Q}_p^{*2}$, i.e. $y^2 = f(x)$ has a solution, then $N(x - \theta) = 1$ in $\mathbb{Q}_p^*/\mathbb{Q}_p^{*2}$. Hence, we have the following diagram:

$$\begin{array}{ccc} & K_p^*/K_p^{*2} & \\ \nearrow & & \searrow N \\ E(\mathbb{Q}_p)/2E(\mathbb{Q}_p) & \xrightarrow{1} & \mathbb{Q}_p^*/\mathbb{Q}_p^{*2} \end{array}$$

Suppose that $(a, b) \in E(\mathbb{Q}_p)$, then we seek an element $z := w\theta^2 + v\theta + u$ such that

$$\delta z^2 = a - b\theta. \quad (5.2)$$

Our first goal is to find another representative δ that is of the form $a - b\theta$ since this will determine a curve that is simpler to test for solubility. Expanding the left-hand side of (5.2) gives us an expression $q_2\theta^2 + q_1\theta + q_0$ such that each q_i is a quadric form in the variables u, v, w satisfying

$$\begin{aligned} q_0(u, v, w) &= a \\ q_1(u, v, w) &= -\theta^2 \\ q_2(u, v, w) &= 0. \end{aligned}$$

As in [Sim02], we can compute that

$$\begin{aligned} q_2 &= u^2e_2 + (e_0 - Be_2)v^2 + (B^2e_2 - Be_0 - Ce_1)w^2 \\ &\quad + 2e_1uv + 2(e_0 - Be_2)uw - 2(Be_1 + Ce_2)vw, \end{aligned}$$

which we also verified in Magma. Using the following linear transformation computed in [Sim02]:

$$\begin{aligned} U &= e_2u + e_1v + (e_0 - e_2B)w \\ V &= \alpha v - \beta w \\ W &= rw, \end{aligned}$$

we obtain the conic:

$$\alpha U^2 + e_2 W^2 - V^2 = 0. \quad (5.3)$$

The existence of a solution is quickly decidable using the Hilbert symbol $(\frac{a, e_2}{p})$. If the symbol turns out to be equal to 1, then we can obtain integral coefficients and search using Hensel's lemma for an integral solution up to some finite precision. Then via the formulas:

$$\begin{aligned} w &:= W/r \\ v &:= (V + \beta w)/a \\ u &:= (U - e_1v - (e_0 - e_2B)w)/e_2 \end{aligned}$$

we get back a solution of $q_2 = 0$. In practice, we can take a and $-b$ to be the coefficients of δz^2 modulo p^n for an appropriate value n . We can now take $\delta = a - b\theta$. The next step is to expand δz^2 again, this time we have from [Sim02] the following expressions, which we verified in Magma,

$$\begin{aligned} q_2 &= -2buv + 2auw + av^2 + 2bBvw - aBw^2 + bCw^2 \\ q_1 &= -u^2b + 2uva + 2Buwb + Bv^2b - 2Bvwa + 2Cvwb - Cw^2a - B^2w^2b. \end{aligned}$$

Using another linear transformation:

$$\begin{aligned} w &:= l\mu/(ry) \\ v &:= (aw - \mu^2/y)/b \\ u &:= (abv + (a^2 + 2d_4b^2)w - l^2/y)/(2b^2), \end{aligned}$$

we reduce to solving the conic $W^2 - UV = 0$, which has solutions given by points $(U, V, W) = (\lambda^2/y, \mu^2/y, \lambda\mu/y)$. Now substituting these value for U, V, W into u, v, w and back into q_1 , we obtain a hyperelliptic curve:

$$y^2 = \frac{1}{4b^3}(\lambda^4 - 6a\lambda^2\mu^2 + 8r\lambda\mu^3 + (-4Bb^2 - 3a^2)\mu^4). \quad (5.4)$$

In theory, we can find a solution to this again by obtaining integral coefficients and searching and lifting via Hensel's lemma. In doing so, we will obtain values for u, v, w such that $q_1 \equiv 1 \pmod{2^{n-1}}$ and $q_2 \equiv 0 \pmod{2^{n-1}}$. Finally, letting $x = q_0$, we obtain a point (x, y) on the Jacobian. Sending it through all the appropriate maps, we will ultimately obtain a point on X that is a basis element of $\text{Pic}(X_p)/2\text{Pic}(X_p)$.

In the algorithms we implemented in Magma, we opt to use the `IsLocallySolvable()` function, which takes as input a projective scheme C/K , with K number field, and a prime v and decides using algorithms devised in [Bru07] if it is solvable. If it indeed has solutions, then Magma picks one of the completions K_v and returns a point. We will primarily use this function to determine if the hyperelliptic curves of (5.1) and (5.4) are indeed soluble and to find a solution to the quadric (5.3) if its Hilbert symbol is 1.

We all make usage of a function called `LiftPoint()`. It takes as input a point from the `IsLocallySolvable()` function and raises the current precision of its coordinates to a desired level. This is important since the `IsLocallySolvable()` function often returns points that have very low precision, sometimes as low as $O(p^1)$. However, this function does not always behave well. We noticed that if the point has sufficiently high precision then it will sometimes return errors. Hence, we needed to account for this in some the code in [Roz].

The Isomorphism

It should be mentioned that Magma does not have schemes defined over p -adic numbers in general. As such, there is no built-in method for determining maps between two schemes over a p -adic field. In fact, as of writing this, one can go into Magma and give it an elliptic curve E over $\mathbb{Q}_p$ not in short Weierstrass form. If one requests Magma to return the elliptic curve E' defined by the Weierstrass model and subsequently asks `IsIsomorphic(E, E')`, Magma will return `false`. Therefore, in order to compute the isomorphism:

$$\varphi : E(\mathbb{Q}_p) \longrightarrow X(\mathbb{Q}_p),$$

we devised a simple work-around in the form of an algorithm that we named: `find_map()`.

We can take a large value M so that we check for all $a \leq M$ if

$$f(a) = c_4 a^4 + c_3 a^3 + c_2 a^2 + c_1 a + c_0$$

is a square in $\mathbb{Q}_p$. If so, then we consider the number field $K := \mathbb{Q}(\sqrt{f(x)})$. Since, $(a, \sqrt{f(a)}) \in X(K)$, we therefore, have an isomorphism

$$\psi : E(K) \longrightarrow X(K),$$

which Magma computes automatically. Therefore, our algorithm extracts the defining equations of the morphism and lifts them into $\mathbb{Q}_p$. Hence, we obtain the desired isomorphism.

If the upper bound $x = M$ is reached and no point is found, we leave it to the devices of Magma and let it compute an automatic point $(x : y : z) \in X(\mathbb{Q}_p)$ via the previously mentioned function `IsLocallySolvable()` function. In some sense, this might be seen as less desirable since we have less control over the algorithm, especially if it returns a point with $z = 0$. However, in this case, we recall we have the gluing maps

$$u = \frac{1}{x}, \quad v = \frac{y}{x^2}$$

so that our rational function becomes

$$f(u, v) = \frac{1 + u\alpha}{1 + u\beta},$$

but $z = 0$ implies $u = 0$ so that $f = 1$ and the corresponding Hilbert symbol is trivial. Hence, we can exclude this value and move on.

On Approximations and p -Adic Precision

Using the decomposition, $K_v = \mathbb{Z} \times \mathcal{O}_v^*$, the p -adic numbers $\mathbb{Q}_p$ and its extensions K_v have elements that are computed up to some finite precision in practice. Thus we comment on the validity of the results to come regarding precision; it is natural to ask what the minimal precision must be. In practice, we didn't find any limiting factors on precision beyond applying what is necessary to apply Hensel's lemma. We selected precision 50, i.e. we computed every quantity to $O(p^{50})$, including in Section 5.2, where we compute the coefficients of algebraic numbers over $\mathbb{Q}_p$ up to precision 50. In any case, the algorithms that required a choice for precision ran fast enough, so it never felt necessary to select a lower precision. The slowest algorithm that was implemented, computing functions $f = \frac{x-\alpha}{x-\beta}$ in both 2 and 3-coverings, actually had nothing to do with p -adic precision

We have the following lemma, which confirms that the Artin symbol will still provide valid answers if we take p -adic approximations that are sufficiently close.

Proposition 5.1.4. Let K be a number field, $v \in \mathfrak{M}(K)$. Let $L = K_v[x]/(g)$ with $g \in K_v[x]$ irreducible and $\alpha, \beta \in K_v^*$. Let $m := [L : K_v]$. If $|\alpha - \beta|_v$ is sufficiently small, i.e. $v(\alpha - \beta) > v(\alpha) + 2m$, and $|\alpha|_v, |\beta|_v$ are not smaller than their difference: $v(\alpha), v(\beta) > v(\alpha - \beta)$, then $[\alpha, L/K_v] = [\beta, L/K_v]$.

Proof. Suppose that $v(\alpha - \beta) = n$. If $v(\alpha) \neq v(\beta)$, then, supposing that $v(\alpha) > v(\beta)$, $v(\alpha - \beta) = v(\beta) + v(\alpha' - 1) = v(\beta)$ where $\alpha = \alpha'\beta$ for $\alpha' \in \mathcal{O}_v$. However this implies that $v(\beta) = n$, which contradicts the assumption that $v(\beta) > n$. It follows that $v(\alpha) = v(\beta)$, which means that $v(\alpha\beta^{-1}) = 0$.

We note here that this makes sense since two approximately close p -adic numbers should have a difference with a higher p -adic valuation.

We have that $[\cdot, L/K_v]$ is a homomorphism so that

$$[\alpha, L/K_v][\beta, L/K_v]^{-1} = [\alpha\beta^{-1}, L/K_v]$$

Now $\alpha - \beta = u\pi^n$ implies $\alpha\beta^{-1} - 1 = u\beta^{-1}\pi^n \in \mathcal{O}_v^*$. Let $n' := n - v(\beta) > 0$.

$$[\alpha\beta^{-1}, L/K_v] = [1 + u'\pi^{n'}, L/K_v]$$

Since $L_w = K_v(\theta)$, it has basis $1, \theta, \dots, \theta^{m-1}$. For an arbitrary element $\gamma = a_0 + a_1\theta + \dots + a_{m-1}\theta^{m-1} \in L$, multiplication by γ defines a linear transformation $M_\gamma : L_w \rightarrow L_w$. We have that $N_{w/v}(\gamma) = \det M_\gamma = f(a_0, a_1, \dots, a_{m-1}) \in K_v[a_0, \dots, a_{m-1}]$. Now we see that multiplying $\gamma \cdot \theta^i$ yields the term $a_0\theta^i$ for $0 \leq i \leq m-1$. Therefore, we find in each entry of the diagonal of M_γ the element a_0 . This means that when we compute $\det M_\gamma$, there is a term a_0^m . Hence $f = a_0^m + f_1$ for some homogeneous polynomial $f_1 \in K_v[a_0, \dots, a_{m-1}]$. Now we have

$$f - 1 - u'\pi^{n'} = a_0^m + f_1 - 1 - u'\pi^{n'} \equiv a_0^m + f_1 - 1 = 0 \pmod{\mathfrak{p}_v^{n'}},$$

where we take $(a_0, \dots, a_{m-1}) = (1, 0, \dots, 0)$ to obtain the last inequality. Taking the derivative with respect to a_0 gives $f_{a_0} = ma_0^{m-1} + f_{1,a_0}$, which means that $f_{a_0}(1, 0, \dots, 0) = m$. Therefore if we have that $n' > 2 \cdot m$, then there exists a tuple $(r_0, \dots, r_{m-1}) \equiv (1, 0, \dots, 0) \pmod{\mathfrak{p}_v^{n'}}$ such that $f(r_0, \dots, r_{m-1}) = 1 + u'\pi^{n'}$ in K_v , i.e. there exists element $\gamma = r_0 + r_1\theta + \dots + r_{m-1}\theta^{m-1}$ such that $N_{w/v}(\gamma) = \alpha\beta^{-1}$. From Artin reciprocity, we conclude that $[\alpha\beta^{-1}, L/K_v] = 1$. $\square$

Applying the Method

Now we consider the hyperelliptic curve:

$$X : y^2 = -727x^4 - 104x^3 + 92x^2 + 4x - 4$$

defined over $\mathbb{Q}$, originally covered in both [Sik01] and [BS08]. It seems appropriate to test the method on an example that is only briefly described in Sections 8-9 of the latter paper.

X is a two covering of the elliptic curve 571A E from Cremona's tables, and it has the short Weierstrass equation

$$E : y^2 = x^3 - \frac{44608}{3}x - \frac{18842960}{27}.$$

Just as in [BS08], we utilise the function

$$f = \frac{x + \frac{16}{53}}{x}.$$

We compute that $\alpha = 16/53$ and $\beta = 0$ both have residue field $\mathbb{Q}(i)$. The discriminant of $\mathbb{Q}(i)$ is -4 , and so we have that the only ramified finite prime is 2. Then for the prime $p = 53$, we have that

$$x^2 + 1 \equiv (x - 23)(x - 30) \pmod{53}$$

so that 53 is split in $\mathbb{Q}(i)$. As

$$f(-16/53) = \frac{-1 \cdot 2^2}{53} \quad \text{and} \quad f(0) = -1 \cdot 2^2$$

there are no other primes arising in the numerators and denominators of $f(\alpha)$, $f(\beta)$. Therefore, the primes that we must check are 2 and ∞ .

Now we wish to compute $E(\mathbb{Q}_2)/2E(\mathbb{Q}_2)$. We can show that $x^3 + Ax + B$ has no roots in $\mathbb{Q}_2$, and as such we conclude that $E(\mathbb{Q}_2)[2] = 1$ via the identity $P = -P$. Therefore, using the formula from Proposition 5.1.3,

$$\#E(\mathbb{Q}_2)/2E(\mathbb{Q}) = 2 \cdot \#E(\mathbb{Q}_2)[2]$$

we know that this group has order 2 and, accordingly, only one generator.

Now $K_2 \cong \mathbb{Q}_2(\sqrt[3]{2})$, and here Magma finds 3 nontrivial elements $2(\frac{\sqrt[3]{2}}{2})^2 + 1$, $7(\frac{\sqrt[3]{2}}{2}) + 7$, $14(\frac{\sqrt[3]{2}}{2})^2 + 7(\frac{\sqrt[3]{2}}{2}) - 21$ of norms 1, 9, and 9 respectively. Magma provides an isomorphism which takes $K_2^*/K_2^{*2} \rightarrow \mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}(\sqrt[3]{2})^{*2}$, which sends some $\alpha \in K_2$ to the element $\sqrt[3]{2}/2$. Now pulling these elements back, we get the following elements in K_2 of the form $\delta := e_2\theta^2 + e_1\theta + e_0$.

For $\delta = 14(\frac{\sqrt[3]{2}}{2})^2 + 7(\frac{\sqrt[3]{2}}{2}) - 21$, the Hilbert symbol is $(\alpha, e_2)_2 = -1$. Therefore, it is not possible to find any elements of $E(\mathbb{Q}_2)$ that will correspond to this δ value since this means that $\delta z^2 = x - \theta$ is not possible. We have for the value $\delta = 7\sqrt[3]{2} + 7$, the corresponding Hilbert symbol evaluates to 1. Hence, we run through the procedure and obtain a hyperelliptic curve as in (5.4), and we use `IsLocallySolvable()`, which determines that the curve is insoluble. Thus, we again find that there is no point in this case. For the last value of δ , we already know that will return a point from previous remarks. We now write out the steps of the computation for this last case.

For the generator $2\sqrt[3]{2}^2 + 1$, this corresponds to the element

$$\delta = 1/2 \cdot (276199282665 \cdot \tilde{\theta}^2 + 14316577975740 \cdot \tilde{\theta} + 63391622789330) \in K,$$

which defines a generator of K_2^*/K_2^{*2} with norm $r = 208318319100109 + O(2^{49})$. One computes that $\alpha = -473433464463272902252917225 \cdot 2^{-1} + O(2^{49})$ and $\beta = 13082879544914273626140115950 + O(2^{50})$. Then, we return the point:

$$(U, V, W) = (150033675361025 + O(2^{49}), O(2^{49}), 1 + O(2^{50}))$$

lying on (5.3), which via the above transformation gives the point:

$$(u, v, w) = (46108802264983 \cdot 2^3 + O(2^{50}), 32402240323493 \cdot 2^2 + O(2^{50}), -225221322806267 + O(2^{49})).$$

We may take our new approximate delta value to be $\delta' := \delta \cdot (u + v\tilde{\theta} + w\tilde{\theta}^2)$ and lift this K_2 . We write $v_2(\delta') = 2l + 1$, for some l , and we divide each coefficient by 2^l to improve tractability. Thus our new δ element ends up being

$$\delta := (-1002304247369853 + O(2^{50})) \cdot \theta + 807997980197124 + O(2^{50});$$

with norm $r = -3038325048371 \cdot 2^2 + O(2^{51})$. Then we obtain a hyperelliptic curve as in (5.4). This is soluble, with point

$$(\lambda, y, \mu) = (3628169 \cdot 2^2 + O(2^{50}), -31453598764707 \cdot 2 + O(2^{50}), 1 + O(2^{53})).$$

Putting this point through the morphisms gives

$$(u, v, w) = (109929047810727 + O(2^{49}), 63316225526203 \cdot 2^{-1} + O(2^{48}), 75186400236177 \cdot 2^{-1} + O(2^{47}))$$

This means that

$$x = q_0(u, v, w) = -55317322457291 \cdot 2^3 + O(2^{50}).$$

Plugging this value into $x^3 - 44608/3 - 18842960/27$ returns the value

$$y = -18043224011623 \cdot 2^2 + O(2^{48}),$$

so $(x, y) \in E(\mathbb{Q}_2)/2E(\mathbb{Q}_2)$ is the generator of this group. Then we run our algorithm that computes the map $E(\mathbb{Q}_2) \rightarrow X(\mathbb{Q}_2)$ described above, and we find the point

$$(15627936868297 \cdot 2^{-1} + O(2^{47}), -270018571802787 \cdot 2^{-2} + O(2^{47})) \in X(\mathbb{Q}_2).$$

For this point we compute that

$$f(P) = 27062438278817 + O(2^{50}).$$

Our method for computing the map also returns the image of $\mathcal{O}$

$$(2 + O(2^{51}) : 92983690021369 \cdot 2 + O(2^{50}) : 1 + O(2^{50}))$$

with

$$f(\mathcal{P}) = 446111283843305.$$

Then we compute using `NormResidueSymbol()` in Magma that

$$\left(\frac{f(\mathcal{P}), -1}{2} \right) = 1$$

for both points. For the point at ∞ , we found the following point to real precision 50

$$(-0.30186037207441488297659531906, 0.00011629329154499122338354589)$$

via a short search. We quickly note that the y -value we find is the conjugate root to the one found in [BS08]. We compute that

$$f(\mathcal{P}) = -8.75251634845017942658514322924 \cdot 10^{-5} < 0.$$

Therefore, the Hilbert symbol evaluates to

$$\left(\frac{f(\mathcal{P}), -1}{\infty} \right) = -1.$$

If we write $\mathcal{P} = a_1\mathcal{P}_{2,1} + a_2\mathcal{P}_{2,2}$ in $E(\mathbb{Q}_2)/2E(\mathbb{Q}_2)$ and $\mathcal{P} = a_3\mathcal{P}_\infty$, then, using Theorem 4.4.1, we have that

$$\prod_v \varphi_v(f(\mathcal{P})) = \left(\frac{f(\mathcal{P}_{2,1}), -1}{2} \right)^{a_1} \left(\frac{f(\mathcal{P}_{2,2}), -1}{2} \right)^{a_2} \left(\frac{f(\mathcal{P}_\infty), -1}{\infty} \right)^{a_3} = (-1)^{a_3} = 1,$$

which implies that $a_3 \not\equiv 1 \pmod{2}$. Hence there are no points of degree 1 and accordingly no rational divisors of degree 1.

5.1.3 Lind's Counterexample Revisited

We return to Lind's counterexample:

$$X : 2y^2 = x^4 - 17,$$

which is well-known for presenting a Brauer-Manin obstruction to the Hasse principle, which previously discussed in Section 4.3. By multiplying by 2, we obtain the standard form for a hyperelliptic curve:

$$y^2 = 2x^4 - 34.$$

Via a large search, we find the following function:

$$f = \frac{x + 143/21}{x - 3}.$$

One computes that

$$F(-143, 21) = 2 \cdot 20368 \quad \text{and} \quad F(3, 1) = 8.$$

Hence, we have that both points correspond to the residue field $\mathbb{Q}(\sqrt{2})$. The curve X has Jacobian:

$$E : y^2 = x^3 + 272x, \tag{5.5}$$

which means that $E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)$ is computable using descent by 2-isogeny. Let $\varphi_v : E(\mathbb{Q}_v) \rightarrow X(\mathbb{Q}_v)$ be an isomorphism. Running through the computations returns that we must check the primes $p = 2, 3, 19, 67$ via Proposition 5.1.1. We note that $2 > 0$ implies it is unnecessary to check the archimedean prime since then $\mathbb{Q}_\infty = \mathbb{R}$ and $N : \mathbb{R}^* \rightarrow \mathbb{R}^*$ is clearly surjective.

For the purpose of illustrating how the algorithm works, we explicitly show the computation for $p = 2$. Let $\mathcal{Q}_i$ denote the generators of $E'(\mathbb{Q}_2)/\psi(E(\mathbb{Q}_2))$ and $\mathcal{P}_i$ their images under $\hat{\psi}$. Running the 2-descent method returns the images of the points in $E'(\mathbb{Q}_2)/\psi(E(\mathbb{Q}_2))$:

$$\begin{aligned}\mathcal{P}_1 &:= [O(2^{51}), O(2^{52}), 1 + O(2^{50})], \\ \mathcal{P}_2 &:= [450359962737041 + O(2^{50}), 136584826800865 + O(2^{49}), 1 + O(2^{50})], \\ \mathcal{P}_3 &:= [59227271370617 \cdot 2^4 + O(2^{51}), 95177111903803 \cdot 2^4 + O(2^{52}), 1 + O(2^{50})], \\ \mathcal{P}_4 &:= [187649984473777 + O(2^{49}), -97805794110225 + O(2^{48}), 1 + O(2^{50})], \\ \mathcal{P}_5 &:= [-66147708369911 \cdot 2^4 + O(2^{51}), -14752760619981 \cdot 2^4 + O(2^{52}), 1 + O(2^{50})]\end{aligned}$$

associated to the generators g_i of $G := \mathbb{Q}_2/\mathbb{Q}_2^* : g_1, g_2, g_1 + g_2, g_3, g_1 + g_3$, respectively. The presence of the first point implies that $\#E(\mathbb{Q}_2)[2] = 2$ so that $\#E(\mathbb{Q}_2)/2E(\mathbb{Q}_2) = 4$ from the formula from Proposition 5.1.3. Note that for the generator $g_2 + g_3$, it corresponds to the point $(0, 0) \in E'(\mathbb{Q}_2)$, which will not directly contribute to the basis. However, we can use $\mathcal{Q}_2 + \mathcal{Q}_4 = (0, 0)$ in $E'(\mathbb{Q}_2)/\psi(E(\mathbb{Q}_2))$ to derive that

$$[\mathcal{P}_2] = [\mathcal{P}_4]$$

in $E(\mathbb{Q}_2)/2E(\mathbb{Q}_2)$, which implies the basis is given by $\{\mathcal{P}_1, \mathcal{P}_4\}$.

We see that for $y = 0$, $x^4 = 17$ admits a root in $\mathbb{Q}_2$. Hence, the map $E(\mathbb{Q}_2) \rightarrow X(\mathbb{Q}_2)$ may be written as $(X, -8\sqrt[4]{17}YZ, 1/17\sqrt[4]{17}^3X - 8\sqrt[4]{17}Z)$ and sends $\mathcal{O} \mapsto (\sqrt[4]{17} : 0 : 1) =: \mathcal{P}$. Now, we compute $(f(\mathcal{P}), 2)_2$ for our basis elements of $\text{Pic}(\mathbb{Q}_2)/2\text{Pic}(\mathbb{Q}_2)$. Hence we have

$$(f(\mathcal{P}), 2)_2 = 1, \quad (f(\mathcal{P}_1), 2)_2 = 1, \quad (f(\mathcal{P}_4), 2)_2 = 1.$$

Our algorithm computes that for $p \neq 3$ that

$$(f(\mathcal{P}), 2) = 1.$$

However, for $p = 3$, there are 3 points to check, and the algorithm computes that

$$(f(\mathcal{P}), 2) = -1$$

for each point. Then using theorem 4.4.1 and writing the image of our rational point $\mathcal{P} = \sum_{i=1}^3 a_{3,i} \mathcal{P}_{3,i}$ we have

$$\prod_{v \in B} \varphi_v(f(\mathcal{P})) = \varphi_3(f(\mathcal{P})) = (-1)^{\sum a_{3,i}} = 1$$

Which implies that $\sum a_{3,i} \equiv 0$ and $1 \pmod{2}$, which is obviously false. Hence there are indeed no rational points. On the other hand, considering the original proof, we may instead try using the rational function:

$$f = y. \tag{5.6}$$

The divisor of this function is

$$\text{div } y = \mathcal{P} - 2\mathcal{O},$$

where $\mathcal{P}$ represents the 4 Galois conjugates of $(\sqrt[4]{17}, 0)$. Now the corresponding residue fields are $\mathbb{Q}(\sqrt[4]{17})$ and $\mathbb{Q}(\sqrt{2})$, respectively. The former is not Galois, but it contains the maximal Galois subextension $\mathbb{Q}(\sqrt{17})$, which via Theorem 3.3.4 implies that $N_{\mathbb{Q}(\sqrt[4]{17})}(C_{\mathbb{Q}(\sqrt[4]{17})}) = N_{\mathbb{Q}(\sqrt{17})}C_{\mathbb{Q}(\sqrt{17})}$ since these are both open subgroups. Then we have that

$$C_L = N_{\mathbb{Q}(\sqrt{17})}C_{\mathbb{Q}(\sqrt{17})}(N_{\mathbb{Q}(\sqrt{2})}C_{\mathbb{Q}(\sqrt{2})})^2$$

Now we have that

$$(N_{\mathbb{Q}(\sqrt{2})}C_{\mathbb{Q}(\sqrt{2})})^2 \subseteq C_{\mathbb{Q}}^2 \subseteq N_{\mathbb{Q}(\sqrt{17})}\mathbb{Q}(\sqrt{17}),$$

where the last inclusion follows from the fact that C_K^2 is the intersection of all norm subgroups $N_{L/K}C_L$ for L a degree 2 abelian extension by [Neu99, Exercise VI.6.3 pg 404]. Therefore, we conclude that $L = \mathbb{Q}(\sqrt{17})$. Hence y is harmful, and the only ramified prime is 17.

We claim that the only primes that might not have $\varphi_v = 1$ are 2 and 17. Indeed, suppose that $v(y) > 0$, then this implies that $x^4 \equiv 17 \pmod{p}$, which implies that p is split in L . If $v(y) < 0$, then (5.5) implies $2v(x) = v(y) < 0$. Hence from the gluing formulas $w = y/x^2$ and $u = 1/x$, we obtain that $v(w) = v(y/x^2) = 0$, so w is a unit. We also have that $y = w/u^2$, hence $(\frac{y,17}{p}) = (\frac{w,17}{p})(\frac{u^{-2},17}{p}) = (\frac{u^{-1},17}{p})^2 = 1$ since w is a p -adic unit and p is unramified. Also there is no need to compute for the prime over ∞ since L is real field.

We saw above that there are 3 points generating $\text{Pic}(X_2)/2\text{Pic}(X_2)$, and we have for each point that

$$\left(\frac{y(\mathcal{P}), 17}{2}\right) = 1,$$

and for the 2 points corresponding to $\text{Pic}X_{17}/2\text{Pic}X_{17}$ that

$$\left(\frac{y(\mathcal{P}), 17}{17}\right) = -1.$$

Therefore, writing the image of $\mathcal{P}$ in $\text{Pic}X_{17}/2\text{Pic}X_{17}$ as $a_{17,1}\mathcal{P}_{17,1} + a_{17,2}\mathcal{P}_{17,2}$ and using Theorem 4.4.1 we obtain that

$$\varphi_{17}(y(\mathcal{P})) = \varphi_{17}(y(\mathcal{P}_{17,1}))^{a_{17,1}}\varphi_{17}(y(\mathcal{P}_{17,2}))^{a_{17,2}} = (-1)^{a_{17,1}+a_{17,2}} = 1,$$

which implies $a_{17,1} + a_{17,2}$ is equal to both 1 and 0 modulo 2 simultaneously. Hence, using the same rational function arising from the original proof, we obtain the Brauer-Manin obstruction again.

The Brauer-Manin Obstruction Revisited

To tie up some loose ends, we now explain how the obstruction produced from our computations above is the same as that from the original from Lind-Reichardt. We recall that in Section 4.3 that the quaternion algebra $(\frac{Y,17}{\mathbb{Q}(X)})$ defined an element of $\text{Br}(X)$ and produced a Brauer-Manin obstruction ultimately coming from the condition that $(\frac{Y}{17}) = 1$. On the other hand, we discussed at the end of Chapter 4 how the rational function Y also gives rise to an element of $\mathcal{A}_\chi = Y \cup \delta\chi \in \text{Br}(X) \subseteq \text{Br}(F)$, where we take χ to be the unique character arising from the non-trivial character $\chi : G(\mathbb{Q}(\sqrt{17})/\mathbb{Q}) \rightarrow \mathbb{Q}/\mathbb{Z}$. We had the cup-product pairing:

$$H^0(F, \bar{F}^*) \times \times H^2(F, \mathbb{Z}) \longrightarrow H^2(F, \bar{F}^*),$$

where $F = K(X)$. We commented there that one had that $f \cup \delta\chi = (\frac{f, \chi}{F})$ in $\text{Br}(F)$, a cyclic algebra. Since χ is a character coming from the Galois group $G(\mathbb{Q}(\sqrt{17})/\mathbb{Q})$, then $(\frac{Y, \chi}{F}) = (\frac{Y, 17}{F})$ by [GS06, Corollary 2.5.5 p 36]. Therefore, we see that the Brauer-Manin obstruction produced by this method is indeed identical to the original given by Lind.

5.1.4 Further Computation

We use our algorithm to compute many examples for the following parameters: Let $N \leq 504$ be the conductor of an elliptic curve; we choose the rational function

$$f = \frac{x + \alpha_1/\beta_1}{x + \alpha_2/\beta_2}$$

such that $|\alpha_i| \leq 1000$ and $0 \leq \beta_i \leq 1000$. We applied the method to 114 2-coverings, which took around 12 hours. We produced approximately 2300 functions in total and computed for each a list consisting of $(\frac{f(\mathcal{P})}{p}, \gamma)$ for each p as in Proposition 5.1.1. For many of these curves, we indeed found functions that produce a Brauer-Manin obstruction. A handful of curves produced no functions with the parameters above.

There were also two 3 exceptions that arose. For conductor $N = 435$, there was a function that $f = \frac{9x-9}{9x-7}$ associated to the curve $C : y^2 = -28x^4 + 164x^3 - 203x^2 - 108x - 12$ such that at one of the p -adic points $\mathcal{P}$ in the basis, we had $f(\mathcal{P}) = 0$. Assuming this is not an error, this would be the only examples computed thus far that actually requires that moving lemma. The other two exceptions occurred at $N = 438$ and $N = 510$, where the descent by 2-isogeny method returns hyperelliptic curves that have singular reduction at some primes p . Therefore, the method `LiftPoint()` fails, which meant the computation does not go through for now.

5.2 3-Coverings of Elliptic Curves

We recall that in Section 4.2, we discussed that the 3-coverings of an elliptic curve could all be embedded into $\mathbb{P}^2$ as a plane cubic. The original methods of [Sik01] proved to be useful for 2-coverings and found a type of generalisation in [BS08]. The method in the latter paper, has been extensively tested on 2-coverings. The original authors also discussed plans in the paper to expand the method to 3-coverings. We discuss one famous example of a 3-covering and use Theorem 4.4.1 to find a Brauer-Manin obstruction. To this end, we will describe the algorithms we wrote in Magma and the results of the computations using them.

5.2.1 The Selmer Curve

Three Torsion in the Tate-Shafarevich Group

The following non-singular curve $X/\mathbb{Q}$, known as the Selmer curve,

$$X : 3x^3 + 4y^3 + 5z^3 = 0, \tag{5.7}$$

is non-trivial of order 3 in the Tate-Shafarevich group of the elliptic curve

$$E : x^3 + y^3 + 60z^3 = 0$$

of conductor $24300 = 2^2 \cdot 3^5 \cdot 5^2$. In [Sel51], Selmer considers the general class of curves

$$aX^3 + bY^3 + cZ^3 = 0.$$

He created tables for many different choices of a, b, c on pp 347-360, documenting which of the curves are soluble or insoluble. Of particular interest for us is table 4^b on p 352. For the a, b, c as in (5.7), he computes that $X(\mathbb{Q}) = \emptyset$ using infinite descent on the Jacobian $X^3 + Y^3 = 60Z^3$ and some computations involving class groups and units of the cubic extension $\mathbb{Q}(\sqrt[3]{6})$ [Sel51, Theorem IX p

306 and Chapter VII p 277]. We recommend the reader to look through [Con] if one wishes to see the explicit computations.

For our purposes, it will be natural to append the primitive third root of unity $\rho \in \mu_3$ to $\mathbb{Q}$ since we will be working with cubic extensions of $\mathbb{Q}$; otherwise, those fields will not be Galois. As mentioned in [Sel51, p 211], the curve X remains insoluble over $K = \mathbb{Q}(\rho)$.

Now the first step is to devise a method to compute $\text{Pic}(X_v)/3\text{Pic}(X_v)$ for any $v \in \mathfrak{M}(K)$. By Proposition 5.1.2, we have that

$$\text{Pic}(X_v)/3\text{Pic}(X_v) \cong \mathbb{Z}/3\mathbb{Z} \oplus E(K_v)/3E(K_v).$$

Thus problem is reduced to computing $E(K_v)/3E(K_v)$ and K_v -rational points $\mathcal{P}_v \in E(K_v)$. We base our computations for this on the 3-descent method for $\mathbb{Q}$ described in [Top93]. Let $C := -24300$. Now we have that $E(K)[3]$ is nonempty, so $E(K_v)[3]$ is also nonempty. Let $P := (0, 90\sqrt{-3}) \in E(K_v)[3]$. As in [Top93, Section 3 p 3], this determines a 3-isogeny $\psi : E \rightarrow E'$, where

$$E' : v^2 = u^3 - 27C$$

and

$$\psi := \left(\frac{y^2 + 3C}{x^2}, \frac{y(x^3 - 8C)}{x^3} \right).$$

Then repeating the process from above we obtain that the dual isogeny is given by

$$\hat{\psi} := \left(\frac{v^2 - 3 \cdot 27C}{9u^2}, \frac{v(u^3 + 8 \cdot 27C)}{27u^3} \right)$$

We know that ψ is surjective as it is a nonconstant map of curves; suppose that $E'(K_v) \ni Q' = (u, v) = \psi(Q) = \psi(x, y)$, where $Q \in E$ is not necessarily rational. Our goal is to determine the map $\delta : E'(K_v)/\psi(E(K_v)) \rightarrow K_v^*/K_v^{*3}$. First, we map

$$\begin{aligned} E'(K_v)/\psi(E(K_v)) &\longrightarrow H^1(K_v, E[\psi]) \\ Q' &\longmapsto (\sigma \mapsto Q'^\sigma - Q). \end{aligned}$$

Then the map $E[\psi] \rightarrow \mu_3, P \mapsto \rho$ induces the isomorphism $H^1(K_v, E[\psi]) \rightarrow H^1(K_v, \mu_3)$, which will send

$$(\sigma \mapsto Q'^\sigma - Q) \longmapsto \sigma \mapsto \begin{cases} 1 & \text{if } Q'^\sigma - Q = \mathcal{O} \\ \rho & \text{if } Q'^\sigma - Q = P \\ \rho^2 & \text{if } Q'^\sigma - Q = 2P. \end{cases}$$

Finally, we have that $H^1(K_v, \mu_3) \cong K_v^*/K_v^{*3}$ by the canonical isomorphism from Galois cohomology. This implies that every cocycle is defined by an element $\alpha = \beta^3 \mapsto (\sigma \mapsto \beta^\sigma/\beta)$. Hence, we need to determine the action of $\sigma \in G_{K_v}$ on the element Q to find the image of Q' in K_v^*/K_v^{*3} . First, we assume that $u \neq 0$, then using the definition of ψ we have that

$$u = \frac{y^2 + 3C}{x^2}.$$

Multiplying through by x^2 and using the defining equation for E , we obtain $x^3 - ux^2 + 4C = 0$. The discriminant of this polynomial is $16Cv^2$, a square in K_v^* . Therefore, we have that the splitting

field of this polynomial is obtained by appending any root to K_v . Using the cubic formula and simplifying, we obtain that one of the roots is given by

$$x = \frac{1}{3}\sqrt[3]{e} + \frac{1}{3}\sqrt[3]{\bar{e}} - \frac{u}{3},$$

where $e = (v + \sqrt{-27C})^2$ and $\bar{e} = (v - \sqrt{-27C})^2$, with $\sqrt{-27C} = 2 \cdot 3^4 \cdot 5 \in \mathbb{Q}$. From Cardano's formulas, the other roots are $\rho^{\frac{1}{3}}\sqrt[3]{e} + \rho^{\frac{2}{3}}\sqrt[3]{\bar{e}} - \frac{u}{3}$ and $\rho^{\frac{2}{3}}\sqrt[3]{e} + \rho^{\frac{1}{3}}\sqrt[3]{\bar{e}} - \frac{u}{3}$. We also have that $\sqrt[3]{e}\sqrt[3]{\bar{e}} = u^2$, so that the Galois action of σ on $\sqrt[3]{e}$ determines that of $\sqrt[3]{\bar{e}}$. Therefore, the action of σ on Q is determined by the action on $\sqrt[3]{e}$. Moreover, there is an identity $(v + \sqrt{-27C})^2 = \frac{1}{v + \sqrt{-27C}}$ in K_v^*/K_v^{*3} , which determines the map.

Now $K_v(Q)/K_v$ is thus at worst a degree 3 extension. We can also show that $e \in K_v^{*3}$ if and only if $Q' \in \psi(E(K_v))$. If we consider the three torsion point $(0, \sqrt{-27C})$, then we see using the map φ that $y^2 + 3C = 0$, which implies $y = \sqrt{-3C} \in \mathbb{Q} \subseteq K_v$ and $x = \sqrt[3]{-4C}$, so that the action of σ on x determines that on Q . Therefore, we obtain the map

$$\begin{aligned} E'(K_v)/\psi(E(K_v)) &\longrightarrow K_v^*/K_v^{*3} \\ (u, v) &\longmapsto (v + \sqrt{-27C})^{-1} \\ (0, \sqrt{-27C}) &\longmapsto -4C \end{aligned}$$

If our point (u, v) corresponds to a generator of this group then we have that there exists $z \in K_v^{*3}$ such that

$$v + \sqrt{-27C} = z^3\delta^{-1}.$$

Now since δ is a generator of the Selmer group, we have that δ^{-1} is also a generator, and we may assume that it is also integral, and so we continue to write δ . Substituting $v = z^3\delta - \sqrt{-27C}$ into our equation for E' returns

$$\begin{aligned} u^3 &= (z^3\delta - \sqrt{-27C})^2 + 27C \\ &= z^3\delta(z^3\delta - 2\sqrt{-27C}). \end{aligned}$$

This implies that $\delta(z^3\delta - 2\sqrt{-27C})$ is an element of K_v^{*3} . As such we can instead solve

$$t^3 - \delta(z^3\delta - 2\sqrt{-27C}) = 0,$$

where $t = u/z$. We may homogenise the equation to lift any potential solutions using Hensel's lemma:

$$T^3 - \delta(Z^3\delta - 2\sqrt{-27C}W^3) = 0.$$

Then we have that $z = Z/W$. Hence $u = ZT/W^2$ and $v = (Z/W)^3\delta - \sqrt{-27C}$.

Now, for the dual $\hat{\psi}$ we compute that $u = 9\sqrt{f} + 9\sqrt{\bar{f}} + 3x$ with $f = (y + \sqrt{C})^2$ and $\bar{f} = (y - \sqrt{C})^2$. This implies that

$$\begin{aligned} E(K_v)/\hat{\psi}(E'(K_v)) &\longrightarrow K_v^*/K_v^{*3} \\ (x, y) &\longmapsto (y + \sqrt{C})^{-1} \\ (0, \sqrt{C}) &\longmapsto 4C \end{aligned}$$

This means that $\sqrt[3]{4C} = 2 \cdot 3 \sqrt[3]{-2 \cdot 3^2 \cdot 5^2} \in K_v^{*3}$ if and only if $E[3] \subseteq E(K_v)$. As usual in descent via an isogeny, we have the following short exact sequence of $\mathbb{F}_3$ -vector spaces:

$$0 \longrightarrow \frac{E'(K_v)[\hat{\psi}]}{\psi(E(K_v)[3])} \longrightarrow \frac{E'(K_v)}{\psi(E(K_v))} \longrightarrow \frac{E(K_v)}{3E(K_v)} \longrightarrow \frac{E(K_v)}{\hat{\psi}(E'(K_v))} \longrightarrow 0,$$

As such, a basis of $E(K_v)/3E(K_v)$ is selected by picking the basis elements of $E(K_v)/\hat{\psi}(E'(K_v))$ and the images of $E'(K_v)/\varphi(E(K_v))$ excluding those that are in the kernel of $\hat{\psi}$. Again, $E'[3] \subseteq E(K_v)$ if and only if $\mathcal{P} \in E(K_v)[3]$ is an element of $3E(K_v)$.

Since $X(K_v) \neq \emptyset$, we have that $X_v \cong E_v$ so that there exists a K_v -rational isomorphism $E_v \rightarrow X_v$. We recall from Section 5.1, that Magma does not have general schemes over p -adic fields implemented. Thus, we proceed in a similar way as there: we find an algebraic field extension F of K such that $F\mathbb{Q}_p \subseteq K_v$ and $X/F \cong E/F$. As $X(F) \neq \emptyset$, we will obtain an isomorphism $\varphi : E_1 \rightarrow X$ for some elliptic curve E_1 . The elliptic curves E_1 and E are isomorphic via some map $\varphi_1 : E \rightarrow E_1$, and we compose them $\varphi' := \varphi_1 \circ \varphi$ to obtain an F -rational map which lifts to a map on K_v using the fibre product $(-) \times_k K_v$.

Suppose that $\mathcal{P} \in X(K)$, then $K(\mathcal{P}) = K$, which means that $\deg(\mathcal{P}) = 1$. Thus, we have that for any $v \in \mathfrak{M}(K)$ that $\pi_v^{-1}\mathcal{P}$ corresponds to exactly one point $\mathcal{Q} \in X_v$ with $\pi_v(\mathcal{Q}) = \mathcal{P}$ such that $K_v(\mathcal{Q}) = K_v$ by Corollary 4.1.2. Thus we have that $\mathcal{Q} \in X_v(K_v)$ so that $\varphi_v = \theta_v(\text{Norm}_{\mathcal{Q}}(f(\mathcal{Q}))) = \theta_v(f(\mathcal{Q}))$, and also we note $\mathcal{Q}$ will have the same coordinates as $\mathcal{P}$ when viewed as K_v -projective point. Therefore, for each v , we have that $\mathcal{Q}$ is given as linear combination of the basis elements in $\text{Pic}(X_v)/3\text{Pic}(X_v)$. Since $\mathcal{Q}$ is a degree 1 point of $X(K_v)$ and we have well-defined degree modulo 3, we can thus write $\mathcal{Q} = \sum a_{v,i} \mathcal{Q}_{v,i}$ such that $\sum a_i \equiv 1 \pmod{3}$ in $\text{Pic}(X_v)/3\text{Pic}(X_v)$. Then, via our main theorem, we have that

$$\mathcal{P} \mapsto \prod_v \varphi_v([\mathcal{P}]) = 1.$$

5.2.2 The Obstruction

As in the 2-covering case, we consider functions

$$f = \frac{x - \alpha}{x - \beta}$$

such that $K(\sqrt[3]{h(\alpha)}) = K(\sqrt[3]{h(\beta)})$. We will check the following primes:

1. the primes that ramify in L (and 3 if it is not split)
2. the primes in the denominators of α and β (that are not split)
3. primes dividing the numerator or the denominator of $h(\alpha), h(\beta)$ (that are not split)

We note from our equation for X that 2 should always be a ramified prime. Also, the reason we include 3 is that the corresponding Hilbert symbol $(\frac{\cdot}{\mathfrak{p}_v})$ is not tame, which creates opportunities for $\varphi_v \neq 1$. Generalising [BS08, Lemma 9.1] to the cubic case, we obtain:

Lemma 5.2.1. We have that for $v \notin B$ that $\varphi_v(f([\mathcal{P}])) = 1$ for all $\mathcal{P} \in X(K_v)$.

Proof. let $\pi \in \mathfrak{p}_v$ be a uniformiser. Using the definition of f yields:

$$x - \alpha = f(\mathcal{P})(x - \beta) \implies x = \frac{\alpha - \beta f}{1 - f}.$$

Substituting this into our equation for X and clearing denominators we obtain:

$$3(\alpha - \beta f)^3 + 4y^3(1 - f)^3 + 5(1 - f)^3 = 0.$$

By our assumptions, we have that $v(\alpha), v(\beta) \geq 0$ and $v(4) = 0$. Thus if $v(f) > 0$, then we have that $v(y) \geq 0$. It follows that equation is integral and we may reduce modulo $\mathfrak{p}_v$ to obtain

$$4y^3 \equiv -3\alpha^3 - 5 = h(\alpha) = \gamma\delta_1^3 \pmod{\mathfrak{p}_v}.$$

This would imply that $(\frac{\gamma}{\pi}) = 1$ since $v(h(\alpha)) = 0$, which contradicts the fact that v is not split in L . Likewise, if $v(f) < 0$ then we divide by f and may reduce modulo $\mathfrak{p}_v$ again

$$4y^3 \equiv -3\beta^3 - 5 = \gamma\delta_2^3 \pmod{\mathfrak{p}_v}$$

so that $(\frac{\gamma}{\pi}) = 1$ since $v(h(\beta)) = 0$, which is false for the same reason as before. $\square$

To obtain such functions f , we generalise the ideas described in [Sik01]. We now describe three functions we wrote concerning the Eisenstein integers $\mathbb{Z}[\rho]$. The first is **PrimaryDecomposition()** that outputs a list containing the factorisation information of any integer $t \in \mathbb{Z}[\rho]$ as a primary decomposition as in Section 3.4. The algorithm takes an element $t \in K$ with $t \neq 0, \pm\rho^i$ (note that we also implemented these edge cases in Magma, but we omit the description here for brevity) and begins by computing the prime factorisation of the norm $N(t\mathcal{O}_K)$ in $\mathbb{Q}$. For each $p|N(t\mathcal{O}_K)$, if $p \equiv 2 \pmod{3}$, then we immediately append p and the corresponding exponent to the factorisation list. If $p \equiv 1 \pmod{3}$, we then run a quick search for two integers $0 \leq a, b \leq p$ such that $p = a^2 + 3b^2 = (a + b + 2b\rho)(a - b - 2b\rho) =: \pi_1\pi_2$. We iterate over the third roots of unity until at least one of $\rho^j\pi_i \equiv 2 \pmod{3}$, at which point we select $\pi := \rho^j\pi_i$. Now we compute the valuation of $v_\pi(t)$ and $v_{\bar{\pi}}(t)$. Then we append the respective pairs $(\pi, v_\pi(t))$ and $(\bar{\pi}, v_{\bar{\pi}}(t))$ to the factorisation list if their respective valuation is not zero. If $3|N(t\mathcal{O}_K)$, then we choose $\lambda = 1 - \rho$ to be the prime in K and append $(\lambda, v_\lambda(t)) = (\lambda, 2v_3(N(t\mathcal{O}_K)))$. Finally, we multiply by a root of unity to ensure that the factorisation equals t .

Next, we wrote a function **Cubefree()** that returns the cubefree part and the cubed part, i.e. $t = \gamma\delta^3$ with γ cubefree. Running **PrimaryDecomposition()**, it is a simple matter to iterate through the elements of the list and check if $v_\pi(t) < 3$. If $v_\pi \geq 3$, we do some modulo 3 computations to separate the cube-free part of π .

Lastly, we wrote a function called **GCDeis()**, that will determine the greatest common divisor of two Eisenstein integers a, b . Ignoring edge cases involving zeros and units, our algorithm uses **primaryfactorisation()** on a and b , respectively. Let $gcd := 1$. We iterate over the primes π appearing in the support of a and b . Then we select $n := \min(v_\pi(a), v_\pi(b))$ and we let $gcd := gcd \cdot \pi^n$, which eventually returns the gcd.

For $t_1 = \alpha_1 + \beta_1\rho$ and $t_2 = \alpha_2 + \beta_2\rho$ with $-5 \leq \alpha_i \leq 5$ and $0 \leq \beta \leq 5$, it took 10113.922 seconds (~ 3 hours) to output a list of 9 tuples $[t_1, t_2, \gamma, \delta]$, with γ the cube-free part and δ the cubed component of $h(t_1, t_2)$. Using these tuples we build functions by comparing the γ_i values, and from this we obtain 28 functions.

Consider the function from the procedure:

$$f = \frac{x - \frac{2\rho-1}{3\rho+5}}{x - \frac{2\rho+3}{2\rho+5}}.$$

It also has field $L = K(\sqrt[3]{4})$. The numbers $\beta_1 = 3\rho + 5$ and $\beta_2 = 2\rho + 5$ are conjugates and divide 19. Using the algorithms described, we compute the factorisations of

$$h(\alpha_1, \beta_1) = -4 \cdot (3\rho - 4)^3 \quad \text{and} \quad h(\alpha_2, \beta_2) = 4 \cdot (-3\rho - 7)^3,$$

where the two latter primes appearing in each factorisation are the conjugate primes dividing 37. Therefore, from Lemma 5.2.1, we know we only need to check the primes $v|2, 3, 19, 37$, respectively.

For $v|3$, we compute $\#E(K_v)/3E(K_v) = 27$ from Proposition 5.1.3. Then using the kernel-cokernel sequence, we know that $\dim E(K_v)/\hat{\psi}(E'(K_v)) + \dim E'(K_v)/\psi(E(K_v)) = 3$. For each isogeny, we find that points corresponding to 7 and $\sqrt{-3} \in K_v^*/K_v^{*3}$, respectively: $E(K_v)/\hat{\psi}(E'(K_v))$ is generated by

$$\mathcal{P}_{3,1} := (118097655241 \cdot \sqrt{-3}^2, 37 \cdot \sqrt{-3}^3) \text{ and } \mathcal{P}_{3,2} := (-94019126392 \cdot \sqrt{-3}^4, -34 \cdot \sqrt{-3}^5)$$

and $E'(K_v)/\psi(E(K_v))$ by

$$\mathcal{Q}_{3,3} := (77351169064, -72263420317) \text{ and } \mathcal{Q}_{3,4} := (-73260127802, -231078711691),$$

where we omit the big- O notation. Let $\mathcal{P}_{3,3} := \hat{\psi}(\mathcal{Q}_{3,3})$ and $\mathcal{P}_{3,4} := \hat{\psi}(\mathcal{Q}_{3,4})$. Now we have that $\mathcal{Q} := (0, \sqrt{-27C}) = [\mathcal{Q}_{3,3}] + 2[\mathcal{Q}_{3,4}]$ in $E'(K_v)/\psi(E(K_v))$. Thus this means that there exists an $\mathcal{M} \in E(K_v)$ such that

$$\mathcal{Q} - \mathcal{Q}_{3,3} - 2\mathcal{Q}_{3,4} = \psi(\mathcal{M}).$$

Then since $\mathcal{Q}$ is in the kernel of $\hat{\psi}$, we have

$$-\mathcal{P}_{3,3} - 2\mathcal{P}_{3,4} = [3](\mathcal{M})$$

Therefore, we have that in $E(K_v)/3E(K_v)$ that

$$[\mathcal{P}_{3,3}] = [\mathcal{P}_{3,4}]$$

in $E(K_v)/3E(K_v)$. As such, we have that only one of the points in the basis modulo $\psi(E(K_v))$ as image that corresponds to a basis element modulo $3E(K_v)$. Now suppose that there exists an $M \in E(K_v)$ such that

$$a_1\mathcal{P}_{3,1} + a_2\mathcal{P}_{3,2} + a_3\mathcal{P}_{3,3} = 3M$$

then modulo $\hat{\psi}(E'(K_v))$, we have that

$$a_1[\mathcal{P}_{3,1}] + a_2[\mathcal{P}_{3,2}] = 0.$$

This implies that $a_1 = a_2 = 0$ so that

$$a_3\mathcal{P}_{3,3} = 3M$$

Then modulo $\psi(E(K_v))$, we must have that

$$a_3[\mathcal{P}_{3,3}] = 0$$

which means that $a_1 = 0$ since $\mathcal{Q}_{3,3}$ is a basis element of $E'(K_v)/\psi(E(K_v))$. Therefore, we have found our basis for $E(K_v)/3E(K_v)$. Now we give the map for $E_v \rightarrow X_v$. We have that $x^3 + 5/4$

admits a K_v -rational solution. Therefore, we take $F = K(\sqrt[3]{5/4})$ and obtain the map $E/F \rightarrow X/F$ given by the following rational function:

$$\begin{aligned} X &:= 5/9 \sqrt[3]{5/4}^2 x \\ Y &:= 5/108 \sqrt[3]{5/4} y - 25/2 \sqrt[3]{5/4} \\ Z &:= -5/108 y - 25/2. \end{aligned}$$

Hence our map $\lambda : E_v \rightarrow X_v$ has $\lambda(\mathcal{O}) = (0, -\sqrt[3]{5/4}, 1)$. Thus, we find the following points that generate $\text{Pic}(X_v)/3\text{Pic}(X_v)$:

$$\begin{aligned} \mathcal{Q}_{3,1} &:= (0, -\sqrt[3]{5/4}) \\ \mathcal{Q}_{3,2} &:= \left(\frac{1}{4069}(a_{3,1}\sqrt{-3} + b_{3,1}), \frac{1}{4069}(c_{3,1}\sqrt{-3} + d_{3,1}) \right) \\ \mathcal{Q}_{3,3} &:= (1/91(a_{3,2}\sqrt{-3} + b_{3,2}), 1/91(c_{3,2}\sqrt{-3} + d_{3,2})) \\ \mathcal{Q}_{3,4} &:= (a_{3,3}, b_{3,3}) \end{aligned}$$

where one can find the full values $a_{3,i}, b_{3,i}, c_{3,i}$ and $d_{3,i}$ up to precision 50 in [Roz] or with truncated precision in the table below. We compute that the Hilbert symbol

$$\left(\frac{f(\mathcal{P}_i), 4}{\mathfrak{p}_3} \right) = 1$$

is trivial for each point $\mathcal{P}_{3,i}$, so that this prime contributes nothing.

For $v|2$, $4C \notin K_v^{*3}$, therefore $\#E(K_v) = 3$. It follows that $\#E(K_v)/3E(K_v) = 3$ from Proposition 5.1.3 such that the group is generated by $\mathcal{Q}_v := (0, \sqrt{C})$. We have that $3 \in K_v^{*3}$; therefore, over K , we consider the extension $F = K(\sqrt[3]{3})$, and here X admits the solution $(\sqrt[3]{3}, -1, -1)$. Therefore, there is an isomorphism from $E/F \rightarrow X/F$ defined by the rational functions

$$\begin{aligned} X &:= (-3265173504/62748517 \sqrt[3]{3}^2 x^2 - 816293376/62748517 xy - 220399211520/62748517 x \\ &\quad - 29386561536/62748517 \sqrt[3]{3} y + 7934371614720/62748517 \sqrt[3]{3}) \\ Y &:= 7346640384/62748517 \sqrt[3]{3} x^2 + 272097792/62748517 \sqrt[3]{3}^2 xy - 73466403840/62748517 \sqrt[3]{3}^2 x \\ &\quad + 29386561536/62748517 y + 7934371614720/62748517 \\ Z &:= 272097792/62748517 \sqrt[3]{3}^2 xy - 14693280768/4826809 \sqrt[3]{3}^2 x - 58773123072/62748517 y). \end{aligned}$$

Lifting the root $\sqrt[3]{3}$ into K_v implies that this is also an isomorphism $\lambda : E_v \rightarrow X_v$ with $\lambda(\mathcal{O}) = (\sqrt[3]{3}, -1, -1)$. Therefore, we find the following generators of $\text{Pic}(X_v)/3\text{Pic}(X_v)$:

$$\begin{aligned} \mathcal{Q}_{2,1} &:= (\sqrt[3]{3}, -1, -1) \\ \mathcal{Q}_{2,2} &:= \left(\frac{1}{2} \left((-55740905426565 + O(2^{50}))\sqrt{-3} - 55740905426565 + O(2^{50}) \right), \rho \right). \end{aligned}$$

We then compute that for both points $\mathcal{P}_{2,1}$ and $\mathcal{P}_{2,2}$ that Hilbert symbol has:

$$\left(\frac{f(\mathcal{P}_{2,i}), 4}{\mathfrak{p}_v} \right) = \rho^2.$$

Let $v|37$ be one of the conjugate primes. First, the element $5/3$ is a cube in K_v , so we have that $\mathcal{P}_{37,1} = (-\sqrt[3]{5/3}, 0, 1) \in X(K_v)$. Thus, there is a map $E_v \rightarrow X_v, \mathcal{O} \mapsto \mathcal{P}_{37,1}$. We have that $-4C \in K_v^{*3}$ so that $\#E(K_v)/3E(K_v) = 9$. Using the Mordell-Weil sieve, we obtain 2 points in $E(K_v)/3E(K_v)$, each one coming from one of the dual isogenies and both corresponding to the element $2 \in K_v^*/K_v^{*3}$. For each point, we compute that $f(\mathcal{P}_{19,i})$ is a unit. Since v is unramified, it follows that the Artin symbol is trivial in this case.

For $v|19$, we admit that $\sqrt[3]{5/3} \in K_v^{*3}$. So that as in the previous case, we have the point $\mathcal{P}_{19,1} = (-\sqrt[3]{5/3}, 0, 1) \in X(K_v)$ with $E_v \rightarrow X_v$ defined by $\mathcal{O} \mapsto \mathcal{P}$. This time $-4C \notin K_v^{*3}$, so that $\#E(K_v)/3E(K_v) = 3$. Thus, $\text{Pic}(X_v)/3\text{Pic}(X_v)$ is generated by 2 points, the second $\mathcal{P}_{19,2}$ being the image of the point $(0, \sqrt{C}) \in E(K_v)/3E(K_v)$. For these points, we compute that $f(\mathcal{P}_{19,i}) = u_i 19^{-1}$. Again v is unramified, so the Hilbert symbol is determined by

$$\left(\frac{19, 4}{\mathfrak{p}_v} \right) = \left(\frac{4}{\pi} \right),$$

where π is a uniformiser of $\mathcal{O}_v$. To compute the latter symbol, we can use the congruence from Proposition 3.4.4:

$$\left(\frac{4}{\pi} \right) \equiv 4^{(N(\pi)-1)/3} = 4^6 \equiv \rho^2 \pmod{\mathfrak{p}_v}$$

Letting v be replaced by its conjugate we have obtain the same computation so that

$$\left(\frac{f(\mathcal{P}_{19,i}), 4}{\mathfrak{p}_v} \right) = \rho, \left(\frac{f(\mathcal{P}_{19,i}), 4}{\bar{\mathfrak{p}}_v} \right) = \rho \implies \left(\frac{f(\mathcal{P}_{19,i}), 4}{\mathfrak{p}_v} \right) \left(\frac{f(\mathcal{P}_{19,i}), 4}{\bar{\mathfrak{p}}_v} \right) = \rho^2,$$

which determines the symbol for both primes dividing 19.

Writing the image of $\mathcal{P}$ in the local schemes as $\mathcal{P} = \alpha_{2,1}\mathcal{P}_{2,1} + \alpha_{2,2}\mathcal{P}_{2,2}$ and $\mathcal{P} = \alpha_{19,1}\mathcal{P}_{19,1} + \alpha_{19,2}\mathcal{P}_{19,2}$ and $\mathcal{P} = \beta_{19,1}\mathcal{Q}_{19,1} + \beta_{19,2}\mathcal{Q}_{19,2}$, where we account for the conjugate primes over 19. We have via Artin reciprocity that

$$\mathcal{P} \mapsto \varphi_2([\mathcal{P}])\varphi_{19}([\mathcal{P}]) = \rho^{2(\alpha_{2,1}+\alpha_{2,2})+\alpha_{19,1}+\alpha_{19,2}+\beta_{19,1}+\beta_{19,2}} = 1$$

Therefore, we have the following system:

$$\begin{aligned} \alpha_{2,1} + \alpha_{2,2} &\equiv 1(3) \\ \alpha_{19,1} + \alpha_{19,2} &\equiv 1(3) \\ \beta_{19,1} + \beta_{19,2} &\equiv 1(3) \\ 2(\alpha_{2,1} + \alpha_{2,2}) + \alpha_{19,1} + \alpha_{19,2} + \beta_{19,1} + \beta_{19,2} &\equiv 0(3) \end{aligned}$$

This implies that $4 \equiv 0 \pmod{3}$, which is clearly false. Therefore, we have a contradiction. We summarise the results of computations above to the truncated precision p^{10} , and we omit the big- O notation. We only list the points over $\mathbb{Q}_{19}$ and $\mathbb{Q}_{37}$ since they end up being the same values as in K_v .

	Basis points $\mathcal{P} \in X(K_p)$ of $\text{Pic}(X_p)/3\text{Pic}(X_p)$	$f(\mathcal{P})$	$\psi_v(\mathcal{P})$
2	$(-645\rho - 645, 1)$	$619\rho - 125$	ρ^2
	$(645, 1)$	$1015\rho - 155$	ρ^2
3	$(-43\rho - 55133, -48277 \cdot 3\rho + 51859)$	$55519 \cdot 3^2\rho - 7253$	1
	$(1724\rho - 47006, -10981\rho - 25672)$	$-6070 \cdot 3^2\rho + 39862$	1
	$(-34001 \cdot 3, 11914)$	$49823 \cdot 3^2\rho - 54476$	1
	$(0, 50659)$	$-36850 \cdot 3^2\rho + 52390$	1
19	$(5744078164122, 0)$	$1058620915515 \cdot 19^{-1}$	ρ
	$(-2981990687163, 0)$	$-2869796508377 \cdot 19^{-1}$	ρ
37	$(-3728219474759723, 2484853001284800)$	671879641876174	1
	$(-1104923496690079, -3561026092002563)$	3141170520181596	1
	$(-2189108858361115, 0)$	1967355039647726	1

6. FURTHER DISCUSSION

We review what we have done thus far. Given a nice curve over a number field K , one selects a rational function $f \in K(X)$ such that its vanishing behaviour gives rise to an abelian extension L/K via class field theory. Using the homomorphisms $\text{Pic } X_v \rightarrow G(L/K)$ arising from local reciprocity, we may be able to use Artin reciprocity to conclude there are no rational divisors in $\text{Pic } X$ of a given degree. By implementing descent methods to compute $E(K_v)/nE(K_v)$ and using functions of the form $f = \frac{x-\alpha}{x-\beta}$, we have shown that this method is quite computable and capable of producing obstructions to the Hasse principle. We did this for many examples in the case of 2-coverings, and one special example of a 3-covering.

Now, we discuss what we view as the next steps of the method discussed. A good starting point is to apply the method to more 3-coverings in general. For this, we need to implement algorithms that would be capable of computing $E(K_v)/3E(K_v)$, which seems straightforward enough in the case that $E(K_v)[3] \neq \emptyset$. If that condition fails, perhaps one may proceed as in Section 5.1 where we had no 2-torsion or as in [Top93], and find an extension of K_v that does have 3 torsion and then use the restriction map from cohomology and a norm map to determine $E(K_v)/3E(K_v)$. The next logical step is to proceed as we did in the case of 2-coverings and see how many Hasse principle obstructions can be obtained from functions $f = \frac{x+\alpha}{x+\beta}$.

It is also of interest to find an example where the method completely fails to produce an obstruction for all rational functions $f \in K(X)$. There are of course known examples of curves of genus greater than one that violate the Hasse principle, but which have no Brauer-Manin obstruction. For example see [CVV18, Theorem 4.1 p 11] where they construct a genus 3 curve with this property. In particular, we what like to know if there is an example of a curve with a Brauer-Manin obstruction, but such that this method fails to produce any such obstruction. One possible idea is to find a Brauer-Manin obstruction that cannot arise from a cyclic algebra in $\text{Br}(F)$, which would then prove the result based on what we showed in Section 4.4.

One of the goals, we had starting this thesis was to give a concrete connection between Siksek's method in [Sik01] and the method of [BS08] discussed in this thesis. Beyond the obvious connection of both methods involving a form of quadratic reciprocity, we believe that the connection lies in the Brauer-Manin obstruction. We demonstrated that the method from [BS08] could return the exact same obstruction as what Lind originally had demonstrated. Thus, we would like to explain the Brauer-Manin obstruction of the example in [Sik01, Section 4], without directly using Artin reciprocity as in this thesis. We presume that the methods should independently produce the same obstruction.

Moreover, we would also like to explore the possibility of generalising Siksek's method of generating congruences to the cubic case, which seems to have some promise from the counterexample involving Selmer's curve in 5.2. The starting point is working out the rest of the details of the cubic symbol, as in the case of the kronecker symbol from the quadratic case.

On a related note, in [CS09], Chen and Siksek use quadratic reciprocity to prove that the generalised Fermat equations $a^3 + b^3 = c^n$ have no non-trivial primitive solutions if n is divisible by $d \equiv 51, 103$, or $105 \pmod{106}$. They show that solutions (a, b, c) give rise to a hyperelliptic curve, which reduces the problem to checking if this curve is soluble. They pick the rational function $c_2 - c_1^2$ on a certain model of this curve and show using reciprocity that $(\frac{c_2 - c_1^2}{107}) = 0$ or 1 . This ultimately implies that the curve is insoluble. This paper showcases a nice crossover of the modularity method and obstructions to the Hasse principle arising via reciprocity laws.

Beyond the realm of elliptic curves, the method perhaps still has some potential for abelian varieties of genus greater than or equal to 2. In this case perhaps one would use the Mordell-Weil sieve, as we did in this paper for n -coverings. The idea, as described in [BS10], is to select the Jacobian J of the given curve and to compute $J(K_v)/nJ(K_v)$ using the n -coverings of J . This latter group is finite from Theorem 5.1.3. From [BT21, Section 7.6 p 82], the Jacobian variety is identified with the Picard group of divisor classes with degree 0 $\text{Pic}^0(X)$. Thus a significant portion of the computation would be completed.

7. POPULAR SUMMARY

Elliptic curves are a hot topic of discussion for their interesting number theoretic properties. One may think of an elliptic curve as being given by a so-called Weierstrass equation:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

with a_i rational numbers. Number theorists are often interested in finding all the rational solutions to this equation. To do this, one considers certain other equations, which we call n -coverings; one can find a solution to the Weierstrass equation if the corresponding n -covering also has a solution. If we suspect that one of these n -coverings has no solutions, we can often show this if we can demonstrate that there are no solutions in a so-called finite field, denoted $\mathbb{F}_p$, where p is a prime number. On the other hand, if a curve is soluble in rational numbers, then there are always solutions over finite fields. The idea of the Hasse principle is that the existence of solutions in finite fields would always ideally correspond to a solution in the rational numbers.

This turns out not to be the case, and it is the interest of many number theorists to describe these special curves that are everywhere locally soluble, but fail to admit rational solutions. There are a multitude of ways that one can approach such a problem. In this thesis, we take interest in a specific method that uses a generalisation of the law of quadratic reciprocity and rational functions. It is sometimes capable of producing a conflict that results in a lack of points.

We produced an algorithm based on the latter method that is able to check if the certain functions can produce such a conflict. The algorithm is decently fast and only takes a few minutes to produce hundreds of potential functions. It then checks if these functions give rise to an obstruction to the Hasse principle.

APPENDIX

A. SCHEME THEORY

A.1 Basic notions

For the sake of completeness, we comment on the usage of schemes and varieties. We provide motivation for important concepts related to varieties, but in the language of schemes.

Given a ring R , we consider the topological space $\text{Spec } R$ of prime ideals in R . We may associate a sheaf of groups $\mathcal{O}_{\text{Spec } R}$ that satisfies $\mathcal{O}_{\text{Spec } R}(\text{Spec } R) = R$ and $\mathcal{O}_{\text{Spec } R}(D_f) = R[1/f]$. Hence we may define:

Definition. An **affine scheme** is a tuple $(\text{Spec } R, \mathcal{O}_{\text{Spec } R})$. A morphism of affine schemes, usually written as $\text{Spec } R \rightarrow \text{Spec } S$, is a map of topological spaces such that there is also a ring map $S \rightarrow R$.

A morphism of rings $R \rightarrow S$ induces a natural map from $\text{Spec } S \rightarrow \text{Spec } R$. There is in fact a contravariant functor from **Ring** $\rightarrow$ **Aff** given by mapping $R \mapsto \text{Spec } R$ by [Mum04, Proposition I.3.2 p 21]. It is an equivalence of categories, i.e. is fully faithful and essentially surjective.

Definition. Then a **scheme** is a pair $(X, \mathcal{O}_X)$ such that X is a topological space and $\mathcal{O}_X$ is a sheaf of groups that satisfies the following: (1) for each point $\mathcal{P} \in X$, the stalks $\mathcal{O}_{X, \mathcal{P}} := \varinjlim_{U \ni \mathcal{P}} \mathcal{O}_X(U)$ are local rings and (2) there exists an open covering $\bigcup_{i \in I} U_i$ such that there are rings R_i with $(U_i, \mathcal{O}_{U_i}) \cong (\text{Spec } R_i, \mathcal{O}_{\text{Spec } R_i})$ that glue together accordingly.

We may map schemes $f : X \rightarrow S$ by first mapping their underlying topological spaces continuously. Then we define the pushforward of a sheaf of abelian groups $\mathcal{F}$ of on X : given an open subset $V \subseteq S$, let $f_*\mathcal{F}(V) := \mathcal{F}(f^{-1}V)$. Then a morphism of schemes $f : X \rightarrow S$ is given by a continuous mapping $X \rightarrow S$ as topological spaces together with a natural transformation $f^\# : \mathcal{O}_S \rightarrow f_*\mathcal{O}_X$ such that for any open $V \subseteq S, x \in f^{-1}V, a \in \mathcal{O}_S(V)$, if $a(f(x)) = 0$ then $(f^\#(a))(x) = 0$. We call a scheme X together with a morphism of schemes $X \rightarrow S$ an **S -scheme** and write X/S . The morphisms $X \rightarrow Y$ correspond to commutative triangles so that the map $X \rightarrow S$ factors through $X \rightarrow Y$. In particular, given a field k , a scheme is called a **k -scheme** if there exists a morphism $X \rightarrow \text{Spec } k$.

Definition. A morphism of schemes $f : X \rightarrow S$ is called a **closed immersion** if for all affine opens $V \subseteq S$, $f^{-1}V$ is affine and the map $\mathcal{O}_S(V) \rightarrow \mathcal{O}_X(f^{-1}V)$ is surjective. A scheme U is called a **closed subscheme** of X if there exists a closed immersion $U \hookrightarrow X$.

Definition. A scheme X is called **reduced** if for all $x \in X$, $\mathcal{O}_{X, x}$ is a reduced ring, i.e. admits no nilpotents. A scheme X is also called **irreducible** if the underlying topological space is irreducible, which means that if $X = Z_1 \cup Z_2$ with Z_1, Z_2 closed subsets, then $X = Z_1$ or $X = Z_2$. A scheme is called **integral** if and only if it is a connected topological space and for each point $x \in X$, the stalk $\mathcal{O}_{X, x}$ is an integral domain. It can be shown that a scheme is integral if it is irreducible and reduced [Har77, Proposition II.3.1 p 82].

Definition. A scheme X is called **quasi-compact** if for every open covering, there exists a finite subcovering. We call a scheme morphism $f : X \rightarrow S$ **finite type** if for all affine opens $V \subseteq S$, $f^{-1}(V)$ is a quasi-compact topological space and every affine open $U \subseteq f^{-1}(V)$ implies $\mathcal{O}_X(U)$ is a finite type $\mathcal{O}_S(V)$ -algebra, i.e. there exists a variables $x_1, \dots, x_s$ such that $\mathcal{O}_S(V)[x_1, \dots, x_s] \rightarrow \mathcal{O}_X(U)$ is a surjective map. A scheme $f : X \rightarrow S$ is called **universally closed** if for any other S -scheme $Y \rightarrow S$, we have that the projection map $X \times_S Y \rightarrow Y$ is closed. Finally, we call a scheme **proper** if it is finite type, separated, and universally closed.

The scheme $\mathbb{P}^n$, referred to as **projective n -space**, can be obtained from gluing the schemes $\text{Spec } R_i = \text{Spec } \mathbb{Z}[X_{0,i}, \dots, \widehat{X_{i,i}}, \dots, X_{n,i}]$, where $\widehat{X_{i,i}}$ is omitted and $0 \leq i \leq n$, along the isomorphism $R_i[X_{j,i}^{-1}] \rightarrow R_j[X_{i,j}^{-1}]$ with $X_{l,i} \mapsto X_{l,j}X_{i,j}^{-1}$ and $X_{j,i} \mapsto X_{i,j}^{-1}$. Note that this makes $\mathbb{P}^n = \text{Proj}(\mathbb{Z}[x_0, \dots, x_n])$, and the points correspond to the homogeneous prime ideals of $\mathbb{Z}[x_0, \dots, x_n]$ excluding the irrelevant one: $(x_0, \dots, x_n)$ [Vak25, 4.5.7 p 150]. We also attach to $\mathbb{P}^n$ the sheaf $\mathcal{O}(1)$ characterised by $\Gamma(\mathbb{P}^n, \mathcal{O}(1)) = \mathbb{Z}x_0 \oplus \dots \oplus \mathbb{Z}x_n$ such that for all $y \in \mathbb{P}^n$ the global sections $x_0, \dots, x_n$ always have at least one of the germs $x_{0,y}, \dots, x_{n,y}$ generating $\mathcal{O}(1)_y$ [Vak25, 15.1 and 15.1.D pp 421 and 423]. More generally, the sheaves $\mathcal{O}(m)$ have the property that they are generated by global sections of degree m [Vak25, 15.1.F p 424]. In particular, we have that $\mathbb{Z}[x_0, \dots, x_n] = \bigoplus_{n \geq 0} \Gamma(\mathbb{P}^n, \mathcal{O}(n))$ returns the usual graded ring.

Taking the fibre product with another scheme S yields $\mathbb{P}_S := \mathbb{P} \times_{\mathbb{Z}} S$. For a field k , we denote $\mathbb{P}_k := \mathbb{P} \times_{\mathbb{Z}} k$. The k -scheme $\mathbb{P}_k \rightarrow \text{Spec } k$ turns out to be proper by [Har77, Theorem II.4.9 p 103]. A scheme X is called a **projective S -scheme** if there exists a closed immersion $X \rightarrow \mathbb{P}_S$. Since closed immersions are proper, we also obtain that projective schemes in general are also proper S -schemes by the same theorem.

Now we define a few more important but intuitive notions:

A.2 Schemes and Varieties

Now, we state the relationship between schemes over an algebraically closed field $\bar{k}$ and $\bar{k}$ -varieties. Namely we admit the following equivalence of categories from [Mum04, Theorem II.2 p 123]:

$$\left\{ \begin{array}{l} \text{Integral schemes of finite type} \\ \text{over } \bar{k}, \text{ together with } \bar{k}\text{-morphisms} \end{array} \right\} \longleftrightarrow \{\text{Varieties over } \bar{k}, \text{ with morphisms of varieties}\}$$

It should be noted that one can loosen or strengthen the idea of what a variety is by either adding or dropping conditions on the left-hand side. The classic example is the notion of separatedness, which originally was what Grothendieck included in the definition in EGA. On the idea of loosening restrictions, Liu's definitions of affine and projective varieties mean that an algebraic variety is one that is a Noetherian scheme [LE02, Definition 3.47 and Remark 3.48 p 56].

Given X a projective k -scheme, using the results of scheme theory, there is a unique homogeneous ideal of $S := k[x_0, \dots, x_n]$ associated to X , which generalises the homogeneous Nullstellensatz over algebraically closed fields. Here we have that $X \cong \text{Proj}(S/I)$ by [Har77, Corollary II.5.16 pp 119-120 and Exercise II.5.10 p 125] for the notion of uniqueness.

Let X/K be a variety for K a non-algebraically closed field. Then we know that X is an integral scheme (i.e. reduced and irreducible) with $X \rightarrow \text{Spec } K$ a finite type morphism from the above equivalence. In [Sil09, Sections I.1 and I.2], there is the assumption that an affine (or projective) variety X/K has that its associated (homogeneous) ideal $I(X)$ is a prime ideal with K -rational

generators in $\bar{K}[X]$. To translate this to scheme theory, one requires that X/K is **geometrically integral**, i.e. that $\bar{X} = X \times_{\text{Spec } K} \text{Spec } \bar{K}$ is also an integral scheme, see [Vak25, Definition 3.2.8 p 90]. This follows since if we are given a ring morphism $A \rightarrow B$ and a graded A -algebra S , then $\text{Proj } S \times_{\text{Spec } A} \text{Spec } B \cong \text{Proj } (S \otimes_A B)$ [Vak25, Exercise 17.2A], which holds in the affine case and accordingly here via a gluing argument. We have that $K[X]/I \otimes_K \bar{K} \cong \bar{K}[X]/I$ so that we require that I is a prime ideal in $\bar{K}[X]$, reflecting the idea of Silverman.

The notion of properness serves as a generalisation of complete varieties. We recall a variety X/K is complete if for any other variety Y/K , the (fibre) product $X \times Y \rightarrow Y$ is a closed map. For our purposes, we may call a K -scheme **complete**, if for all K -varieties Y , then $X \times_K Y \rightarrow Y$ is a closed map.

Therefore, going forward, we will be particularly interested in projective K -schemes X that are non-singular, complete and geometrically integral over a number field K .

A.3 Points

Definition. Let $X \rightarrow \text{Spec } K$ be a K -scheme. For any ring R , we let $X(R)$ denote the set of morphisms $\text{Spec } R \rightarrow \text{Spec } K$ that factor through X , i.e. satisfy

$$\begin{array}{ccc} \text{Spec } R & \xrightarrow{\quad} & X \\ & \searrow \quad \swarrow & \\ & \text{Spec } K & \end{array}$$

We call the set $X(R)$ the **R -valued points** of X/K .

We characterise the K -rational points of our scheme:

$$X(K) \cong \left\{ \text{Points } \mathcal{P} \in X \text{ together with an injection } K(\mathcal{P}) \hookrightarrow K \right\}$$

from [Mum04, p 116]. Now suppose there is an embedding $K \hookrightarrow L$, with L a field. We characterise the L -points of a scheme X/K in the following bijection:

$$X(L) \cong \{ \mathcal{P} \in X \mid \text{there exists } K \rightarrow L \text{ factoring through } K(\mathcal{P}) \}$$

by [Mum04, p 116]. Now we recall from [Har77, Proposition II.7.1 p 150], there is a natural transformation on the Y -valued points of projective space and certain invertible sheaves:

$$\begin{aligned} \mathbb{P}_K^n(Y) &\cong \{ \text{Invertible sheaves } \mathcal{L} \text{ over } Y \text{ such that for any } y \in Y, \\ &\quad \mathcal{L}_y \text{ is generated by } (n+1) \text{ sections } s_{y,n}, \dots, s_{y,n} \} / \sim \end{aligned}$$

Using this natural isomorphism, we recover many geometric results of variety theory in the language of schemes. In particular for $Y = \text{Spec } K$, we have that $\mathcal{L}$ corresponds to a vector space K^n and basis elements $s_1, \dots, s_n \in K$. Thus, we may for example recover the classic Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$: we obtain a map $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$ from the above since $p_1^*(\mathcal{O}(1)) \otimes p_2^*(\mathcal{O}(1))$ is an invertible sheaf over $\mathbb{P}^1 \times \mathbb{P}^1$ generated by 4 sections, see [Har77, Exercise 5.11].

Let K be a number field and $\bar{K}$ an algebraic closure. Then for a scheme $X \rightarrow \text{Spec } K$ and $\bar{X} \rightarrow \text{Spec } \bar{K}$, there are series of set bijections, from [Mum04, p 117]:

$$X(\bar{K}) \cong \bar{X}(\bar{K}) \cong \left\{ \text{closed points of } \bar{X} \right\} \tag{A.1}$$

Therefore, we see that all the closed points on a K -scheme are geometric points. From this bijection, we also obtain a natural action of G_K on the closed points of X . Following [LE02, pp 92-93], for any extension L/K and K -automorphism $\sigma \in G(L/K)$, there is a scheme isomorphism $\text{Spec } \sigma : \text{Spec } L \rightarrow \text{Spec } L$. Thus, if we have a map $P : \text{Spec } L \rightarrow X$, then the action sends this map to $P \circ \text{Spec } \sigma : \text{Spec } L \rightarrow X$. Therefore, there is a group action of $G(L/K)$ on $X(L)$. Since, $\text{Div } X$ is generated by the closed points $\mathcal{P}$, we obtain a natural induced action on this group as well.

A.4 Schemes Normalisation

We prove for any irreducible reduced (i.e. integral) scheme that there exists some scheme X^v that is **normal**, i.e. for all $x \in X^v$, $\mathcal{O}_{X^v, x}$ is integrally closed. This scheme will be birationally equivalent to our scheme X , i.e. it maps the generic point $\bar{\eta} \in X^v$ to the generic point $\eta \in X$. In particular, it must extend the canonical map $\text{Spec } \kappa(\eta) \rightarrow X$. We have the following proposition, see [Hartshorne, Exercise II.3.8]

Proposition A.4.1. - Let X be an irreducible reduced scheme (i.e. integral). Then there exists a scheme X^v that is normal such that $X \rightarrow X^v$ is birational.

A classical and relevant example of resolving the singularities of a curve, we consider a hyperelliptic curve: let $f \in K[X]$ be a homogenous polynomial of degree 4 such that f has no repeated roots. Then

$$C : y^2 = f(x)$$

defines a projective curve with a singularity at $(0 : 1 : 0)$ after taking the usual projective closure. However, as in [Sil09, p. II.2.5.1], it is shown that C can be glued together with the curve

$$C' : v^2 = g(u)$$

to obtain a new projective curve C^v , where $x^4 g(1/x) = f(x)$ and $\psi = [\frac{1}{x}, \frac{y}{x^2}]$ is the gluing map. We obtain an open immersion of $C \rightarrow C^v$. We call the points on C^v with $u = 0$ the points at infinity. Then one verifies that the points are smooth: if $f = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$, then the corresponding ideal in $\text{Spec } (K[u, v]/(v^2 - g(u)))$ is $(u, v^2 - a_4)$ if $a_4 \notin K^{*2}$. Then in the local ring $\mathcal{O}_{C', \mathcal{P}} = K[u, v]_{(u, v^2 - a_4)} / (v^2 - g(u)) K[u, v]_{(u, v^2 - a_4)}$ we have that $v^2 - a_4 = u(a_3 + a_2 u + a_1 u^2 + a_0 u^3)$, with $(a_3 + a_2 u + a_1 u^2 + a_0 u^3) \notin (u, v^2 - a_4)$. Therefore, we have that the maximal ideal $\mathfrak{m}_{C', \mathcal{P}} = (u)$, and $\mathcal{O}_{C', \mathcal{P}}$ is therefore a discrete valuation ring by [AM69, Proposition 9.2. p 94].

A.5 Scheme Models

For the following material, one may refer to Stacks 55.8. Let K be a field and R a discrete valuation ring with fraction field K and maximal ideal $\mathfrak{m} \in \text{Spec } R$. We also denote $\kappa := R/\mathfrak{m}$. We briefly recall that there exist an ideal $\eta \in \text{Spec } R$ called the generic point such that the localisation $R_\eta \cong K$. Let $X \rightarrow \text{Spec } K$ be a K -scheme.

First we briefly, define what a flat scheme morphism is from [Sta25, tag 01U3]:

Definition. A morphism of schemes $f : X \rightarrow S$ is **flat** at $x \in X$, if $\mathcal{O}_{X, x}$ is a flat $\mathcal{O}_{S, y}$ -algebra where $y := f(x)$. We say $f : X \rightarrow S$ is **flat** if this holds for all $x \in X$.

If we have a morphism of affine varieties $f^* : \text{Spec } A \rightarrow \text{Spec } B$, then this criterion simply means that $A_{f^{-1}\mathfrak{p}}$ is a flat $B_{\mathfrak{p}}$ -algebra. For example, if we take $A = \mathbb{Z}[i]$ and $B = \mathbb{Z}$, then we have that $\mathbb{Z}[i]_{\mathfrak{p}_2}$ is a flat $\mathbb{Z}_{(2)}$ -algebra. One sees that $\mathbb{Z}[i]$ is a flat $\mathbb{Z}$ -algebra since as a $\mathbb{Z}$ -module $\mathbb{Z}[i] \cong \mathbb{Z} \oplus \mathbb{Z}$,

so that by [AM69, Proposition 3.10] $\mathbb{Z}[i]_{(2)}$ is a flat $\mathbb{Z}_{(2)}$ -algebra. Then $\mathbb{Z}[i]_{\mathfrak{p}_2}$ is a localisation of $\mathbb{Z}_{(2)}$, so it must be a flat $\mathbb{Z}_{(2)}$ -algebra by [AM69, Corollary 3.6] and accordingly via a transitive property a flat $\mathbb{Z}$ -algebra.

Definition. A morphism of schemes $f : X \rightarrow S$ is called **smooth** at $x \in X$ if

1. f is flat at x
2. for $y := f(x)$, $X \times_S \kappa(s) \rightarrow \kappa(s)$ is regular, where $\kappa(s) := \mathcal{O}_{S,s}/\mathfrak{m}_{S,s}$.

We say $f : X \rightarrow S$ is **smooth** if the above holds for all $x \in X$.

By [LE02, Theorem 4.3.36 p 142], we have that if $f : X \rightarrow S$ is smooth, then X is regular. If $S = \operatorname{Spec} K$ for a K a perfect field, we have that the notion of smooth and regular are equivalent by [LE02, Corollary 4.3.33 p 142]. If $S = \operatorname{Spec} \mathcal{O}_K$, these notions are no longer equivalent. In particular, we will see that the reduction of an integral model of a curve might have singularities. This brings us now to the definition of a model of a curve over a local field.

Definition. A **model** for X is an R -scheme $\mathcal{X} \rightarrow \operatorname{Spec} R$ that is a flat finite type morphism such that there exists an isomorphism with the generic fibre: $X \rightarrow \mathcal{X}_K = \mathcal{X} \times_{\operatorname{Spec} R} \operatorname{Spec} K$. We say that $\mathcal{X}$ is a **proper model** if $\mathcal{X} \rightarrow \operatorname{Spec} R$ is a proper morphism. We also call $\mathcal{X}_\kappa = \mathcal{X} \times_{\operatorname{Spec} R} \operatorname{Spec} \kappa$ the **special fibre**.

Note that the names “generic fibre” and “special fibre” come from the fact that $\mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} K_v \cong f^{-1}\{\eta\}$ and $\mathcal{X} \times_{\operatorname{Spec} R} \operatorname{Spec} \kappa \cong f^{-1}\{\mathfrak{m}\}$, where $f : \mathcal{X} \rightarrow \operatorname{Spec} R$ is the canonical morphism (which follows from a more general scheme theory fact). As a topological space, we may decompose $\mathcal{X} = f^{-1}\{\eta\} \cup f^{-1}\{\mathfrak{m}\} \cong X \cup \mathcal{X}_\kappa$. Then we also have that $\mathcal{X}_\kappa \rightarrow \operatorname{Spec} \kappa$ is also a projective scheme.

By definition we have that $\mathbb{P}_K^n = \mathbb{P}_R^n \times_{\operatorname{Spec} R} \operatorname{Spec} K$ and that $\mathbb{P}_S^n \rightarrow S$ for any scheme S is proper. Thus, we have that $\mathbb{P}_R^n$ is a model for $\mathbb{P}_K^n$. From the Stacks Project, we have the following essential lemma:

Lemma A.5.1. - If $X_1 \rightarrow X_2$ is closed immersion of finite type K -schemes and $\mathcal{X}_2$ is a model for X_2 , then the scheme theoretic image $\mathcal{X}_1$ of $f : X_1 \rightarrow \mathcal{X}_2$ is a model for X_1 , i.e. $\mathcal{X}_1$ is the smallest closed subscheme satisfying the following diagram:

$$\begin{array}{ccc} X_1 & \xrightarrow{\quad} & \mathcal{X}_1 \\ & \searrow & \downarrow \\ & & \mathcal{X}_2 \end{array}$$

Proof. see [Sta25, tag 0C2S]. □

Putting this lemma to use, we have that for any projective K -scheme X , there exists a closed immersion $X \rightarrow \mathbb{P}_K^n$ for some $n \in \mathbb{Z}$. Therefore, we have that the scheme-theoretic image $\mathcal{X}$ of $X \rightarrow \mathbb{P}_R^n$ defines a model for X that is also projective. We will call $\mathcal{X}$ a **projective model** for X . It is also worth mentioning that $X \rightarrow \mathcal{X}$ is an open immersion since $\operatorname{Spec} K \rightarrow \operatorname{Spec} R$ is an open immersion, as it maps $(0) \mapsto \eta = V(\mathfrak{m})^c$, and base change preserves open immersions [LE02, Section 10.4 pp 288-289]. Now we suppose that $X \rightarrow \operatorname{Spec} K$ is an irreducible nonsingular curve. Then from the previous comments, we know that the functions fields are isomorphic.

This brings us to a discussion of good reduction of a scheme.

Definition. We say that X has **good reduction** if there exists a projective model $\mathcal{X} \rightarrow \operatorname{Spec} R$ such that the special fibre $\mathcal{X}_\kappa$ is nonsingular. Otherwise, it has **bad reduction**.

When κ is a finite field, we may think of the special fibre as reduction of the integral curve modulo a prime. This is the familiar theory seen in elliptic curves where one says a curve E/K has bad reduction at v if the special fibre of its minimal model $\tilde{E}$ has singular points. Therefore, in general, a given projective model $\mathcal{X}$ may have the property that $X_\kappa = X \times_R \kappa$ is not regular, but there is a nonempty scheme $\mathcal{C} \rightarrow \operatorname{Spec} R$ (not necessarily an elliptic curve) that does indeed have $\mathcal{C}_\kappa$ is nonsingular by [Sil09, Theorem C.15.1 p 446].

We now need a small lemma. We recall from Galois theory that any extension of finite fields is separable [DF04, Proposition 13.37 p 549]. Hence if L/K is a finite extension of p -adic fields, then the extension of residue fields κ_L/κ_K is separable. From [Neu99, Lemma II.10.4 p 178] we then admit the following useful fact (for the appendix):

Lemma A.5.2. Let L/K be an extension of p -adic fields. Then $\mathcal{O}_L \cong \mathcal{O}_K[x]$ for some $x \in \mathcal{O}_L$, i.e. it is a monogenous.

Now we specialise the situation by taking K to be a number field and working over the completion K_v for $v \in \mathfrak{M}(K)$, so that $R = \mathcal{O}_v$. Now let X be an irreducible K_v -curve and $f \in K_v(X)$. Then pick a projective (and accordingly proper) model $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_v$. We accordingly admit the following proposition.

Proposition A.5.3. - Let L/K_v be an extension of p -adic fields and let $\mathcal{O}_L \subseteq L$ be its ring of integers. There is a bijection $X(L) \rightarrow \mathcal{X}(\mathcal{O}_L)$, so every section $\operatorname{Spec} L \rightarrow X$ extends uniquely to a section $\operatorname{Spec} \mathcal{O}_L \rightarrow \mathcal{X}$.

Proof. Suppose we have a section $\operatorname{Spec} \mathcal{O}_L \rightarrow \mathcal{X}$. From Lemma A.5.2 we have that $\mathcal{O}_L = \mathcal{O}_v[x]$ for some $x \in \mathcal{O}_L$, and it follows that $\mathcal{O}_L \otimes_{\mathcal{O}_v} K_v \cong L$. Thus, since X is isomorphic to the generic fibre of $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_v$ implies that there is a map $\operatorname{Spec} L \rightarrow X$.

On the other hand, let $\operatorname{Spec} L \rightarrow X$. Again we have $X \cong \mathcal{X} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} K_v$; thus there is a morphism $\operatorname{Spec} L \rightarrow \mathcal{X}$. Since $\mathcal{X}$ is $\mathcal{O}_v$ -scheme, we have the following diagram:

$$\begin{array}{ccc} \operatorname{Spec} L & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ \operatorname{Spec} \mathcal{O}_L & \longrightarrow & \operatorname{Spec} \mathcal{O}_v \end{array}$$

As $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_v$ is a proper morphism, we have by the valuative criterion of properness [Har77, Theorem II.4.7] that there is precisely one morphism $\operatorname{Spec} \mathcal{O}_L \rightarrow \mathcal{X}$ making the diagram commute. $\square$

This allows us to define a reduction map. For each closed point $\mathcal{P} \in X$, it follows from this proposition that there is a unique closed point $\tilde{\mathcal{P}} \in \mathcal{X}$ such that the map $\operatorname{Spec} \mathcal{O}_L \rightarrow \mathcal{X}$ has $(0) \mapsto \tilde{\mathcal{P}}$ and $\mathfrak{P}_L \mapsto \tilde{\mathcal{P}}$, and there are ideals $\mathfrak{m}_{\mathcal{X}, \tilde{\mathcal{P}}} \in \operatorname{Spec} \mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}$ and $\mathfrak{p}_{\mathcal{X}, \tilde{\mathcal{P}}} \in \operatorname{Spec} \mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}$ and isomorphisms $\mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}/\mathfrak{m}_{\mathcal{X}, \tilde{\mathcal{P}}} \cong L$ and $\mathcal{O}_{\mathcal{X}, \tilde{\mathcal{P}}}/\mathfrak{p}_{\mathcal{X}, \tilde{\mathcal{P}}} \cong \mathcal{O}_L$. We also have for any open $U \subseteq X$ that if $\tilde{\mathcal{P}} \in U$ then

$\mathcal{P} \in U$ as well since $\mathrm{Spec} \mathcal{O}_v \rightarrow \mathcal{X}$ is a continuous map. Moreover, as a closed immersion, we must have that the closure of $\mathcal{P}$ is $\{\mathcal{P}, \tilde{\mathcal{P}}\}$.

We also have that we can always reduce any given point $\mathcal{P} \in \mathcal{X}$. Let L/K_v be a finite extension and $\kappa_L = \mathcal{O}_L/\mathfrak{P}_L$. We see that any section $\mathrm{Spec} \mathcal{O}_L \rightarrow \mathcal{X}$ yields a section $\mathrm{Spec} \kappa_L \rightarrow \mathcal{X}$ via the ring map $\mathcal{O}_L \rightarrow \kappa_L$. Then $\mathcal{X}_\kappa(\kappa_L) \cong \mathcal{X}(\kappa_L)$ via the universal property of the fibre product. Therefore, each closed point in $\mathcal{X}$ corresponds to a closed κ_L -point of the special fibre $\mathcal{X}_\kappa$ via a map $\mathcal{X}(\mathcal{O}_L) \rightarrow \mathcal{X}_\kappa(\kappa_L)$.

The scheme $\mathcal{X}$ is irreducible. Indeed we have that $X \cong U \subseteq \mathcal{X}$ with $\mathcal{X} = U \cup Z$ for Z closed corresponding to the special fibre. Hence if $\mathcal{X} = Z_1 \cup Z_2$ decomposes into two closed subsets, then it follows that $Z_i \neq Z$ and $Z_i \cap U \neq \emptyset$. Hence, we must have that for example $Z_1 \cap U = U$, and we conclude that $\mathcal{X}$ is irreducible. Therefore, there is a function field $K(\mathcal{X}) \cong \mathcal{O}_{\mathcal{X}, \eta}$ also associated to $\mathcal{X}$, where we view the image of η as the generic point of $\mathcal{X}$, and there is an isomorphism $K(\mathcal{X}) \cong K(X)$. Hence, we may view $f \in K(X)$ as an element of $K(\mathcal{X})$.

Proposition A.5.4. If $X \rightarrow \mathrm{Spec} K$ is a projective curve and $\mathcal{X} \rightarrow \mathrm{Spec} R$ is a projective model, then a point $\mathcal{P} : \mathrm{Spec} K \rightarrow X$ with coordinates $(x_0 : \cdots : x_n)$ reduces to $\tilde{\mathcal{P}} : \mathrm{Spec} \mathcal{O}_{K(\mathcal{P})} \rightarrow \mathcal{X}$ with coordinates $(\tilde{x}_0 : \cdots : \tilde{x}_n)$ with $\tilde{x}_i \in \kappa(\tilde{\mathcal{P}})$.

Proof. See [LE02, Corollary 10.1.34 p 468]. □

B. FIBRED SURFACES AND NÉRON MODELS

In this chapter we delve deeper in projective models of our curves, which ties into the theory of fibred surfaces. The following results are found in [LE02], which contains a deep exposition on the subject.

Definition. Let S be a Dedekind domain. If $\mathcal{X} \rightarrow \operatorname{Spec} S$ is a 2-dimensional integral, projective and flat S -scheme, then we call X a **fibred surface**.

Let $\eta \in \operatorname{Spec} S$ be the generic point, so $\mathcal{X}_\eta := \mathcal{X} \times_{\operatorname{Spec} S} \operatorname{Spec} \kappa(\eta)$ is a scheme over the fraction field $K := Q(S) = \kappa(\eta)$. For each closed point $s \in \operatorname{Spec} S$, put $\kappa(s) := S/s$, and we let $\mathcal{X}_s := \mathcal{X} \times_{\operatorname{Spec} S} \operatorname{Spec} \kappa(s)$. We summarise the important results for use in the following theorem:

Theorem B.0.1. If $\mathcal{X} \rightarrow \operatorname{Spec} S$ is a fibred surface then the following hold:

1. If η is the generic point of $\mathcal{X}$, then $\mathcal{X}_\eta$ is an integral scheme of finite type over the fraction field $Q(S)$ of dimension 1. Similarly, for each closed point $s \in \operatorname{Spec} S$, the closed fibre $\mathcal{X}_s$ is a projective scheme of finite type over $\kappa(s)$ of dimension 1.
2. The prime divisors Z of $\mathcal{X}$ are either given by $Z = \overline{\{P\}}$ for $P \in \mathcal{X}_\eta$ closed or $Z = Y$ with Y an irreducible component of a closed fibre X_s .
3. If Y is an irreducible component of $\mathcal{X}_s$, then there is a unique valuation $\operatorname{ord}_Y : K(\mathcal{X})^* \rightarrow \mathbb{Z}$ that extends the valuation $v_s : K(S)^* \rightarrow \mathbb{Z}$.
4. If $\mathcal{P} \in X$ is a closed point, then $Z = \overline{\{\mathcal{P}\}}$ has valuation $\operatorname{ord}_Z : K(\mathcal{X})^* \rightarrow \mathbb{Z}$ such that $\operatorname{ord}_Z(\mathcal{O}_{S,s}) = 0$

Proof. For (1), this is from [LE02, Lemma 8.3.3 p 348].

For (2), [LE02, Proposition 8.3.4 p 349].

For (3), since Y is a prime divisor of $\mathcal{X}$, there is a valuation $\operatorname{ord}_Y : K(\mathcal{X})^* \rightarrow \mathbb{Z}$ by [Har77, Section II.6 pp 129-130]; the discrete valuation ring $\mathcal{O}_Y$ associated to the valuation by definition satisfies $\mathcal{O}_Y = \mathcal{O}_{\mathcal{P}}$ for $\mathcal{P} \in Y \subseteq \mathcal{X}$ the generic point. It follows that there is a scheme morphism $\operatorname{Spec} \mathcal{O}_Y \rightarrow \operatorname{Spec} \mathcal{O}_{S,s}$, and in particular that $\mathcal{O}_Y$ dominates $\mathcal{O}_{S,s}$, i.e. $\mathcal{O}_{S,s} \subseteq \mathcal{O}_Y$ and $\mathfrak{m}_{S,s} \subseteq \mathfrak{m}_{Y_i}$. As such, we conclude that ord_Y is valuation that extends v_s .

(4) The generic point of Z is $\mathcal{P}$, and the valuation is the same as the usual valuation $\operatorname{ord}_{\mathcal{P}} : K(X)^* \rightarrow \mathbb{Z}$. This valuation sends $x \in K(S)^*$ to 0, so that the statement holds. $\square$

Given a projective model $\mathcal{X} \rightarrow \operatorname{Spec} R$ for the K -scheme X , we now prove that $\mathcal{X}$ is a fibred surface.

Lemma B.0.2. Let R be a DVR and K its field of fractions. Let $X \rightarrow \operatorname{Spec} K$ be a complete, regular curve with projective model $\mathcal{X} \rightarrow \operatorname{Spec} R$. Then we have that $\mathcal{X}$ is a fibred surface.

Proof. Using [LE02, Remark 5.6 p 69], which states that the dimension of a scheme is the supremum of the dimensions of an open covering, together with the fact that every affine scheme has $\dim \operatorname{Spec} A = \dim A$ by [LE02, Proposition 5.8], implies it suffices to show that $\sup_{\mathcal{P} \in \mathcal{X}} (\dim \mathcal{O}_{\mathcal{P}}) = 2$.

Let $\mathcal{P} \in \mathcal{X}$ be a closed point coming from the closed point $\mathcal{P}' \in X$ by Proposition A.5.3. Then we have that $\mathcal{O}_{\mathcal{P}'} \cong \mathcal{O}_{X, \mathcal{P}'}$ and so

$$\mathcal{O}_{\mathcal{P}'} \cong \mathcal{O}_{\mathcal{P}} \otimes_R K.$$

However, using that $\mathcal{O}_{\mathcal{P}}$ is a finite type R -module, we have that $\mathcal{O}_{\mathcal{P}} \cong R[x_0, \dots, x_l]/I$ for some finitely generated ideal I . Thus, $\mathcal{O}_{\mathcal{P}'}$ is a DVR, in particular it has dimension 1 so that $\mathcal{O}_{\mathcal{P}}$ must then have dimension 2 using that $\mathcal{O}_{\mathcal{P}}$ is a flat R -algebra. Finally, if $\mathcal{P}$ is in the special fibre, then we have a map $\mathcal{O}_{\mathcal{P}} \rightarrow \mathcal{O}_{\tilde{\mathcal{P}}} = \mathcal{O}_{\mathcal{P}}/\pi\mathcal{O}_{\mathcal{P}}$ and so it follows that $\dim \mathcal{O}_{\mathcal{P}} = 2$ since $\mathcal{X}_k$ has dimension 1. Therefore, we conclude that the dimension of $\mathcal{X}$ is 2 since for all points $\mathcal{P} \in \mathcal{X}$ that $\dim \mathcal{O}_{\mathcal{P}} \leq 2$, which it obtains at least once. $\square$

Néron Models of Elliptic Curves

Let E/K be an elliptic curve over a number field K and consider the product $E_v = E \times_{\operatorname{Spec} K} \operatorname{Spec} K_v$. If we let $C_v \rightarrow \operatorname{Spec} \mathcal{O}_v$ be a projective model of E_v , which exists by A.5.1 and satisfies $C_v \otimes_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} K_v \cong E_v$, and normalise C_v , then we obtain a regular projective scheme $\mathcal{E}_v$ such that $C_v \rightarrow \mathcal{E}$ is birational as in Proposition A.4.1. Therefore, we have that $E_v \cong \mathcal{E}_v \otimes_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} K_v$.

These models $\mathcal{E}_v$ are not necessarily smooth. For this, we need the notion of the Néron model of an elliptic curve. This is a smooth $\mathcal{O}_v$ -scheme $\mathcal{N}$ such that $\mathcal{N} \times_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} K_v \cong E_v$ and satisfies $\mathcal{N}(\mathcal{X}) \rightarrow E_v(X)$ is a bijection for any fibred surface $\mathcal{X}$. We have the following result then from [Sil09, Theorem C.15.1 p 446]:

Theorem B.0.3. Let $\mathcal{E}_v$ be a regular projective model of E_v/K_v , then $\mathcal{E}_v$ is 2-dimensional scheme. And there exists a unique minimal subscheme $\mathcal{X}_v \subseteq \mathcal{E}_v$ obtained by discarding all the the singular points from special fibre $C \otimes_{\operatorname{Spec} \mathcal{O}_v} \operatorname{Spec} \mathcal{O}_v/\mathfrak{p}_v$. The model $\mathcal{X}_v$ is called the **Néron minimal model** and it satisfies two important properties:

- (1) $\mathcal{X}_v(\mathcal{O}_v) \rightarrow E_v(K_v)$ is a group isomorphism
- (2) $\mathcal{X}_v$ contains an open subscheme $\mathcal{X}_v^0$ such that in the special fibre $\tilde{\mathcal{X}}_v$ contains the subscheme $\tilde{\mathcal{X}}_v^0 \rightarrow \operatorname{Spec} \mathcal{O}_v/\mathfrak{p}_v$, and it satisfies $\tilde{\mathcal{X}}_v^0(\mathcal{O}_v/\mathfrak{p}_v) \cong E_0(K_v)/E_1(K_v)$, where we recall that E_0 denotes the points of $P \in E$ that have nonsingular reduction in the special fibre and $E_1 = \{P \in E(K_v) | \tilde{P} = \tilde{O}\}$ is the points reducing to the zero point.

We note that these Néron models are not necessarily projective, so that they are not projective models, which means that problem of determining bad reduction of a curve is not trivialised.

Using Néron models we can exactly compute the size of $\#E(K_v)/nE(K_v)$ in the following proposition.

Proposition B.0.4. (Shaeffer) - For p prime and $v|p$, let $r := \operatorname{ord}_p(n)$, we have that

$$\#E(K_v)/nE(K_v) = p^{r[K_v:\mathbb{Q}_p]} \cdot \#E(K_v)[n]$$

We cover the highlights of the proof, following along with [Sch96]. From Proposition A.5.3 and Theorem B.0.3 (2), it follows that there an open subscheme $E_v^0 \subseteq E_v$ such that $E_v^0(K_v) \cong \mathcal{X}_v^0(\mathcal{O}_v)$. From [Sch96, p 10], there is a natural morphism $\mathcal{X}_v^0(\mathcal{O}_v) \rightarrow \mathcal{X}_v^0(\mathcal{O}_v/\mathfrak{p}_v^i)$. Let $\mathcal{X}_v^i(\mathcal{O}_v)$ be the kernel of

this morphism, then it is isomorphic to some subgroup $E_v^i(K_v) \subseteq E_v(K_v)$ that fits into the following commutative diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & E_v^i(K_v) & \longrightarrow & E_v(K_v) & \longrightarrow & E_v(K_v)/E_v^i(K_v) & \longrightarrow & 0 \\
& & \downarrow [n] & & \downarrow [n] & & \downarrow [n] & & \\
0 & \longrightarrow & E_v^i(K_v) & \longrightarrow & E_v(K_v) & \longrightarrow & E_v(K_v)/E_v^i(K_v) & \longrightarrow & 0
\end{array}$$

Via the snake lemma we have an exact sequence linking the kernels and cokernels:

$$0 \longrightarrow E_v(K_v)[n] \longrightarrow H^1 \longrightarrow E_v^i(K_v)/nE_v^i(K_v) \longrightarrow E_v(K_v)/nE_v(K_v) \longrightarrow H^2 \longrightarrow 0.$$

Finally, we can compute the size of $E_v(K_v)/nE_v(K_v)$ by computing the other quantities in this exact sequence.

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